

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250281284

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Cartledge; Richard George et al.

ACTIVELY CONTROLLABLE STENT, STENT GRAFT, HEART VALVE AND METHOD OF CONTROLLING SAME

Abstract

A prosthetic heart valve can include a stent frame radially movable between a contracted configuration and a maximum expanded configuration. The stent assembly can include a plurality of longitudinally extending jack strut assemblies. Each jack strut assembly can include a proximal jack strut, a distal jack strut with a first connection part, a second connection part coupled to the first connection part of the distal jack strut, and a jack screw connecting the proximal jack strut, the distal jack strut, and the second connection portion. The distal jack strut can include a smooth bore extending longitudinally at least partially therethrough. The second connection part can include a threaded bore extending longitudinally therethrough. The jack screw can extend through the threaded bore of the second connection part and at least partially through the smooth bore of the distal jack strut.

Inventors: Cartledge; Richard George (Boca Raton, FL), Smith; Kevin W. (Coral Gables, FL), Bales, JR.; Thomas O. (Miami, FL), Deville; Derek Dee (Coral Gables, FL), Kline; Korey (Miami, FL), Mendez; Max Pierre (Miami, FL), Palmer; Matthew A. (Miami, FL), Kirk; Michael Walter (Miami, FL), Rivera; Carlos (Cooper City, FL), Petersen; Eric (Homestead, FL), McBrayer; M. Sean (Miami, FL)

Applicant: Edwards Lifesciences CardiAQ LLC (Irvine, CA)

Family ID: 88416411

Assignee: Edwards Lifesciences CardiAQ LLC (Irvine, CA)

Appl. No.: 19/214526

Filed: May 21, 2025

Related U.S. Application Data

parent US continuation 18198169 20230516 parent-grant-document US 12364596 child US 19214526

parent US continuation 17118382 20201210 parent-grant-document US 11707356 child US 18198169
parent US continuation 15822985 20171127 parent-grant-document US 10874508 child US 17118382
parent US continuation 14278594 20140515 parent-grant-document US 9827093 child US 15822985
parent US continuation-in-part 13772203 20130220 parent-grant-document US 9814611 child US 14278594
parent US continuation-in-part 13656717 20121021 parent-grant-document US 9566178 child US 13772203
parent US continuation-in-part 13656717 20121021 parent-grant-document US 9566178 child US 13656717
us-provisional-application US 61550004 20111021
us-provisional-application US 61585937 20120112
us-provisional-application US 61591753 20120127
us-provisional-application US 61601961 20120222
us-provisional-application US 61601961 20120222
us-provisional-application US 61682558 20120813
us-provisional-application US 61682558 20120813
us-provisional-application US 61717037 20121022
us-provisional-application US 61739711 20121219
us-provisional-application US 61824264 20130516

Publication Classification

Int. Cl.: **A61F2/24** (20060101); **A61B5/00** (20060101); **A61B5/021** (20060101); **A61B5/026** (20060101); **A61F2/844** (20130101); **A61F2/95** (20130101); **A61F2/966** (20130101)

U.S. Cl.:

CPC **A61F2/2418** (20130101); **A61B5/6862** (20130101); **A61F2/2412** (20130101); **A61F2/243** (20130101); **A61F2/2436** (20130101); **A61F2/2439** (20130101); **A61F2/844** (20130101); **A61F2/95** (20130101); A61B5/021 (20130101); A61B5/026 (20130101); A61F2/2427 (20130101); A61F2002/9505 (20130101); A61F2002/9511 (20130101); A61F2/9517 (20200501); A61F2002/9534 (20130101); A61F2002/9665 (20130101); A61F2220/0041 (20130101); A61F2220/0075 (20130101); A61F2220/0091 (20130101); A61F2250/0006 (20130101); A61F2250/0007 (20130101); A61F2250/001 (20130101); A61F2250/0018 (20130101); A61F2250/0039 (20130101); A61F2250/006 (20130101); A61F2250/0065 (20130101); A61F2250/0069 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/198,169, filed May 16, 2023, which is a continuation of U.S. patent application Ser. No. 17/118,382, filed Dec. 10, 2020, now U.S. Pat. No. 11,707,356, which is a continuation of U.S. patent application Ser. No. 15/822,985, filed Nov. 27, 2017, now U.S. Pat. No. 10,874,508, which is a continuation of U.S. patent application Ser. No. 14/278,594, filed May 15, 2014, now U.S. Pat. No. 9,827,093. U.S. patent application Ser. No. 14/278,594 claims the benefit of U.S. Provisional Patent Application No. 61/824,264, filed May 16, 2013, and is a

continuation-in-part of U.S. patent application Ser. No. 13/772,203, filed Feb. 20, 2013, now U.S. Pat. No. 9,814,611, and Ser. No. 13/656,717, filed Oct. 21, 2012, now U.S. Pat. No. 9,566,178. U.S. patent application Ser. No. 13/772,203 claims the benefit of U.S. Provisional Patent Application No. 61/739,711, filed Dec. 19, 2012, 61/717,037, filed Oct. 22, 2012, 61/682,558, filed Aug. 13, 2012, and 61/601,961, filed Feb. 22, 2012. U.S. patent application Ser. No. 13/656,717 claims the benefit of U.S. Provisional Patent Application No. 61/682,558, filed Aug. 13, 2012, 61/601,961, filed Feb. 22, 2012, 61/591,753, filed Jan. 27, 2012, 61/585,937, filed Jan. 12, 2012, and 61/550,004, filed Oct. 21, 2011. The prior applications are incorporated by reference herein in their entireties.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

FIELD OF THE INVENTION

[0003] The present invention lies in the field of stents, stent grafts, heart valves (including aortic, pulmonary, mitral and tricuspid), and methods and systems for controlling and implanting stents, stent grafts and heart valves.

BACKGROUND OF THE INVENTION

[0004] Medical and surgical implants are placed often in anatomic spaces where it is desirable for the implant to conform to the unique anatomy of the targeted anatomic space and secure a seal therein, preferably without disturbing or distorting the unique anatomy of that targeted anatomic space.

[0005] While the lumens of most hollow anatomic spaces are ideally circular, in fact, the cross-sectional configurations of most anatomic spaces are, at best, ovoid, and may be highly irregular. Such luminal irregularity may be due to anatomic variations and/or to pathologic conditions that may change the shape and topography of the lumen and its associated anatomic wall. Examples of anatomic spaces where such implants may be deployed include, but are not limited to, blood vessels, the heart, other vascular structures, and vascular defects (such as thoracic and abdominal aortic aneurysms).

[0006] For a patient to be a candidate for existing endograft methods and technologies, to permit an adequate seal, a proximal neck of, ideally, at least 12 mm of normal aorta must exist downstream of the left subclavian artery for thoracic aortic aneurysms or between the origin of the most inferior renal artery and the origin of the aneurysm in the case of abdominal aneurysms. Similarly, ideally, at least 12 mm of normal vessel must exist distal to the distal extent of the aneurysm for an adequate seal to be achieved. The treatment of Aortic Stenosis through Transcatheter Aortic Valve Replacement (TAVR) is becoming more common. The limitations of current TAVR techniques do not allow for repositioning of the implant once it has been deployed in place. Further, the final expanded diameter of the current devices is fixed making pre-sizing a critical and difficult step.

[0007] Migration of existing endografts has also been a significant clinical problem, potentially causing leakage and profusion of aneurysms and/or compromising necessary vascular supplies to arteries such as the coronary, carotid, subclavian, renal, or internal iliac vessels. This problem only has been addressed partially by some existing endograft designs, in which barbs or hooks have been incorporated to help retain the endograft at its intended site. However, most existing endograft designs are solely dependent on radial force applied by varying length of stent material to secure a seal against the recipient vessel walls.

[0008] Because of the limitations imposed by existing vascular endograft devices and endovascular techniques, a significant number of abdominal and thoracic aneurysms repaired in the U.S. are still managed through open vascular surgery, instead of the lower morbidity of the endovascular approach.

[0009] Pre-sizing is required currently in all prior art endografts. Such pre-sizing based on CAT-scan measurements is a significant problem. This leads, many times, to mis-sized grafts. In such

situations, more graft segments are required to be placed, can require emergency open surgery, and can lead to an unstable seal and/or migration. Currently there exists no endograft that can be fully repositioned after deployment.

[0010] Thus, a need exists to overcome the problems with the prior art systems, designs, and processes as discussed above.

SUMMARY OF THE INVENTION

[0011] The invention provides surgical implant devices and methods for their manufacture and use that overcome the hereinafore-mentioned disadvantages of the heretofore-known devices and methods of this general type and that provide such features with improvements that increase the ability of such an implant to be precisely positioned and sealed, with better in situ accommodation to the local anatomy of the targeted anatomic site. The invention provides an adjustment tool that can remotely actuate an adjustment member(s) that causes a configuration change of a portion(s) of an implant, which configuration change includes but is not limited to diameter, perimeter, shape, and/or geometry or a combination of these, to create a seal and provide retention of an implant to a specific area of a target vessel or structure even when the cross-sectional configuration of the anatomic space is non-circular, ovoid, or irregular.

[0012] The invention provides an actively controllable stent, stent graft, stent graft assembly, heart valve, and heart valve assembly, and methods and systems for controlling and implanting such devices that overcome the hereinafore-mentioned disadvantages of the heretofore-known devices and methods of this general type and that provide such features with control both in opening and closing and in any combination thereof even during a surgical procedure or after completion of a surgical procedure.

[0013] One exemplary aspect of the present invention is directed towards novel designs for endovascular implant grafts, and methods for their use for the treatment of aneurysms (e.g., aortic) and other structural vascular defects. An endograft system for placement in an anatomic structure or blood vessel is disclosed in which an endograft implant comprises, for example, a non-elastic tubular implant body with at least an accommodating proximal end. Accommodating, as used herein, is the ability to vary a configuration in one or more ways, which can include elasticity, expansion, contraction, and changes in geometry. Both or either of the proximal and distal ends in an implant according to the present invention further comprise one or more circumferential expandable sealable collars and one or more expandable sealing devices, capable of being expanded upon deployment to achieve the desired seal between the collar and the vessel's inner wall. Exemplary embodiments of such devices can be found in co-pending U.S. patent application Ser. No. 11/888,009, filed Jul. 31, 2007, and Ser. No. 12/822,291, filed Jun. 24, 2010, which applications have been incorporated herein in their entireties. Further embodiments of endovascular implants and delivery systems and methods according to the present invention may be provided with retractable retention tines or other retention devices allowing an implant to be repositioned before final deployment. In other embodiments, the implant can be repositioned after final deployment. An endograft system according to the present invention further comprises a delivery catheter with an operable tubular sheath capable of housing a folded or compressed endograft implant prior to deployment and capable of retracting or otherwise opening in at least its proximal end to allow implant deployment. The sheath is sized and configured to allow its placement via a peripheral arteriotomy site, and is of appropriate length to allow its advancement into, for example, the aortic valve annulus, ascending aorta, aortic arch, and thoracic or abdominal aorta, as required for a specific application. Sheath movement is provided in a novel manner by manual actuation and/or automatic actuation.

[0014] While some post-implantation remodeling of the aortic neck proximal to an endovascular graft (endograft) has been reported, existing endograft technology does not allow for the management of this condition without placement of an additional endograft sleeve to cover the remodeled segment. Exemplary prostheses of the present invention as described herein allow for

better accommodation by the implant of the local anatomy, using an actively controlled expansion device for the sealing interface between the prosthesis collar and the recipient vessel's inner wall. Furthermore, exemplary prostheses of the present invention as disclosed herein are provided with a controllably releasable disconnect mechanism that allows remote removal of an adjustment tool and locking of the retained sealable mechanism after satisfactory positioning and sealing of the endograft. In some exemplary embodiments according to the present invention, the controllably releasable disconnect mechanism may be provided in a manner that allows post-implantation re-docking of an adjustment member to permit post-implantation repositioning and/or resealing of a prostheses subsequent to its initial deployment.

[0015] Certain aspects of the present invention are directed towards novel designs for sealable endovascular implant grafts and endovascular implants, and methods for their use for the treatment of aortic aneurysms and other structural vascular defects and/or for heart valve replacements. Various embodiments as contemplated within the present invention may include any combination of exemplary elements as disclosed herein or in the co-pending patent applications referenced above.

[0016] In an exemplary embodiment according to the present invention, a sealable vascular endograft system for placement in a vascular defect is provided, comprising an elongated main implant delivery catheter with an external end and an internal end for placement in a blood vessel with internal walls. In such an exemplary embodiment, the main implant delivery catheter further comprises a main implant delivery catheter sheath that may be openable or removable at the internal end and a main implant delivery catheter lumen containing within a compressed or folded endovascular implant. Further, an endovascular implant comprises a non-elastic tubular implant body with an accommodating proximal end terminating in a proximal sealable circumferential collar that may be expanded by the operator to achieve a fluid-tight seal between the proximal sealable circumferential collar and the internal walls of the blood vessel proximal to the vascular defect. Moreover, an endovascular implant may further comprise a non-elastic tubular implant body with an accommodating distal end terminating in a distal sealable circumferential collar controlled by a distal variable sealing device, which may be expanded by the operator to achieve a fluid-tight seal between the distal sealable circumferential collar and the internal walls of the blood vessel distal to the vascular defect.

[0017] In a further exemplary embodiment according to the present invention, an implant interface is provided for a sealable attachment of an implant to a wall within the lumen of a blood vessel or other anatomic conduit.

[0018] In a yet further exemplary embodiment according to the present invention, an implant gasket interface is provided for a sealable attachment of an implant to a wall within the lumen of a blood vessel or other anatomic conduit, wherein the sealable attachment provides for auto-adjustment of the seal while maintaining wall attachment to accommodate post-implantation wall remodeling.

[0019] Still other exemplary embodiments of endografts and endograft delivery systems according to the present invention serve as universal endograft cuffs, being first placed to offer their advantageous anatomic accommodation capabilities, and then serving as a recipient vessel for other endografts, including conventional endografts.

[0020] Furthermore, exemplary embodiments of endografts and endograft delivery systems according to the present invention may be provided with a mechanism to permit transfer of torque or other energy from a remote operator to an adjustment member comprising a sealable, adjustable circumferential assembly controlled by an adjustment tool, which may be detachable therefrom and may further cause the assembly to lock upon detachment of the tool. In some exemplary embodiments of the present invention, the variable sealing device may be provided with a re-docking element that may be recaptured by subsequent operator interaction, allowing redocking and repositioning and/or resealing of the endograft at a time after its initial deployment.

[0021] Moreover, the various exemplary embodiments of the present invention as disclosed herein may constitute complete endograft systems, or they may be used as components of a universal endograft system as disclosed in co-pending patent applications that may allow the benefits of the present invention to be combined with the ability to receive other endografts.

[0022] Additionally, the present invention encompasses sealable devices that may be used in other medical devices such as adjustable vascular cannulas or other medical or surgical devices or implants, such as heart valves.

[0023] With the foregoing and other objects in view, there is provided, in accordance with the invention, a method for implanting a stent includes contracting a self-expanding/forcibly-expanding stent of a shape-memory material set to a given shape to a reduced implantation size with a delivery system having drive wires. The stent has a selectively adjustable assembly with adjustable elements operatively connected to the drive wires such that, when the adjustable elements are adjusted by the drive wires, a configuration change in at least a portion of the self-expanding stent occurs. The contracted stent is inserted into a native annulus in which the stent is to be implanted. The drive wires are rotated with the delivery system to forcibly expand the stent into the native annulus. While rotating the drive wires, a torque applied to the drive wires is determined with the delivery system. Rotation of the drive wires is stopped based upon a value of the determined torque.

[0024] With the objects of the invention in view, there is also provided a method for implanting a stent includes contracting a stent to a reduced implantation size with a delivery system having drive wires. The stent has a selectively adjustable assembly with adjustable elements operatively connected to the drive wires such that, when the adjustable elements are adjusted by the drive wires, a configuration change in at least a portion of the stent occurs. The contracted stent is inserted into a native annulus in which the stent is to be implanted. The drive wires are rotated with the delivery system to forcibly expand the stent into the native annulus. While rotating the drive wires, a torque applied to the drive wires is determined with the delivery system. Rotation of the drive wires is stopped based upon a value of the determined torque.

[0025] With the objects of the invention in view, there is also provided a method for implanting a stent includes contracting a stent to a reduced implantation size with a delivery system having drive wires. The stent has a selectively adjustable assembly with adjustable elements operatively connected to the drive wires such that, when the adjustable elements are adjusted by the drive wires, a configuration change in at least a portion of the stent occurs. The contracted stent is inserted into a native annulus in which the stent is to be implanted. The drive wires are move with the delivery system to forcibly expand the stent into the native annulus. While moving the drive wires, a torque applied to the drive wires is determined with the delivery system. Movement of the drive wires is stopped based upon a value of the determined torque.

[0026] In accordance with another mode of the invention, a user is provided with a dynamic value of the torque and permitting the user to change the expansion and contraction of the stent.

[0027] In accordance with a further mode of the invention, the stent is disconnected from the delivery system to implant the stent in the native annulus.

[0028] In accordance with an added mode of the invention, the delivery system has at least one drive wire motor connected to the drive wires for rotating the drive wires and the stopping step is carried out by measuring a current required to drive the at least one drive wire motor and stopping the at least one drive wire motor and thereby the rotation of the drive wires based upon a value of the current.

[0029] In accordance with an additional mode of the invention, an outward radial force imposed by the expanding stent lattice on the native annulus is calculated with the value of the current and the at least one drive wire motor and thereby the rotation of the drive wires is stopped based upon a value of the calculated outward radial force.

[0030] In accordance with a concomitant mode of the invention, the delivery system has at least

one drive wire motor connected to the drive wires for rotating the drive wires and the stopping step is carried out by determining an outward radial force imposed by the expanding stent lattice based upon a current required to drive the at least one drive wire motor and stopping the at least one drive wire motor and thereby the rotation of the drive wires based upon a value of the a value of the calculated outward radial force.

[0031] Although the invention is illustrated and described herein as embodied in an actively controllable stent, stent graft, stent graft assembly, heart valve, and heart valve assembly, and methods and systems for controlling and implanting such devices, it is, nevertheless, not intended to be limited to the details shown because various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

[0032] Additional advantages and other features characteristic of the present invention will be set forth in the detailed description that follows and may be apparent from the detailed description or may be learned by practice of exemplary embodiments of the invention. Still other advantages of the invention may be realized by any of the instrumentalities, methods, or combinations particularly pointed out in the claims.

[0033] Other features that are considered as characteristic for the invention are set forth in the appended claims. As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, which are not true to scale, and which, together with the detailed description below, are incorporated in and form part of the specification, serve to illustrate further various embodiments and to explain various principles and advantages all in accordance with the present invention. Advantages of embodiments of the present invention will be apparent from the following detailed description of the exemplary embodiments thereof, which description should be considered in conjunction with the accompanying drawings in which:

[0035] FIG. 1 is a fragmentary, partially longitudinally cross-sectional, side elevational view of an exemplary embodiment of an actively controllable stent/stent graft deployment system of the present invention in a non-deployed state with a front half of the outer catheter removed;

[0036] FIG. 2 is a fragmentary, side elevational view of an enlarged distal portion of the stent deployment system of FIG. 1;

[0037] FIG. 3 is a fragmentary, perspective view of the stent deployment system of FIG. 1 from above the distal end;

[0038] FIG. 4 is a fragmentary, perspective view of the stent deployment system of FIG. 1 from

above the distal end with the system in a partially deployed state;

[0039] FIG. 5 is a fragmentary, side elevational view of the stent deployment system of FIG. 2 in a partially deployed state;

[0040] FIG. 6 is a top plan view of a drive portion of the stent deployment system of FIG. 2;

[0041] FIG. 7 is a fragmentary, longitudinally cross-sectional view of a rear half of the stent deployment system of FIG. 6;

[0042] FIG. 8 is a fragmentary, perspective view of the stent deployment system of FIG. 6;

[0043] FIG. 9 is a fragmentary, perspective view of the stent deployment system of FIG. 1 from above the distal end with the system in an expanded state and with the assembly-fixed needles in an extended state;

[0044] FIG. 10 is a fragmentary, longitudinal cross-sectional view of the stent deployment system of FIG. 9 showing the rear half in a partially expanded state of the stent lattice;

[0045] FIG. 11 is a fragmentary, longitudinal cross-sectional view of the stent deployment system of FIG. 10 showing the front half in a further expanded state;

[0046] FIG. 12 is a fragmentary, longitudinal cross-sectional view of the stent deployment system of FIG. 11 with a deployment control assembly in a partially disengaged state;

[0047] FIG. 13 is a fragmentary, longitudinally cross-sectional view of the stent deployment system of FIG. 12 with the deployment control assembly in a disengaged state;

[0048] FIG. 14 is a fragmentary, longitudinally cross-sectional view of an enlarged portion of the stent deployment system of FIG. 12 in the partially disengaged state;

[0049] FIG. 15 is a fragmentary, longitudinally cross-sectional view of an enlarged portion of the stent deployment system of FIG. 13 in a disengaged state;

[0050] FIG. 16 is a fragmentary, partially cross-sectional, side elevational view of the stent deployment system of FIG. 9 rotated about a longitudinal axis, with the deployment control assembly in the disengaged state, and showing a cross-section of a portion of the deployment control assembly;

[0051] FIG. 17 is a fragmentary, longitudinally cross-sectional view of the stent deployment system of FIG. 16 showing a cross-section of a drive portion of a stent assembly with a fixed needle;

[0052] FIG. 18 is a fragmentary, perspective view of the stent deployment system of FIG. 16;

[0053] FIG. 19 is a fragmentary, perspective view of an enlarged portion of the stent deployment system of FIG. 18;

[0054] FIG. 20 is a fragmentary, perspective view of the stent deployment system of FIG. 18 with a diagrammatic illustration of paths of travel of strut crossing points as the stent is moved between its expanded and contracted states;

[0055] FIG. 21 is a fragmentary, side elevational view from an outer side of an alternative exemplary embodiment of a jack assembly according to the invention in a stent-contracted state with a drive sub-assembly in a connected state and with a needle sub-assembly in a retracted state;

[0056] FIG. 22 is a fragmentary, cross-sectional view of the jack assembly of FIG. 21;

[0057] FIG. 23 is a fragmentary, cross-sectional view of the jack assembly of FIG. 21 in a partially stent-expanded state;

[0058] FIG. 24 is a fragmentary, cross-sectional view of the jack assembly of FIG. 23 with a needle pusher in a partially actuated state before extension of the needle;

[0059] FIG. 25 is a fragmentary, cross-sectional view of the jack assembly of FIG. 24 with the needle pusher in another partially actuated state with the needle pusher in another partially actuated state with an extension of the needle;

[0060] FIG. 26 is a fragmentary, cross-sectional view of the jack assembly of FIG. 25 with the drive sub-assembly in a partially disconnected state without retraction of the needle pusher;

[0061] FIG. 27 is a fragmentary, cross-sectional view of the jack assembly of FIG. 26 with the drive sub-assembly in a further partially disconnected state with partial retraction of the needle pusher;

[0062] FIG. **28** is a fragmentary, cross-sectional view of the jack assembly of FIG. **27** with the drive sub-assembly in a still a further partially disconnected state with further retraction of the needle pusher;

[0063] FIG. **29** is a fragmentary, cross-sectional view of the jack assembly of FIG. **23** with the drive sub-assembly and the needle pusher in a disconnected state;

[0064] FIG. **30** is a fragmentary, cross-sectional view of another alternative exemplary embodiment of a jack assembly according to the invention in a stent-contracted state with a drive sub-assembly in a connected state and with a needle sub-assembly in a retracted state;

[0065] FIG. **31** is a fragmentary, cross-sectional view of the jack assembly of FIG. **30** in a partially stent-expanded state;

[0066] FIG. **32** is a fragmentary, cross-sectional view of the jack assembly of FIG. **31** with the needle sub-assembly in an actuated state with extension of the needle;

[0067] FIG. **33** is a fragmentary, cross-sectional view of the jack assembly of FIG. **32** with the drive sub-assembly in a disconnected state and the needle sub-assembly in a disconnected state;

[0068] FIG. **34** is a fragmentary, perspective view of the jack assembly of FIG. **33** with the extended needle rotated slightly to the right of the figure.

[0069] FIG. **35** is a fragmentary, perspective view of the jack assembly of FIG. **34** rotated to the right by approximately 45 degrees;

[0070] FIG. **36** is a fragmentary, partially cross-sectional, perspective view from above the jack assembly of FIG. **30** showing the interior of the distal drive block;

[0071] FIG. **37** is a fragmentary, enlarged, cross-sectional view of the jack assembly of FIG. **33**;

[0072] FIG. **38** is a photograph of a perspective view from above the upstream end of another exemplary embodiment of an actively controllable stent graft according to the invention in a substantially contracted state;

[0073] FIG. **39** is a photograph of a perspective view of the stent graft of FIG. **38** in a partially expanded state;

[0074] FIG. **40** is a photograph of a perspective view of the stent graft of FIG. **38** in an expanded state;

[0075] FIG. **41** is a photograph of a side perspective view of the stent graft of FIG. **38** in an expanded state;

[0076] FIG. **42** is a photograph of a perspective view of another exemplary embodiment of an actively controllable stent for a stent graft according to the invention in a substantially expanded state with integral upstream anchors;

[0077] FIG. **43** is a photograph of a perspective view of the stent of FIG. **42** in a partially expanded state;

[0078] FIG. **44** is a photograph of a perspective view of the stent of FIG. **42** in another partially expanded state;

[0079] FIG. **45** is a photograph of a perspective view of the stent of FIG. **42** in a substantially contracted state;

[0080] FIG. **46** is a photograph of a side perspective view of another exemplary embodiment of an actively controllable stent for a stent graft according to the invention in a substantially expanded state with a tapered outer exterior;

[0081] FIG. **47** is a photograph of a top perspective view of the stent of FIG. **46**;

[0082] FIG. **48** is a photograph of a perspective view of the stent of FIG. **46** from above a side;

[0083] FIG. **49** is a photograph of a perspective view of the stent of FIG. **46** from above a side with the stent in a partially expanded state;

[0084] FIG. **50** is a photograph of a perspective view of the stent of FIG. **46** from above a side with the stent in a substantially contracted state;

[0085] FIG. **51** is a photograph of an exemplary embodiment of a low-profile joint assembly for actively controllable stents/stent grafts according to the invention;

[0086] FIG. **52** is a photograph of struts of the joint assembly of FIG. **51** separated from one another;

[0087] FIG. **53** is a photograph of a rivet of the joint assembly of FIG. **51**;

[0088] FIG. **54** is a fragmentary, side perspective view of another exemplary embodiment of an actively controllable stent system for a stent graft according to the invention in a substantially expanded state with a tapered outer exterior;

[0089] FIG. **55** is a side perspective view of the stent system of FIG. **54**;

[0090] FIG. **56** is a side elevational view of the stent system of FIG. **54**;

[0091] FIG. **57** is a side elevational view of the stent system of FIG. **54** in a substantially contracted state;

[0092] FIG. **58** is a side elevational view of another exemplary embodiment of a portion of an actively controllable stent system for a stent graft according to the invention in a substantially contracted state;

[0093] FIG. **59** is a perspective view of the stent system portion of FIG. **58**;

[0094] FIG. **60** is a top plan view of the stent system portion of FIG. **58**;

[0095] FIG. **61** is a side perspective view of the stent system portion of FIG. **58** in a partially expanded state;

[0096] FIG. **62** is a top plan view of the stent system portion of FIG. **61**;

[0097] FIG. **63** is a side elevational view of the stent system portion of FIG. **61**;

[0098] FIG. **64** is a perspective view of a downstream side of an exemplary embodiment of a replacement valve assembly according to the invention in an expanded state;

[0099] FIG. **65** is a side elevational view of the valve assembly of FIG. **64**;

[0100] FIG. **66** is a fragmentary, perspective view of a delivery system according to the invention for the aortic valve assembly of FIG. **64** with the aortic valve assembly in the process of being implanted and in the right iliac artery;

[0101] FIG. **67** is a fragmentary, perspective view of the delivery system and aortic valve assembly of FIG. **66** with the aortic valve assembly in the process of being implanted and in the abdominal aorta;

[0102] FIG. **68** is a fragmentary, perspective view of the delivery system and aortic valve assembly of FIG. **66** with the aortic valve assembly in the process of being implanted and being adjacent the aortic valve implantation site;

[0103] FIG. **69** is a fragmentary, perspective view of the delivery system and aortic valve assembly of FIG. **66** with the aortic valve assembly implanted in the heart;

[0104] FIG. **70** is a fragmentary, enlarged, perspective view of the delivery system and the aortic valve assembly of FIG. **69** implanted at an aortic valve implantation site;

[0105] FIG. **71** is a perspective view of a side of another exemplary embodiment of a replacement aortic valve assembly according to the invention in an expanded state;

[0106] FIG. **72** is a perspective view of the replacement aortic valve assembly of FIG. **71** from above a downstream side thereof;

[0107] FIG. **73** is a perspective view of the replacement aortic valve assembly of FIG. **71** from above a downstream end thereof;

[0108] FIG. **74** is a perspective view of the replacement aortic valve assembly of FIG. **71** from below an upstream end thereof;

[0109] FIG. **75** is a perspective view of an enlarged portion of the replacement aortic valve assembly of FIG. **74**;

[0110] FIG. **76** is a perspective view of the replacement aortic valve assembly of FIG. **71** from a side thereof with the graft material removed;

[0111] FIG. **77** is a perspective view of the replacement aortic valve assembly of FIG. **76** from above a downstream side thereof;

[0112] FIG. **78** is a side elevation, vertical cross-sectional view of the replacement aortic valve

assembly of FIG. **76**;

[0113] FIG. **79** is a perspective view of the replacement aortic valve assembly of FIG. **76** from a side thereof with the valve material removed, with the stent lattice in an expanded state;

[0114] FIG. **80** is a perspective view of the replacement aortic valve assembly of FIG. **79** with the stent lattice in an intermediate expanded state;

[0115] FIG. **81** is a perspective view of the replacement aortic valve assembly of FIG. **79** with the stent lattice in an almost contracted state;

[0116] FIG. **82** is a downstream plan view of the replacement aortic valve assembly of FIG. **79** in an intermediate expanded state;

[0117] FIG. **83** is an enlarged downstream plan view of a portion of the replacement aortic valve assembly of FIG. **79** in an expanded state;

[0118] FIG. **84** is a side elevational view of the replacement aortic valve assembly of FIG. **79** in an expanded state, with graft material removed, and with distal portions of an exemplary embodiment of a valve delivery system;

[0119] FIG. **85** is a perspective view of an exemplary embodiment of a jack assembly of the replacement aortic valve assembly of FIG. **84** from a side thereof with the valve delivery system sectioned;

[0120] FIG. **86** is a perspective view of the replacement aortic valve assembly of FIG. **79** in an expanded state, with graft material removed, and with distal portions of another exemplary embodiment of a valve delivery system;

[0121] FIG. **87** is a fragmentary, enlarged perspective view of the replacement aortic valve assembly of FIG. **86** with graft material shown;

[0122] FIG. **88** is a fragmentary, enlarged, perspective view of the delivery system and the aortic valve assembly of FIG. **71** implanted at an aortic valve implantation site;

[0123] FIG. **89** is a fragmentary, side elevational view of another exemplary embodiment of an actively controllable and tiltable stent graft system according to the invention in a partially expanded state and a non-tilted state;

[0124] FIG. **90** is a fragmentary, side elevational view of the system of FIG. **89** in a partially tilted state from a front thereof;

[0125] FIG. **91** is a fragmentary, side elevational view of the system of FIG. **90** in another partially tilted state;

[0126] FIG. **92** is a fragmentary, side elevational view of the system of FIG. **90** in yet another partially tilted state;

[0127] FIG. **93** is a fragmentary, perspective view of the system of FIG. **90** in yet another partially tilted state;

[0128] FIG. **94** is a fragmentary, partially cross-sectional, side elevational view of another exemplary embodiment of an actively controllable and tiltable stent graft system according to the invention in an expanded state and a partially front-side tilted state

[0129] FIG. **95** is a fragmentary, perspective view of the system of FIG. **94** in a non-tilted state;

[0130] FIG. **96** is a fragmentary, side elevational view of the system of FIG. **94** in a non-tilted state;

[0131] FIG. **97** is a fragmentary, side elevational view of the system of FIG. **96** rotated approximately 90 degrees with respect to the view of FIG. **96**;

[0132] FIG. **98** is a fragmentary, longitudinally cross-sectional, side elevational view of the system of FIG. **94** showing the rear half of the system and a tubular graft material in a non-tilted state and partially expanded state;

[0133] FIG. **99** is fragmentary, partially cross-sectional, perspective view of the system of FIG. **94** showing the rear half of the tubular graft material and in a non-tilted state and a partially expanded state;

[0134] FIG. **100** is a fragmentary, partially cross-sectional, side elevational view of the system of FIG. **94** showing the rear half of graft material for a bifurcated vessel and in a non-tilted state;

[0135] FIG. **101** is a fragmentary, partially cross-sectional, side elevational view of the system of FIG. **100** in an expanded state and a partially tilted state;

[0136] FIG. **102** is a fragmentary, partially cross-sectional, side elevational view of the system of FIG. **101** rotated approximately 45 degrees with respect to the view of FIG. **101**;

[0137] FIG. **103** is a fragmentary, side perspective view of another exemplary embodiment of an actively controllable stent graft system according to the invention in an expanded state;

[0138] FIG. **104** is a fragmentary, side elevational view of the system of FIG. **103**;

[0139] FIG. **105** is a fragmentary, front elevational and partially cross-sectional view of a self-contained, self-powered, actively controllable stent graft delivery and integral control system according to the invention with the prosthesis in an expanded state with the graft material in cross-section showing a rear half thereof;

[0140] FIG. **106** is a perspective view of the control portion of the system of FIG. **105** as a wireless sub-system;

[0141] FIG. **107** is a fragmentary, front elevational view of another exemplary embodiment of a self-contained, self-powered, actively controllable stent graft delivery and separate tethered control system according to the invention with different controls and with the prosthesis in an expanded state;

[0142] FIG. **108** is a fragmentary, perspective view of a control handle of an exemplary embodiment of a self-contained, self-powered, actively controllable prosthesis delivery device according to the invention from above a left side thereof with the upper handle half and power pack removed;

[0143] FIG. **109** is a fragmentary, vertically cross-sectional view of the handle of FIG. **108** with the power pack removed;

[0144] FIG. **110** is a fragmentary, enlarged, vertically cross-sectional and perspective view of a sheath-movement portion of the handle of FIG. **108** from above a left side thereof;

[0145] FIG. **111** is a fragmentary, further enlarged, vertically cross-sectional view of the sheath-movement portion of FIG. **110** from below a left side thereof;

[0146] FIG. **112** is a fragmentary, enlarged, vertically cross-sectional view of a power portion of the handle of FIG. **108** viewed from a proximal side thereof;

[0147] FIG. **113** is a fragmentary, perspective view of a needle control portion of the handle of FIG. **108** from above a distal side with the upper handle half and power pack removed and with the needle control in a lattice-contracted and needle-stowed position;

[0148] FIG. **114** is a fragmentary, perspective view of the needle control portion of the handle of FIG. **113** with the needle control in a lattice-expanded and needle-stowed position;

[0149] FIG. **115** is a fragmentary, perspective view of the needle control portion of the handle of FIG. **114** with the needle control in a needle-extended position;

[0150] FIG. **116** is a fragmentary, perspective view of an engine portion of the handle of FIG. **108** from above a left side thereof with the upper handle half removed;

[0151] FIG. **117** is a fragmentary, enlarged, vertically cross-sectional view of the engine portion of FIG. **116** viewed from a proximal side thereof;

[0152] FIG. **118** is a fragmentary, enlarged, vertically cross-sectional view of the engine portion of the handle portion of FIG. **117** viewed from a distal side thereof;

[0153] FIG. **119** is a flow diagram of an exemplary embodiment of a procedure for implanting an abdominal aorta prosthesis according to the invention;

[0154] FIG. **120** is a perspective view of an exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having nine lattice segments in a native, self-expanded position with jack screw assemblies disposed between adjacent pairs of repeating portions of the lattice, with jack screws through a wall of the lattice, and with each jack screw backed out in a thread-non-engaged state to allow crimp of lattice for loading into a stent delivery system;

[0155] FIG. **121** is a perspective view of the lattice of FIG. **120** in a contracted/crimped state for loading into the stent delivery system with each jack screw in a thread-non-engaged state;

[0156] FIG. **122** is a perspective view of the lattice of FIG. **121** after being allowed to return to the native position of the lattice in a deployment site with each jack screw in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0157] FIG. **123** is a perspective view of the lattice of FIG. **122** partially expanded from the state shown in FIG. **122** with each jack screw in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0158] FIG. **124** is a tilted perspective view of the lattice of FIG. **123** partially expanded from the state shown in FIG. **123** with each jack screw in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0159] FIG. **125** is a perspective view of the lattice of FIG. **124** further expanded near a maximum expansion of the lattice with each jack screw in a thread-engaged state;

[0160] FIG. **126** is a fragmentary, enlarged perspective and longitudinal cross-sectional view of a portion of two adjacent halves of repeating portions of an alternative exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly with a separate jack screw assembly connecting the two adjacent halves and with a lattice-disconnect tube of a stent delivery system in an engaged state covering a pair of drive screw coupler parts therein and with the jack screw in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0161] FIG. **127** is a fragmentary, further enlarged portion of the two adjacent halves of the repeating portions and intermediate jack screw assembly of FIG. **125** with the disconnect tube in a disengaged state with respect to the pair of drive screw coupler parts;

[0162] FIG. **128** is a fragmentary enlarged portion of the two adjacent halves of the repeating portions and intermediate jack screw assembly of FIG. **125** with the disconnect tube in a disengaged state and with the pair of drive screw coupler parts disconnected from one another;

[0163] FIG. **129** is a perspective view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having nine separate lattice segments with an exemplary embodiment of a proximal disconnect block of a stent delivery system as an alternative to the disconnect tube of FIGS. **126** to **128** with the proximal disconnect block in an engaged state covering a pair of drive screw coupler parts therein and with each jack screw in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0164] FIG. **130** is a perspective view of the lattice of FIG. **129** with the proximal disconnect blocks of the delivery system disconnected from the lattice with the proximal disconnect block in a disengaged state with respect to the pair of drive screw coupler parts and illustrating how all of the pairs of drive screw coupler parts can be coupled for simultaneous release;

[0165] FIG. **131** is a perspective view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having nine separate lattice segments connected to intermediate tubes for jack screws with each jack screw in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0166] FIG. **132** is a top plan view of the lattice of FIG. **131**;

[0167] FIG. **133** is a perspective view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having nine lattice segments with locally thicker sections of lattice to accommodate and connect to non-illustrated jack screw assemblies;

[0168] FIG. **134** is a perspective view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having nine lattice segments with bent-over tabs for connecting to non-illustrated jack screw assemblies;

[0169] FIG. **135** is a perspective view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable valve assembly having six lattice segments

in an expanded position with jack screw assemblies disposed between adjacent pairs of repeating portions of the lattice and having three valve leaflets and jack screws through a wall of the lattice in a thread-non-engaged state of the jack screw;

[0170] FIG. **136** is a plan view of the valve assembly of FIG. **135**;

[0171] FIG. **137** is a perspective view of the valve assembly of FIG. **135** in a partially compressed state of the lattice without the valve leaflets and with each jack screw in a thread-non-engaged state;

[0172] FIG. **138** is a perspective view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable valve assembly having six lattice segments in a native, self-expanded position with jack screw assemblies attached at an interior surface between adjacent pairs of segments of the lattice without the valve leaflets and with each of the jack screws in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0173] FIG. **139** is a perspective view of the lattice of FIG. **138** in a contracted/crimped state for loading into the stent delivery system with each jack screw in a thread-non-engaged state;

[0174] FIG. **140** is a tilted perspective view of the lattice of FIG. **138**;

[0175] FIG. **141** is a perspective view of the lattice of FIG. **138** partially expanded from the state shown in FIG. **138** with each jack screw in an engaged state for further outward expansion or inward contraction of the lattice;

[0176] FIG. **142** is a perspective view of the lattice of FIG. **138** further expanded near a maximum expansion of the lattice with each jack screw in an engaged state for further outward expansion or inward contraction of the lattice;

[0177] FIG. **143** is a side elevational view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having nine lattice segments in a native, self-expanded position with jack screw assemblies integral with the stent assembly and with each of the jack screws in a thread-engaged state for outward expansion and inward contraction of the lattice and with a portion of the stent assembly delivery system having connector control tubes with one connector control tube shown in transparent form;

[0178] FIG. **144** is a top plan view of the lattice of FIG. **143**;

[0179] FIG. **145** is a perspective view of the lattice of FIG. **143** from above;

[0180] FIG. **146** is a side elevational view of the lattice of FIG. **143** with the connector control tubes of the delivery system in a non-engaged state and the respective connector portions of the jack screw assemblies and the delivery system shown in a disconnected state after implantation;

[0181] FIG. **147** is an enlarged, fragmentary, perspective view of a portion of the lattice of FIG. **143** from outside a side thereof;

[0182] FIG. **148** is an enlarged, fragmentary, perspective view of a portion of the lattice of FIG. **143** from above a top thereof;

[0183] FIG. **149** is a perspective view of the lattice of FIG. **143** from above a side thereof with the lattice expanded by the jack screw assemblies almost to a fullest expanded extent;

[0184] FIG. **150** is a perspective view of the lattice of FIG. **143** from a side thereof with the lattice contracted by the jack screw assemblies almost to a fullest contracted extent;

[0185] FIG. **151** is a perspective view of the lattice of FIG. **150** from a side thereof tilted with respect to FIG. **150**;

[0186] FIG. **152** is a fragmentary, enlarged, perspective view of an upper portion of the lattice of FIG. **151**;

[0187] FIG. **153** is a fragmentary, enlarged, perspective and vertical cross-sectional view of an intermediate portion of the lattice of FIG. **150**;

[0188] FIG. **154** is a perspective view of the lattice of FIG. **143** before manufacture of the stent assembly and illustrating one exemplary embodiment for manufacturing the lattice of the stent assembly;

[0189] FIG. **155** is a side elevational view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having six lattice segments in a partially expanded state with each of the jack screws in a thread-engaged state for further outward expansion and in a slacked state for inward contraction of the lattice, with jack screw assemblies integral with the stent assembly through key-hole slots in the lattice, and with an alternative exemplary embodiment of outer lattice fixation paddles bent outwards to shape the lattice into a longitudinal hourglass;

[0190] FIG. **156** is a top plan view of the implantable stent assembly of FIG. **155** showing the key-hole slots in the lattice for the jack screw assemblies;

[0191] FIG. **157** is a perspective view of the lattice of FIG. **156** from above a side of the top thereof;

[0192] FIG. **158** is a top plan view of the lattice of FIG. **156**;

[0193] FIG. **159** is an enlarged, fragmentary, perspective view of a portion of a top of the lattice of FIG. **156**;

[0194] FIG. **160** is a perspective view of an upper portion of the lattice of FIG. **156** from the outside of a side thereof;

[0195] FIG. **161** is a side elevational view of the lattice of FIG. **156** in a self-expanded, natural state with each of the jack screws in a thread-engaged state for outward expansion and in a slacked state for inward contraction of the lattice;

[0196] FIG. **162** is a side elevational view of the lattice of FIG. **161** in a self-expanded, natural state with each of the jack screws in a thread-engaged state for inward contraction and in a slacked state for outward expansion of the lattice;

[0197] FIG. **163** is a side elevational view of the lattice of FIG. **162** in a forcibly contracted state with each of the jack screws in a thread-engaged state for further inward contraction or outward expansion of the lattice;

[0198] FIG. **164** is a side elevational view of the lattice of FIG. **163** in a forcibly contracted state by a delivery sheath;

[0199] FIG. **165** is a side elevational view of the lattice of FIG. **161** in a forcibly expanded state with each of the jack screws in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0200] FIG. **166** is a side elevational view of the lattice of FIG. **165** in a further forcibly expanded state with each of the jack screws in a thread-engaged state for further outward expansion or inward contraction of the lattice;

[0201] FIG. **167** is a side elevational view of another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly having six lattice segments in a forcibly expanded position with jack screw assemblies having intermediate jack screw nuts longitudinally staggered about the circumference of the lattice;

[0202] FIG. **168** is a fragmentary, perspective view of the lattice of FIG. **167** in a forcibly contracted position showing the staggered positions of the jack screw nuts;

[0203] FIG. **169** is a fragmentary, perspective view of a distal end of an exemplary embodiment delivery system containing the lattice of FIGS. **156** to **166** in a forcibly expanded and implantation-ready state;

[0204] FIG. **170** is a fragmentary, side elevational view of the delivery system and lattice of FIG. **169** with the connector control sub-assembly in a lattice-connected state;

[0205] FIG. **171** is a fragmentary, side elevational view of the delivery system and lattice of FIG. **169** with the connector control sub-assembly in a lattice-disconnected state with each of the disconnect tubes respectively retracted proximally from each of the jack-screw-connector pairs but before the jack-screw-connector pairs disconnect from one another;

[0206] FIG. **172** is a fragmentary, side elevational view of the delivery system and lattice of FIG. **169** with the connector control sub-assembly in a lattice-disconnected state with each of the jack-

screw-connector pairs disconnected from one another and with the jack-screw-connector portion of the delivery system separated from the lattice;

[0207] FIG. **173** is a fragmentary, enlarged perspective view of the connector-control portion of the delivery system and lattice of FIGS. **169** to **172** with two control coils for two connector tubes removed to show distal and proximal sleeves residing in respective counter-bores of a tube-control puck and with a distal portion of the respective two jack-screw-control wires removed;

[0208] FIG. **174** is a photograph of a fragmentary, perspective view from a side of an exemplary embodiment of a delivery system and lattice of FIGS. **167** to **168** in a forcibly expanded state of the lattice;

[0209] FIG. **175** is a photograph of a fragmentary, perspective view from a side of the delivery system and lattice of FIG. **174** rotated with respect to FIG. **174**;

[0210] FIG. **176** is a photograph of a fragmentary, perspective view from a side of the delivery system and lattice of FIG. **174**;

[0211] FIG. **177** is a photograph of a fragmentary, perspective view from a side of the delivery system and lattice of FIG. **174** rotated with respect to FIG. **174**;

[0212] FIG. **178** is a photograph of a fragmentary, perspective view from a side of a lattice control portion of the delivery system of FIG. **174**;

[0213] FIG. **179** is a photograph of a fragmentary, enlarged, perspective view from a side of a distal portion of the lattice control portion of the delivery system of FIG. **178**;

[0214] FIG. **180** is a photograph of a fragmentary, perspective view from a side of a proximal portion of the lattice control portion of the delivery system of FIG. **178**;

[0215] FIG. **181** is a photograph of a perspective view from a side of an exemplary embodiment of a self-expanding/forcibly-expanding implantable heart valve assembly having nine lattice segments in an expanded state and with valve leaflets in an open state;

[0216] FIG. **182** is a photograph of a side view of the heart valve assembly of FIG. **181**;

[0217] FIG. **183** is a photograph of a side view of the heart valve assembly of FIG. **181** rotated with respect to the view shown in FIG. **182**;

[0218] FIG. **184** is a photograph of a side view of the heart valve assembly of FIG. **183**;

[0219] FIG. **185** is a photograph of an upstream plan view of the heart valve assembly of FIG. **181**;

[0220] FIG. **186** is a photograph of a downstream plan view of the heart valve assembly of FIG. **181**;

[0221] FIG. **187** is a photograph of a downstream plan view of an exemplary embodiment of a self-expanding/forcibly-expanding implantable heart valve assembly having six lattice segments in an expanded state and with valve leaflets in an open state;

[0222] FIG. **188** is a photograph of a valve leaflet assembly of the heart valve assembly of FIG. **187**;

[0223] FIG. **189** is a photograph of a downstream perspective view of the heart valve assembly of FIG. **187**;

[0224] FIG. **190** is a photograph of a side perspective view of the heart valve assembly of FIG. **187**;

[0225] FIG. **191** is a photograph of a downstream perspective view of the heart valve assembly of FIG. **187** forcibly expanded in an exemplary embodiment of a delivery system;

[0226] FIG. **192** is a photograph of a perspective view of the heart valve assembly of FIG. **187**;

[0227] FIG. **193** is a photograph of an enlarged, perspective view from a side of an exemplary embodiment of a self-expanding/forcibly-expanding implantable heart valve assembly;

[0228] FIG. **194** is a photograph of an enlarged perspective view from a side of the heart valve assembly of FIG. **193** rotated with respect to the view of FIG. **193**;

[0229] FIG. **195** is a photograph of an enlarged portion of an exemplary embodiment of a graft portion of a heart valve assembly in an unstretched state;

[0230] FIG. **196** is a photograph of a further enlarged first portion of the graft of FIG. **195**;

[0231] FIG. **197** is a photograph of a further enlarged second portion of the graft of FIG. **195**;

[0232] FIG. **198** is a photograph of the graft portion of the heart valve assembly of FIG. **195** in stretched with a 100% extension;

[0233] FIG. **199** is a photograph of the graft portion of the heart valve assembly of FIG. **198** after the stretch is removed;

[0234] FIG. **200** is a cross-sectional view of an exemplary embodiment of an adjustable valve leaflet sub-assembly of a heart valve assembly;

[0235] FIG. **201** is a cross-sectional view of another exemplary embodiment of an adjustable valve leaflet sub-assembly of a heart valve assembly;

[0236] FIG. **202** is a cross-sectional view of another exemplary embodiment of an adjustable valve leaflet sub-assembly of a heart valve assembly;

[0237] FIG. **203** is a side elevational view of an exemplary embodiment of an adjustment shim that, when moved longitudinally, takes up more or lets out more of the valve leaflet edge to shorten or lengthen the overlap portions of the valve leaflets.

[0238] FIG. **204** is a fragmentary, perspective view of an exemplary embodiment of a self-expanding/forcibly-expanding implantable stent assembly having six lattice segments in an expanded state and with an alternative embodiment of a jack screw assembly having an outer-facing jack screw keyhole;

[0239] FIG. **205** is a side elevational view of the implantable stent assembly of FIG. **204** with a valve sub-assembly;

[0240] FIG. **206** is a fragmentary, enlarged portion of the outer-facing jack screw keyhole of the stent assembly of FIG. **204**;

[0241] FIG. **207** is a fragmentary, enlarged, top plan and partially hidden view of an inner-facing jack screw keyhole;

[0242] FIG. **208** is a fragmentary, enlarged, perspective view of the stent assembly of FIG. **204** with an exemplary embodiment of a valve leaflet commissure connector;

[0243] FIG. **209** is a side perspective view of an exemplary embodiment of a self-expanding/forcibly-expanding implantable valve assembly having six lattice segments in an expanded state and with an alternative embodiment of securing the valve sub-assembly to the lattice and graft;

[0244] FIG. **210** is a perspective view of a downstream side of the self-expanding/forcibly-expanding implantable valve assembly of FIG. **209**;

[0245] FIG. **211** is a perspective view of an upstream side of the self-expanding/forcibly-expanding implantable valve assembly of FIG. **209**;

[0246] FIG. **212** is a photograph of an exemplary embodiment of a self-expanding/forcibly-expanding implantable valve assembly connected to a distal end of an exemplary embodiment of a delivery system with the valve assembly and the delivery approximately in an implantation state;

[0247] FIG. **213** is a photograph of the valve assembly and delivery system of FIG. **212** with the valve assembly in an intermediate re-sheathing state where the lattice-disconnector tubes partially re-sheathed;

[0248] FIG. **214** is a photograph of the valve assembly and delivery system of FIG. **212** with the valve assembly in an intermediate re-sheathing state where the lattice-disconnector tubes are re-sheathed and a proximal portion of the valve assembly is re-sheathed;

[0249] FIG. **215** is a photograph of the valve assembly and delivery system of FIG. **212** with the valve assembly in an intermediate re-sheathing state where the valve assembly is half re-sheathed;

[0250] FIG. **216** is a photograph of the valve assembly and delivery system of FIG. **212** with the valve assembly in an intermediate re-sheathing state where the valve assembly is approximately three-quarters re-sheathed;

[0251] FIG. **217** is a photograph of the valve assembly and delivery system of FIG. **212** with the valve assembly re-sheathed into the delivery catheter;

[0252] FIG. **218** is a photograph of a manufacturing process for creating a distal end of an exemplary embodiment of a delivery catheter sized to fit within an 18-French hole;

[0253] FIG. **219** is a photograph of an end plan view of the distal end of the delivery catheter of FIG. **218** without the remainder of the delivery system;

[0254] FIG. **220** is a photograph of a side perspective view of the distal end of the delivery catheter of FIG. **218** after the valve assembly has been extended and/or re-sheathed;

[0255] FIG. **221** is a photograph of an exemplary embodiment of a self-expanding/forcibly-expanding implantable stent assembly having six lattice segments in an intermediate expanded state without a valve sub-assembly and inside an irregular-shaped implantation site;

[0256] FIG. **222** is a photograph of the stent assembly of FIG. **221** in an further intermediate expanded state;

[0257] FIG. **223** is a photograph of the stent assembly of FIG. **221** in an further intermediate expanded state;

[0258] FIG. **224** is a photograph of the stent assembly of FIG. **221** in an implanted state within the irregular-shaped implantation site;

[0259] FIG. **225** is a process flow diagram of an exemplary embodiment of a method for controlling implantation of a self-expanding and forcibly-expanding device according to the described embodiments

[0260] FIG. **226** is a fragmentary, exploded, perspective view of an exemplary embodiment of a distal control handle for implanting a self-expanding and forcibly-expanding device;

[0261] FIG. **227** is a fragmentary, exploded, perspective view of a distal portion of the distal control handle of FIG. **226** from a side thereof;

[0262] FIG. **228** is a fragmentary, exploded, perspective view of a proximal portion of the distal control handle of FIG. **226** from a side thereof;

[0263] FIG. **229** is a perspective view of an alternative embodiment of the distal control handle of FIG. **226** from above a side thereof;

[0264] FIG. **230** is a fragmentary, perspective view of a proximal portion of the distal control handle of FIG. **229** from above a side thereof;

[0265] FIG. **231** is a side elevational view of an exemplary embodiment of a self-expanding and forcibly-expanding implant;

[0266] FIG. **232** is a fragmentary, side elevational view of the self-expanding and forcibly-expanding implant of FIG. **231** inside a valve orifice;

[0267] FIG. **233** is a fragmentary, side elevational view of the self-expanding and forcibly-expanding implant of FIG. **231** inside a cylindrical vessel;

[0268] FIG. **234** is a side elevational view of an exemplary embodiment of a self-expanding and forcibly-expanding implant with a barbell-shaped extension on a distal end thereof;

[0269] FIG. **235** is a side elevational view of an exemplary embodiment of a self-expanding and forcibly-expanding implant with a bulb-shaped extension on a distal end thereof;

[0270] FIG. **236** is a fragmentary, diagrammatic cross-sectional view of a heart having an atrial septal defect;

[0271] FIG. **237** is a fragmentary, diagrammatic cross-sectional view of the heart of FIG. **236** having an exemplary embodiment of a self-expanding and forcibly-expanding implant implanted within the atrial septal defect;

[0272] FIG. **238** is a fragmentary, diagrammatic cross-sectional view of a heart in which a WATCHMAN® device is implanted within the left atrial appendage;

[0273] FIG. **239** is a fragmentary, diagrammatic, enlarged cross-sectional view of the heart of FIG. **238** with a view of the left atrial appendage;

[0274] FIG. **240** is a fragmentary, diagrammatic cross-sectional view of a left atrium and a left atrial appendage of a heart with a self-expanding and forcibly-expanding implant partially expanded in the appendage;

[0275] FIG. **241** is a fragmentary, diagrammatic cross-sectional view of a heart having a left ventricular aneurysm;

[0276] FIG. **242** is a fragmentary illustration of the arterial and venous circulation of the human legs;

[0277] FIG. **243** is a perspective view of an alternative embodiment of a distal control handle according to the invention from above a right rear thereof and with the distal prosthesis delivery system removed;

[0278] FIG. **244** is a right side elevational view of the distal control handle of FIG. **243**;

[0279] FIG. **245** is a left side elevational view of the distal control handle of FIG. **243**;

[0280] FIG. **246** is a top plan view of the distal control handle of FIG. **243**;

[0281] FIG. **247** is a bottom plan view of the distal control handle of FIG. **243**;

[0282] FIG. **248** is a rear elevational view of the distal control handle of FIG. **243**;

[0283] FIG. **249** is a perspective view of the distal control handle of FIG. **243** from above the right rear corner;

[0284] FIG. **250** is a perspective view of the distal control handle of FIG. **243** from above the left rear;

[0285] FIG. **251** is a perspective view of the distal control handle of FIG. **243** from above the right upper front side with a top half of a casing removed and with the distal prosthesis delivery system removed;

[0286] FIG. **252** is a perspective view of a motor and transmission assembly of the distal control handle of FIG. **251** from above;

[0287] FIG. **253** is a perspective view of the front of the motor assembly of the distal control handle of FIG. **251** with a distal transmission cover removed;

[0288] FIG. **254** is a perspective view of the front of the motor assembly of the distal control handle of FIG. **251** with a proximal transmission cover and motors removed;

[0289] FIG. **255** is a perspective view of the rear of the motor assembly of the distal control handle of FIG. **251** with the proximal transmission cover and motors removed;

[0290] FIG. **256** is a photograph of the rear of the distal control handle of FIG. **243**;

[0291] FIG. **257** is a photograph of the distal control handle of FIG. **243** from above with the top half of the casing removed;

[0292] FIG. **258** is a photograph of a delivery system with the distal control handle of FIG. **243** from in front of a left side thereof with the top half of the casing removed;

[0293] FIG. **259** is a photograph of the delivery system of FIG. **258** from above the left side thereof;

[0294] FIG. **260** is a photograph of the delivery system of FIG. **258** from behind a left side thereof;

[0295] FIG. **261** is a multi-variable display of an exemplary embodiment of a method for determining a characteristic curve of an implant for detecting a native annulus size according to the invention;

[0296] FIG. **262** is a multi-variable display of the method for determining the characteristic curve of FIG. **261** with an different implant;

[0297] FIG. **263** is a multi-variable display of the method for determining the characteristic curve of FIG. **261** with an different implant;

[0298] FIG. **264** is a multi-variable display of a method for detecting a native annulus size according to the invention;

[0299] FIG. **265** is a fluoroscopic image of an exemplary embodiment of a delivery system with a self-expanding and forcibly-expanding aortic valve implant in a partially expanded state;

[0300] FIG. **266** is a fluoroscopic image of the delivery system and valve implant of FIG. **265** rotated counter-clockwise;

[0301] FIG. **267** is a fluoroscopic image of the delivery system and valve implant of FIG. **265** rotated clockwise;

[0302] FIG. **268** is a fluoroscopic image of the delivery system and valve implant of FIG. **265** expanded to an implantation position;

[0303] FIG. **269** is a fragmentary perspective view of an exemplary embodiment an angular correction device of a distal end of a stent delivery system;

[0304] FIG. **270** is a fluoroscopic image of an exemplary embodiment of an angular correction device at a distal end of a delivery system with a self-expanding and forcibly-expanding aortic valve implant in a partially expanded state;

[0305] FIG. **271** is a fluoroscopic image of the angular correction device, the delivery system, and the valve implant of FIG. **270** expanded to an implantation position; and

[0306] FIG. **272** is a fluoroscopic image of the angular correction device, the delivery system, and the valve implant of FIG. **270** with the angular correction device actuated to rotate the implant clockwise.

DETAILED DESCRIPTION OF THE INVENTION

[0307] As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

[0308] Alternate embodiments may be devised without departing from the spirit or the scope of the invention. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

[0309] Before the present invention is disclosed and described, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an”, as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically.

[0310] Relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

[0311] As used herein, the term “about” or “approximately” applies to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure.

[0312] The terms “program,” “programmed”, “programming,” “software,” “software application,” and the like as used herein, are defined as a sequence of instructions designed for execution on a computer system. A “program,” “software,” “computer program,” or “software application” may

include a subroutine, a function, a procedure, an object method, an object implementation, an executable application, an applet, a servlet, a source code, an object code, a shared library/dynamic load library and/or other sequence of instructions designed for execution on a computer system. [0313] Herein various embodiments of the present invention are described. In many of the different embodiments, features are similar. Therefore, to avoid redundancy, repetitive description of these similar features may not be made in some circumstances. It shall be understood, however, that description of a first-appearing feature applies to the later described similar feature and each respective description, therefore, is to be incorporated therein without such repetition.

[0314] Described now are exemplary embodiments of the present invention. Referring now to the figures of the drawings in detail and first, particularly to FIGS. **1** to **19**, there is shown a first exemplary embodiment of an actively controllable stent deployment system **100** according to the invention. Even though this exemplary embodiment is illustrated as a stent deployment system without the presence of a stent graft, this embodiment is not to be considered as limited thereto. Any stent graft embodiment according to the invention as disclosed herein can be used in this embodiment. The stent graft is not shown in these figures for clarity. Further, as used herein, the terms “stent” and “stent graft” are used herein interchangeably. Therefore, any embodiment where a stent is described without referring to a graft should be considered as referring to a graft additionally or in the alternative, and any embodiment where both a stent and a graft are described and shown should be considered as also referring to an embodiment where the graft is not included.

[0315] In contrast to prior art self-expanding stents, the actively controllable stent deployment system **100** includes a stent lattice **110** formed by interconnected lattice struts **112**, **114**. In this exemplary embodiment, pairs of inner and outer struts **114**, **112** are respectively connected to adjacent pairs of inner and outer struts **114**, **112**. More particularly, each pair of inner and outer struts **114**, **112** are connected pivotally at a center point of each strut **114**, **112**. The ends of each inner strut **114** of a pair is connected pivotally to ends of adjacent outer struts **112** and the ends of each outer strut **112** of a pair is connected pivotally to ends of adjacent inner struts **114**. In such a configuration where a number of strut pairs **114**, **112** are connected to form a circle, as shown in each of FIGS. **1** to **19**, a force that tends to expand the lattice **110** radially outward will pivot the struts **114**, **112** at each pivot point and equally and smoothly expand the entire lattice **110** from a closed state (see, e.g., FIG. **3**) to any number of open states (see FIGS. **4** to **13**). Similarly, when the stent lattice **110** is at an open state, a force that tends to contract the stent lattice **110** radially inward will pivot the struts **114**, **112** at each pivot point and equally and smoothly contract the entire stent lattice **110** towards the closed state. This exemplary configuration, therefore, defines a repeating set of one intermediate and two outer pivot points about the circumference of the stent lattice **110**. The single intermediate pivot point **210** is, in the exemplary embodiment shown in FIGS. **1** to **19**, located at the center point of each strut **112**, **114**. On either side of the single intermediate pivot point **210** is a vertically opposing pair of outer pivot points **220**.

[0316] To provide such expansion and contraction forces, the actively controllable stent deployment system **100** includes at least one jack assembly **700** that is present in each of FIGS. **1** to **19** but is described, first, with regard to FIG. **7**. Each jack assembly **700** has a distal drive block **710**, a proximal drive block **720**, and a disconnecter drive block **730**. A drive screw **740** connects the distal drive block **710** to the proximal drive block **720**. The drive screw **740** has a distal threaded drive portion **742** having corresponding threads to a threaded drive bore **712** of the distal drive block **710**. The drive screw **740** has an intermediate unthreaded portion **744** that rotates freely within a smooth drive bore **722** of the proximal drive block **720**. In the embodiment shown, the inner diameter of the smooth drive bore **722** is slightly larger than the outer diameter of the unthreaded portion **744** so that the unthreaded portion **744** can freely rotate within the smooth drive bore **722** substantially without friction. As used here, and in any of the other exemplary embodiments, substantially without friction means that the drive screw **740** can be turned when intended by a drive screw motor (as explained below) but does not turn when the lattice is

disconnected from the drive motor. This characteristic is a result of having the lead angle of the thread on the drive screw **740** be very small, for example, between approximately 1 and approximately 10 degrees, in particular, between approximately 3 and approximately 7 degrees, further, between approximately 4 and approximately 5 degrees. This low angle makes turning the drive screw **740** to impart motion as described herein very easy but back-driving the screw almost impossible without damaging the drive screw **740**. Based upon this attribute, the stent lattice (and other stent lattices described herein) become self-locking. The drive screw **740** also has an intermediate collar **746** just proximal of the proximal drive block **720**. The outer diameter of the intermediate collar **746** is greater than the inner diameter of the smooth drive bore **722**. Lastly, the drive screw **740** has a proximal key portion **748** extending from the intermediate collar **746** in a proximal direction. The jack assembly **700** is configured to retain the drive screw **740** within the distal drive block **710** and the proximal drive block **720** in every orientation of the stent lattice **110**, from the closed state, shown in FIG. 3, to a fully open state, shown in FIG. 11, where the distal drive block **710** and the proximal drive block **720** touch one another.

[0317] Each jack assembly **700** is attached fixedly to the stent lattice **110** at a circumferential location thereon corresponding to the vertically opposing pair of outer pivot points **220**. In one exemplary embodiment of the jack assembly **700** shown in FIGS. 1 to 19, the outer surface **714** of the distal drive block **710** and the outer surface **724** of the proximal drive block **720** each have a protruding boss **716**, **726** having an outer shape that is able to fixedly connect to a respective one of the outer pivot points **220** of the stent lattice **110** but also rotationally freely connect thereto so that each of the inner and outer struts **114**, **112** connected to the boss **716**, **726** pivots about the boss **716**, **726**, respectively. In this exemplary embodiment, each boss **716**, **726** is a smooth cylinder and each outer pivot point **220** is a cylindrical bore having a diameter corresponding to the outer smooth surface of the cylinder but large enough to pivot thereon without substantial friction. The materials of the boss **716**, **726** and the outer pivot points **220** of the inner and outer struts **114**, **112** can be selected to have substantially frictionless pivoting.

[0318] Accordingly, as the drive screw **740** rotates between the open and closed states, the unthreaded portion **744** of the drive screw **740** remains longitudinally stable within the proximal drive block **720**. In contrast, the distal threaded drive portion **742** progressively enters the threaded drive bore **712** from the proximal end to the distal end thereof as the stent lattice **110** expands outwardly. As shown in the progressions of FIG. 2 to FIG. 4 and FIGS. 5 to 7 to 8 to 9, as the drive screw **740** rotates within the proximal drive block **720**, the distal drive block **710** moves closer and closer to the proximal drive block **720**, thereby causing a radial expansion of the stent lattice **110**.

[0319] To implant the stent lattice **110** in a tubular anatomic structure (such as a vessel or a valve seat), the stent lattice **110** needs to be disconnected from the delivery system. Delivery of the stent lattice **110** to the anatomic structure will be described in further detail below. When the stent lattice **110** enters the implantation site, it will be most likely be in the closed state shown in FIG. 3, although for various reasons, the stent lattice **110** can be expanded partially, if desired, before reaching the implantation site. For purposes of explaining the disconnect, the extent of expansion is not relevant. When at the implantation site, the stent lattice **110** will be expanded by rotating the drive screw **740** in a corresponding expansion direction (the direction of threads of the drive screw **740** and the drive bore **712** will determine if the drive screw **740** needs to be rotated clockwise or counter-clockwise). The stent lattice **110** is expanded to a desired expansion diameter, for example as shown in the progression of FIGS. 4 to 9 or FIGS. 10 to 11, so that it accommodates to the natural geometry of the implantation site, even if the geometry is non-circular or irregular. When the implantation diameter is reached, e.g., in FIGS. 9 and 11, the jack assemblies **700** need to be disconnected from the remainder of the stent deployment system **100**.

[0320] To accomplish disconnect of the jack assemblies **700**, the disconnecter drive block **730** is provided with two lumens. A first lumen, the drive lumen **732**, accommodates a drive wire **750** that is able to rotationally engage the proximal key portion **748**. Use of the word wire for the drive wire

750 does not mean that this structure is a solid cord. The drive wire **750** can also be a hollow tube, a coil, or any other structure that can perform the functions described herein. In the exemplary embodiment shown, which is most clearly illustrated in FIG. **19**, the proximal key portion **748** has a square cross-sectional shape. A drive wire bushing **734** rotationally freely but longitudinally fixedly resides in the drive lumen **732**. The drive wire bushing **734** is connected to the drive wire **750** either as an integral part thereof or through a connection sleeve **752**. Regardless of the connection design, any rotation of the drive wire **750** in either direction will cause a corresponding rotation of the drive wire bushing **734**. A key hole **738** at the distal end of the disconnect drive block **730** and having an internal shape corresponding to a cross-section of the proximal key portion **748** allows a rotationally fixed but longitudinally free connection to occur with the proximal key portion **748**. In the exemplary embodiment shown in FIG. **19**, the key hole **738** also has a square cross-sectional shape.

[0321] The disconnect drive block **730** also has a second lumen, a disconnect lumen **731**, which is best shown in FIGS. **14** and **16**. Residing in the disconnect lumen **731** in a rotationally free but longitudinally fixed manner is a retainer screw **760**. The retainer screw **760** has a distal threaded portion **762**, an intermediate shaft **764**, and a proximal connector **766**. The distal threaded portion **762** has an exterior thread corresponding to an internal thread of a connect lumen **1631**, which is located in the proximal drive block **720** and is coaxial with the disconnect lumen **731**. The intermediate shaft **764** has a smooth exterior surface and a cross-sectional shape that is slightly smaller than the cross-sectional shape of the disconnect lumen **731** so that it can be rotated freely within the disconnect lumen **731** substantially without friction (as above, this turns with the controlling motor but remains fixed when disconnected, i.e., it is self-locking). The proximal connector **766** has a flange with an outer diameter greater than the inner diameter of the disconnect lumen **731**. The proximal connector **766** is connected at a proximal end thereof to a disconnect wire **770**, which connection can either be an integral part thereof or through a secondary connection, such as a weld or connection sleeve. Use of the word wire for the disconnect wire **770** does not mean that this structure is a solid cord. The disconnect wire **770** can also be a hollow tube, a coil, or any other structure that can perform the functions described herein.

[0322] With such a configuration of the proximal drive block **720** and the disconnect drive block **730** of a jack assembly **700**, rotation in a securing direction will longitudinally secure the proximal drive block **720** to the disconnect drive block **730** so that the stent lattice **110** remains connected to the drive wire **750** and the disconnect wire **770**. In the connected state, the stent lattice **110** may be extended outward and retracted inward as many times as needed until implantation alignment occurs according to the surgeon's desire. Likewise, rotation in a disconnecting direction will longitudinally release the proximal drive block **720** from the disconnect drive block **730** so that the stent lattice **110** disconnects entirely from the drive wire **750** and the disconnect wire **770**.

[0323] This process is illustrated with regard to FIGS. **10** to **19**. In the exemplary illustration of FIG. **10**, the stent lattice **110** is not fully expanded. Because the distal threaded portion **762** of the retainer screw **760** is threaded within the connect lumen **1631** of the proximal drive block **720**, the disconnect drive block **730** remains longitudinally fixed to the proximal drive block **720**—ideally, a configuration that exists from the time that the stent deployment system **100** first enters the patient and at least up until implantation of the stent lattice **110** occurs. Expansion of the stent lattice **110** is finished in the configuration of FIG. **11** and, for purposes of this example, it is assumed that the stent lattice **110** is correctly implanted at the implantation site. Therefore, disconnection of the delivery system can occur. It is noted that this implantation position just happens to be at a circumferential extreme of the stent lattice **110** because the distal drive block **710** and the proximal drive block **720** are touching. In actual use, however, it is envisioned that such touching does not occur when expanded for implantation and, in such a state, there is a separation distance between the distal drive block **710** and the proximal drive block **720** to give the stent lattice **110** room to expand further into the implantation site if needed. Disconnection of the stent

lattice **110** begins by rotating the disconnect wire **770** in a direction that unscrews the threaded portion **762** of the retainer screw **760** from the connect lumen **1631**. As the stent lattice **110** is implanted with expansive force at the implantation site, the disconnect drive block **730** moves proximally as unthreading occurs. Complete unthreading of the retainer screw **760** is shown in FIGS. **12** and **14**. In a configuration with more than one jack assembly **700** (the configuration of FIGS. **1** to **19** has 4, for example), each disconnect wire **770**, **770'** will rotate synchronously to have each disconnect drive block **730** disconnect from its respective proximal drive block **720** substantially simultaneously, as shown in FIG. **12**. Such synchronous movement will be described in greater detail below. With the stent lattice **110** implanted, as shown in FIGS. **13**, **15**, **18**, and **19**, the delivery system for the stent lattice **110** can be withdrawn proximally away from the implantation site and be retracted out from the patient.

[0324] It is noted that the exemplary embodiment of FIGS. **1** to **19** shows the actively controllable stent deployment system **100** as having four jack assemblies **700** equally spaced around the circumference of the lattice **110**. This configuration is merely exemplary and any number of jack assemblies **700** can be used to expand and contract the lattice **110**, including a minimum of one jack assembly **700** in total and a maximum of one jack assembly **700** for each intersection between each inner and outer strut pair **112**, **114**. Herein, three and four jack assemblies **700** are depicted and used to show particularly well performing configurations. By using an even number, counter-rotating screws can be used to null the torque.

[0325] FIG. **20** is provided to further explain how the stent lattice **110** moves when it is expanded and contracted. As set forth above, the actively controllable stent deployment system **100** is based upon the construction of the stent lattice **110** and the attachment of the proximal and distal drive blocks **720**, **710** of at least one jack assembly **700** to at least one set of the vertically opposing upper and lower pivot points **220** of the stent lattice **110**. With the exemplary connections **716**, **726** and pivot points **210**, **220** shown in FIGS. **1** to **19**, a longitudinal vertical movement of one of the proximal or distal drive blocks **720**, **710** with respect to the other will expand or contract the stent lattice **110** as described herein. FIG. **20** illustrates with solid cylinders **2000** a radial path of travel that each intermediate pivot point **210** will traverse as the stent lattice **110** is moved between its expanded state (e.g., FIG. **9**) and its contracted state (e.g., FIG. **2**). Because the travel path is linear, the stent lattice **110** expands and contracts smoothly and equally throughout its circumference.

[0326] It is noted that the struts **112**, **114** shown in FIGS. **1** to **19** appear to not be linear in certain figures. Examples of such non-linearity are the struts in FIGS. **10** and **11**. Therein, each strut **112**, **114** appears to be torqued about the center pivot point such that one end is rotated counter-clockwise and the other is rotated clockwise. This non-linearity can create the hourglass figure that will help fix the graft into an implantation annulus and to create a satisfactory seal at the top edge of the implant. The non-linear illustrations are merely limitations of the computer design software used to create the various figures of the drawings. Such non-linear depictions should not be construed as requiring the various exemplary embodiments to have the rotation be a part of the inventive struts or strut configuration. Whether or not the various struts **112**, **114** will bend, and in what way they will bend, is dependent upon the characteristics of the material that is used to form the struts **112**, **114** and upon how the pivot joints of the lattice **110** are created or formed. The exemplary materials forming the struts **112**, **114** and the pivots and methods for creating the pivots are described in further detail below. For example, they can be stamped, machined, coined or similar from the family of stainless steels and cobalt chromes.

[0327] With the invention, force is applied actively for the controlled expansion of the stent lattice **110**. It may be desirable to supplement the outwardly radial implantation force imposed on the wall at which the stent lattice **110** is implanted. Prior art stent grafts have included barbs and other similar devices for supplementing the outward forces at the implantation site. Such devices provide a mechanical structure(s) that impinge(s) on and/or protrude(s) into the wall of the implantation site and, thereby, prevent migration of the implanted device. The systems and methods of the invention

include novel ways for supplementing the actively applied outward expansion force. One exemplary embodiment includes actively controllable needles, which is described, first, with reference to FIG. 17. In this exemplary embodiment, the distal drive block **710** and the proximal drive block **720** contain a third lumen, a distal needle lumen **1711** and a proximal needle lumen **1721**. Contained within both of the distal and proximal needle lumens **1711**, **1721** is a needle **1700**. In an exemplary embodiment, the needle **1700** is made of a shape memory material, such as Nitinol, for example. The needle **1700** is preset into a shape that is, for example, shown in the upper left of FIG. 12 (or it can form a majority or an entirety of a closed circle). A portion that remains in the distal and proximal needle lumens **1711**, **1721** after implantation of the stent lattice **110** can be preset into a straight shape that is shown in FIG. 17. A tissue-engaging distal portion of the needle **1700**, however, is formed at least with a curve that, when extended out of the distal drive block **710**, protrudes radially outward from the center longitudinal axis of the stent lattice **110**. In such a configuration, as the needle **1700** extends outward, it drives away from the outer circumferential surface **714** (see FIG. 5) of the distal drive block **710** (i.e., towards the viewer out from the plane shown in FIG. 5). The needle **1700** also has a longitudinal extent that places the distal tip **1210** within the distal needle lumen **1711** when the stent lattice **110** is in the closed state, e.g., shown in FIG. 2.

[0328] Deployment of the needles **1700** in each jack assembly **700** (or the number of needles can be any number less than the number of jack assemblies **700**) is illustrated, for example, starting with FIG. 5. In this example, the needles **1700** in each of the four jack assemblies **700** has a length that is shorter than the longitudinal end-to-end distance of the proximal and distal drive blocks **720**, **710** because the needles **1700** have not yet protruded from the distal upper surface **612** of each distal drive block **710** even though the stent lattice **110** is partially expanded. When the stent lattice **110** has expanded slightly further, however, as shown in FIG. 7, the needles **1700** begin protruding from the distal upper surface **612**. As the needles **1700** are pre-bent as set forth above, the needles **1700** immediately begin bending into the natural pre-set curved shape. See also FIGS. 7 and 8. FIG. 10 illustrates two needles **1700** even further extended out from the distal needle lumen **1711** (only two are shown because this is a cross-section showing only the rear half of the stent lattice **110**). FIG. 11 illustrates two needles **1700** in a fully extended position (as the distal and proximal drive blocks **710**, **720** touch one another in the most-expanded state of the stent lattice **110**). FIGS. 9, 13, 16, 17, 18, and 21 also show the needles **1700** in an extended or fully extended state.

[0329] How the needles **1700** each extend from the distal drive block **710** can be explained in a first exemplary embodiment with reference to FIG. 17. A proximal portion of the needle **1700** is connected fixedly inside the proximal needle lumen **1721**. This can be done by any measure, for example, by laser welding. In contrast, the intermediate and distal portions of the needle **1700** is allowed to entirely freely slide within the distal needle lumen **1711**. With the length set as described above, when the distal and proximal drive blocks **710**, **720** are separated completely, as shown in FIG. 3, the needle **1700** resides in both distal and proximal needle lumens **1711**, **1721**. As one of the distal and proximal drive blocks **710**, **720** begins to move towards the other (as set forth above, the exemplary embodiment described with regard to these figures has the distal drive block **710** move towards the proximal drive block **720**), the proximal portion of the needle **1700** remains in the proximal needle lumen **1721** but the distal portion of the needle **1700** begins to exit the distal upper surface **612**, which occurs because the intermediate and distal portions of the needle **1700** are disposed slidably in the distal needle lumen **1711**. This embodiment where the proximal portion of the needle **1700** is fixed in the proximal needle lumen **1721** is referred to herein as dependent control of the needles **1700**. In other words, extension of the needles **1700** out from the distal needle lumen **1711** occurs dependent upon the relative motion of the distal and proximal drive blocks **710**, **720**.

[0330] Alternatively, the supplemental retention of the stent lattice **110** at the implantation site can occur with independent control of the needles. FIGS. 21 to 29 illustrate such an exemplary

embodiment of a system and method according to the invention. Where similar parts exist in this embodiment to the dependently controlled needles **1700**, like reference numerals are used. The jack assembly **2100** is comprised of a distal drive block **710**, a proximal drive block **720**, a disconnect drive block **730**, a drive screw **740**, a drive wire **750** (shown diagrammatically with a dashed line), a retainer screw **760**, and a disconnect wire **770**. Different from the jack assembly **700** of FIGS. **1** to **19**, the jack assembly **2100** also includes a needle **2200** and a needle pusher **2210** and both the proximal drive block **720** and the disconnect drive block **730** each define a co-axial third lumen therein to accommodate the needle pusher **2210**. More specifically, the distal drive block **710** includes a first pusher lumen **2211**, the proximal drive block **720** includes a second pusher lumen **2221** and the disconnect drive block **730** includes a third pusher lumen **2231**. As described above, the retainer screw **760** keeps the proximal drive block **720** and the disconnect drive block **730** longitudinally grounded to one another up until and after implantation of the stent lattice **110** and separation of the delivery system occurs. Rotation of the drive screw **740** causes the distal drive block **710** to move towards the proximal drive block **720**, thereby expanding the stent lattice **110** to the desired implantation diameter. This movement is shown in the transition between FIG. **22** and FIG. **23**. Now that the stent lattice **110** is determined to be properly implanted within the implantation site, it is time to deploy the needles **2200**. Deployment starts by advancing the needle pusher **2180** as shown in FIG. **24**. The needle pusher **2810** can, itself, be the control wire for advancing and retracting the needle **2200**. Alternatively, and/or additionally, a needle control wire **2182** can be attached to or shroud the needle pusher **2180** to provide adequate support for the needle pusher **2180** to function. Continued distal movement of the needle pusher **2180** causes the needle **2200** to extend out from the distal upper surface **612** and, due to the preset curvature of the memory-shaped needle **2200**, the needle tip curves outward and into the tissue of the implantation site. This curvature is not illustrated in FIG. **25** because the curvature projects out of the plane of FIG. **25**.

[0331] Now that the stent lattice **110** is implanted and the needles **2200** are extended, disconnection of the stent lattice **110** occurs. First, as shown in FIG. **26**, the retainer screw **760** is rotated to disconnect the proximal drive block **720** from the disconnect drive block **730** and a proximally directed force is imparted onto one or both of the drive wire **750** and the disconnect wire **770**. This force moves the disconnect drive block **730** distally to remove the proximal key portion **748** of the drive screw **740** out from the keyhole **738**, as shown in the progression from FIGS. **26** to **27**. Simultaneously, distal movement of the disconnect drive block **730** starts the withdrawal of the needle pusher **2180** from the first pusher lumen **2211** (if not retracted earlier). Continued distal movement of the disconnect drive block **730** entirely removes the needle pusher **2180** from the first pusher lumen **2211**, as shown in FIG. **28**. Finally, withdrawal of the stent lattice delivery system entirely from the implantation site removes the needle pusher **2180** out from the second pusher lumen **2221** leaving only the implanted stent lattice **110**, the jack assembly(ies) **2100**, and the needle(s) **2200** at the implantation site.

[0332] FIGS. **30** to **37** illustrate another exemplary embodiment of an independent needle deployment system and method according to the invention. Where similar parts exist in this embodiment to the embodiments described above, like reference numerals are used. The jack assembly **3000** is comprised of a distal drive block **3010**, a proximal drive block **3020**, a disconnect drive block **3030**, a drive screw **3040**, a drive wire **750**, a retainer screw **760**, and a disconnect wire **770**. The jack assembly **3000** also includes a needle **3070** and a needle movement sub-assembly **3090**. The needle movement sub-assembly **3090** is comprised of a needle support **3092**, a needle base **3094**, a needle disconnect nut **3096**, and a needle disconnect wire **3098**.

[0333] The distal drive block **3010** defines three longitudinal lumens. The first is a support rod lumen **3012** and is defined to slidably retain a support rod **3080** therein. As rotational torque is imparted when any screw associated with the jack assembly **3000** rotates, the support rod **3080** is employed to minimize and/or prevent such torque from rotating the distal and proximal drive

blocks **3010**, **3020** and disconnecter drive block **3030** with respect to one another and, thereby, impose undesirable forces on the stent lattice **110**. The longitudinal length of the support rod **3080** is selected to not protrude out from the distal upper surface **3011** of the distal drive block **3010** in any expansion or retracted state of the stent lattice **110**. The second vertically longitudinal lumen is the drive screw lumen **3014**. As in previous embodiments, the drive screw lumen **3014** is configured with internal threads corresponding to external threads of the drive screw **740** and the longitudinal vertical length of the drive screw lumen **3014** is selected to have the drive screw **740** not protrude out from the distal upper surface **3011** of the distal drive block **3010** in any expansion or retracted state of the stent lattice **110**. Finally, the distal drive block **3010** defines a needle assembly lumen that is comprises of a relatively wider proximal needle lumen **3016** and a relatively narrower distal needle lumen **3018**, both of which will be described in greater detail below.

[0334] In comparison to other proximal drive blocks described above, the proximal drive block **3020** of jack assembly **3000** defines two vertically longitudinal lumens. The first lumen is a drive screw lumen **3024**. In this exemplary embodiment, the drive screw **740** is longitudinally fixedly connected to the proximal drive block **3020** but is rotationally freely connected thereto. To effect this connection, a distal drive screw coupler part **3052** is fixedly secured to the proximal end of the drive screw **740** within a central bore that is part of the drive screw lumen **3024** of the proximal drive block **3020**. The distal drive screw coupler part **3052** is shaped to be able to spin along its vertical longitudinal axis (coaxial with the vertical longitudinal axis of the drive screw **740**) freely within the central bore of the drive screw lumen **3024**. A distal portion of the drive screw lumen **3024** is necked down to have a diameter just large enough to allow a portion of the drive screw **740** (e.g., non-threaded) to spin therewithin substantially without friction. Through a circular port **3100** in a side of the proximal drive block **3020**, the distal drive screw coupler part **3052** can be, for example, spot-welded to the proximal non-threaded end of the drive screw **740**. With such a connection, the drive screw **740** is longitudinally fixedly grounded to the proximal drive block **3020** within the central bore of the drive screw lumen **3024**. This means that rotation of the drive screw **740** causes the distal drive block **3010** to move towards the proximal drive block **3020** and, thereby, cause an expansion of the stent lattice **110** connected to the jack assembly **3000**, for example, at bosses **3600** shown in FIG. **36**. Fasteners **3610** in the form of washers, rivet heads, or welds, for example, can hold the stent lattice **110** to the bosses **3600**. Further expansion of the drive screw coupler **3052**, **3054** is made below with regard to the disconnecter drive block **3030**.

[0335] The second lumen within the proximal drive block **3020** of jack assembly **3000** is a retainer screw lumen **3022**. A distal portion of the retainer screw lumen **3022** is shaped to fixedly hold a proximal end of the support rod **3080** therein; in other words, the support rod **3080** is fastened at the distal portion of the retainer screw lumen **3022** and moves only with movement of the proximal drive block **3020**. Fastening can occur by any measures, for example, by corresponding threads, welding, press fitting, or with adhesives. A proximal portion of the retainer screw lumen **3022** has interior threads corresponding to exterior threads of the retainer screw **760**. Accordingly, disconnection of the disconnecter drive block **3030** from the proximal drive block **3020** is carried out by rotation of the retainer screw **760** fixedly connected to disconnecter wire **770**. Connection between the retainer screw **760** and the disconnecter wire **770** can be accomplished by any measures, including for example, a hollow coupler sheath fixedly connected to both the distal end of the disconnecter coupler wire **770** and the proximal end of the retainer screw **760** as shown in FIG. **30**. As described above, the retainer screw **760** keeps the proximal drive block **3020** and the disconnecter drive block **3030** longitudinally grounded to one another until after implantation of the stent lattice **110** and separation of the delivery system occurs.

[0336] This exemplary embodiment also has an alternative to the device and method for uncoupling the drive screw **740** from the remainder of the jack assembly **3000**, in particular, the two-part drive screw coupler **3052**, **3054**. The distal drive screw coupler part **3052** as, at its proximal end, a mechanical coupler that is, in this exemplary embodiment, a semicircular boss

extending in the proximal direction away from the drive screw **740**. The proximal drive screw coupler part **3054**, has a corresponding semicircular boss extending in the distal direction towards the drive screw **740**. These can be seen, in particular, in the enlarged view of FIG. **37**. Therefore, when the two semicircular bosses are allowed to interconnect, any rotation of the proximal drive screw coupler part **3054** will cause a corresponding rotation of the distal drive screw coupler part **3052**. The disconnect drive block **3030** has a screw coupler bore **3031** shaped to retain the distal drive screw coupler part **3052** therein. As in the proximal drive block **3020**, the screw coupler bore **3031** is shaped to surround the proximal drive screw coupler part **3054** and allow the proximal drive screw coupler part **3054** to rotate freely therewithin substantially without friction. A proximal portion of the screw coupler bore **3031** is necked down to a smaller diameter to prevent removal of the proximal drive screw coupler part **3054** after it is fixedly connected to the drive wire **750** either directly or through, for example, a hollow coupler as shown in FIGS. **30** to **37**.

[0337] Implantation of the stent lattice **110** with the jack assembly **3000** is described with regard to FIGS. **30** through **35**. First, rotation of the drive screw **740** causes the distal drive block **3010** to move towards the proximal drive block **3020**, thereby expanding the stent lattice **110** to the desired implantation diameter. This movement is shown in the transition between FIG. **30** and FIG. **31**. Now that the stent lattice **110** is properly within the implantation site, deployment of the needles **3070** can occur. Deployment starts by advancing the needle sub-assembly **3090** as shown in the transition between FIGS. **31** and **32**. Continued distal movement of the needle subassembly **3090** causes the needle **3070** to extend out from the distal upper surface **3011** and, due to the preset curvature of the memory-shaped needle **3070**, the tip of the needle **3070** curves outward and into the tissue of the implantation site. This curvature is not illustrated in FIGS. **32** and **33** because the curvature projects out of the plane of these figures.

[0338] In comparison to previous proximal drive blocks above, the disconnect drive block **3030** does not have a lumen associated with the needle **3070**. Only distal drive block **3010** has a lumen therein to accommodate the needle **3070**. More specifically, the distal drive block **3010** includes a distal needle lumen **3018** and a proximal needle lumen **3016**. The distal needle lumen **3018** is shaped to accommodate the needle **3070** only. In contrast to other needle lumens, however, the proximal needle lumen **3016** is non-circular in cross-section and, in the exemplary embodiment, is ovular in cross-section. This shape occurs because the memory-shaped needle **3070** is supported on its side along its proximal extent by a needle support **3092**, which is fastened side-to-side, for example, by welding. The needle support **3092** has a relatively higher columnar strength than the needle **3070** and, therefore, when fixedly connected to the side of the needle **3070**, the needle support **3092** significantly increases the connection strength to the needle **3070** at its side than if the needle **3070** was controlled only from the very proximal end thereof. A high-strength, exterior threaded needle base **3094** is fixedly attached to the proximal end of the needle support **3092**. This configuration also keeps the needle clocked properly so that its bend direction is away from the center of the lattice and most directly attaches to the vessel wall.

[0339] Control of the needle **3070** is, as above, carried out by a needle disconnect wire **3098** (depicted with dashed lines). Attached to the distal end of the disconnect wire **3098** is a needle disconnect nut **3096** defining a distal bore with interior threads corresponding to the exterior threads of the needle base **3094**. In this configuration, therefore, rotation of the needle disconnect wire **3098** causes the needle disconnect nut **3096** to either secure to the needle base **3094** or remove from the needle base **3094** so that disconnection of the delivery system from the stent lattice **110** can occur. The top side of the distal drive block **3010** is cross-sectioned in FIG. **36** at the boss **3600** to show the shapes of the various lumens therein. As described above, the support rod lumen **3012** is a smooth, circular-cross-sectional bore to allow the support rod **3080** to slide longitudinally vertically therein. Similarly, the distal-portion of the drive screw lumen **3014** is also a smooth, circular-cross-sectional bore to allow the drive screw **740** to move longitudinally vertically therein as it is rotated and the threads engage the proximal threaded portion of the drive screw lumen **3014**.

The proximal needle lumen **3016**, in contrast, is non circular (e.g., ovalar) to accommodate the cylindrical-shaped needle **3070** and the side-by-side-connected, cylindrical-shaped, needle support **3092**. As shown in the view of FIG. **36**, at least the contacting portion of the needle **3070** to the needle support **3092** is shrouded with a connector sleeve **3071**, which has material properties that allow it to be fixedly connected to the needle **3070** and, at the same time, to the needle support **3092**.

[0340] Extension of the needle **3070** out from the distal upper surface **3011** by the distal movement of the disconnect wire **3098** is illustrated by the transition from FIG. **31** to FIG. **32**. Only a small portion of the needle **3070** extends from the distal upper surface **3011** because the views of FIGS. **30** to **33** are vertical cross-sections along a curved intermediate plane shown, diagrammatically, with dashed line X-X in FIG. **36**. As the needle **3070** extends in front of this sectional plane, it is cut off in these figures. FIGS. **34** and **35**, however clearly show the extended needle **3070** curving out and away from the outer side surface **3415**, however, merely for clarity purposes, the needle **3070** is rotated on its longitudinal axis slightly to the right so that it can be seen in FIG. **34** and seen better in FIG. **35**. It is note that another exemplary embodiment of the needle **3070** includes a hooked or bent needle tip **3072**. Correspondingly, the distal drive block **3010** includes a needle tip groove **3013** to catch the bent needle tip **3072** and utilize it in a way to keep tension on the needle **3070** and the needle disconnect wire **3098**. The bend in the needle tip **3072** also allows the needle **3070** to penetrate earlier and deeper than without such a bend. Another advantage for having this bend is that it requires more load to straighten out the tip bend than the overall memory shape of the needle and, thereby, it keeps the needle located distally in the jack assembly **3000**. If sufficient space exists in the distal drive block, a plurality of needles (e.g., a forked tongue) could be used.

[0341] Removal of the delivery system is described with regard to FIGS. **32**, **33**, and **37** after the stent lattice **110** is implanted and the needle **3070** of each jack assembly **3000** is extended. The retainer screw **760** keeps the proximal drive block **3020** and the disconnect drive block **3030** longitudinally grounded to one another up until implantation of the stent lattice **110** and extension of the needles **3070** (if needles **3070** are included). Separation of the delivery system begins by rotation of the disconnect wire **770** to unscrew the retainer screw **760** from the retainer screw lumen **3022**, which occurs as shown in the transition from FIG. **32** to FIG. **33**. Because the two parts of the drive screw coupler **3052**, **3054** are not longitudinally fastened to one another, the drive screw coupler **3052**, **3054** does not hinder disconnection of the disconnect drive block **3030** in any way. Before, at the same time, or after removal of the retainer screw **760** from the retainer screw lumen **3022**, the needle disconnect wire **3098** is rotated to, thereby, correspondingly rotate the needle disconnect nut **3096**. After a number of rotations, a needle disconnect nut **3096** is entirely unscrewed from the threads of the needle base **3094**, which is shown in FIG. **33**, for example. The delivery system, including the disconnect drive block **3030**, its control wires (drive wire **750** and disconnect wire **770**), and the needle disconnect wire **3098** and disconnect nut **3096**, can now be removed from the implantation site.

[0342] Other exemplary embodiments of the stent lattice according to the invention is shown with regard to FIGS. **38** to **50**. In a first exemplary embodiment, the stent lattice is a proximal stent **3810** of a stent graft **3800**. The proximal stent **3810** is connected to and covered on its exterior circumferential surface with a graft **3820**. With the proximal stent **3810** in a partially expanded state in FIG. **39** and other expanded states in FIGS. **40** and **41**, it can be seen that the outer struts **3812** have at least one throughbore **3814**, in particular, a line of throughbores from one end to the other, extending through the outer strut **3812** in a radial direction. These throughbores allow the graft **3820** to be sewn to the outer struts **3812**.

[0343] As described above, it can be beneficial for stents to have barbs, hooks, or other measures that catch and do not release tissue when they contact the tissue at or near an implantation site. FIGS. **42** to **45** illustrate one exemplary embodiment of the invention. When constructing the stent lattice **4200**, attachment of the three pivot points makes each outer strut **4230** curve about its center

pivot point, as can be seen in the lower right corner of FIG. 44, for example. Past the outer two pivot points of each outer strut **4230**, however, there is no curve imparted. The invention takes advantage of this and provides extensions **4210** and barbs **4220** on one or more ends of the outer struts **4230** because the lack of curvature at the ends of the outer strut **4230** means that the outer portion will extend outward from the circumferential outer surface of the stent lattice **4200**. In the expanded configuration of the stent lattice **4200** shown in FIG. 42, it can be seen that the extensions **4210** and barbs **4220** each project radially outward from the outer circumferential surface of the stent lattice **4200** and the points of the barbs **4220** also point radially outward, even if at a shallow angle.

[0344] It is noted that each of the exemplary embodiments of the stent lattices illustrated above has the intermediate pivot point at the center point of each strut. Having the intermediate pivot point in the center is only exemplary and can be moved away from the center of each strut. For example, as shown in FIGS. 46 to 50, the stent lattice **4600** can have the intermediate center pivot **4612** of the struts **4610** be closer to one end **4614** than the other end **4616**. When the center pivot **4612** is off-center, the side closer to the one end **4614** tilts inwards so that the outer circumferential surface of the stent lattice **4600** takes the shape of a cone. FIGS. 48, 49, and 50 illustrate the conical stent lattice **4600** expanded, partially expanded, and almost completely retracted, respectively.

[0345] The exemplary stent lattice embodiments in FIGS. 38 to 50 show the pivot points connected by screws. Any number of possible pivoting connections can be used at one or more or all of the pivot points. One exemplary embodiment of a strut-connection assembly **5100** can be seen in FIGS. 51 to 53. Because the stent lattice of the invention is intended to be small and fit in very small anatomic sites (e.g., heart valve, aorta, and other blood vessels), it is desirable to have the lattice struts be as thin as possible (i.e., have a low profile). The profile of the screws shown in FIGS. 38 to 50 can be reduced even further by the inventive strut-connection system **5100** as shown in FIGS. 51 to 53. FIG. 51 illustrates one such low-profile connection, which is formed using a rivet **5110** and forming the rivet bores in the each of the strut ends with one of a protrusion **5120** and an opposing indentation (the latter not illustrated in FIG. 52). The rivet **5110** is formed with a low-profile rivet head **5112**, an intermediate cylindrical boss **5114**, and a slightly outwardly expanded distal end **5116** (see FIG. 53). By placing two of the ends of the struts next to one another as shown in FIG. 52, with one of the protrusions **5120** placed inside the indentation of the opposing strut, the two strut ends form a pivot that is able to slide about the central pivot axis. The rivet **5110** is merely used to lock to strut ends against one another by having the expanded distal end **5116** enter through one of the non-illustrated indentation sides of the strut and exit through the protrusion-side of the opposing strut. It is the features on the struts that form the pivot and not the features of the rivet **5110**.

[0346] FIGS. 54 to 63 illustrate various alternative configurations of the struts in stent lattices according to exemplary embodiments of the invention. Each of the different lattice configurations provides different characteristics. One issue that occurs with lattices having alternating struts is that expansion and contraction of the adjacent struts can adversely rub against the graft securing measures (e.g., stitchings). With that consideration, the invention provides two separate cylindrical sub-lattices in the embodiment of FIG. 54 to 57. Each of the crossing points of the interior and exterior sub-lattices is connected via fasteners (e.g., rivets, screws, and the like). The outer ends of the struts, however, are not directly connected and, instead, are connected by intermediate hinge plates having two throughbores through which a fastener connects respectively to each of the adjacent strut ends. The intermediate hinge plates translate longitudinally towards each other upon expansion of the stent lattice and never have any portion of stent lattice pass in front or behind them. These hinge plates, therefore, could serve as connection points to the graft or could connect to a band or a rod, the band serving to join the two hinge plates together and, thereby, further spread the expansion forces on the graft. In an exemplary embodiment where the graft material has a transition zone where expansible material transitions to non-expansible material (and back again

if desired), such bands or rods could extend further past the longitudinal end of the lattice and provide an attachment or securing point for a non-expansible portion of the graft material. In this configuration, as shown in FIG. 57, for example, if graft material is attached to the outer sub-lattice, then, there is no interruption and the graft is not damaged with the struts acting as scissors. FIGS. 58 to 63 illustrate another exemplary embodiment of the strut lattices according to the invention in which the inner sub-lattice is shorter in the longitudinally vertical direction than the outer sub-lattice.

[0347] The exemplary actively controllable stent lattices of the invention can be used in devices and methods in which prior art self-expanding stents have been used. In addition to the example of a proximal stent shown in the exemplary stent graft of FIGS. 38 to 41, the technology described herein and shown in the instant stent delivery systems and methods for delivering such devices can be use in any stent graft or implant, such as those used in abdominal or thoracic aneurysm repair. Additionally, the exemplary stent lattices of the invention can be used in replacement heart valves, for example.

[0348] Referring now to the figures of the drawings in detail and first, particularly to FIGS. 64 to 70, there is shown a first exemplary embodiment of an actively controllable aortic valve assembly and methods and systems for controlling and implanting such devices. Even though the exemplary embodiment is shown for an aortic valve, the invention is not limited thereto. The invention is equally applicable to pulmonary, mitral and tricuspid valves.

[0349] The inventive technology used, for example, with regard to aortic valve repair includes a replacement aortic valve assembly 6400 according to the invention. One exemplary aortic valve assembly 6400 is depicted in FIGS. 64 and 65. FIG. 64 illustrates an adjustable lattice assembly 6410 similar to that shown in FIG. 103. In particular, the lattice assembly 6410 includes a number of struts 6412 crossing one another in pairs and pivotally connected to one another in an alternating manner at crossing points 6420 and end points 6422 of the struts 6412. Like the embodiment in FIG. 103, the lattice assembly 6410 is controlled, in this exemplary embodiment, by a set of three jack assemblies 6430 each having a proximal drive block 6432, a distal drive block 6434, and a drive screw 740 connecting the proximal and distal drive blocks 6432, 6434 together. In this exemplary embodiment, the drive screw 740 performs as above, it is longitudinally fixed but rotationally freely connected to the distal and proximal drive blocks 6432, 6434 such that, when rotated in one direction, the distal and proximal drive blocks 6432, 6434 move away from one another and, when rotated in the other direction, the distal and proximal drive blocks 6432, 6434 move towards one another. In such a configuration, the former movement radially contracts the lattice assembly 6410 and the latter movement expands the lattice assembly 6410. The lattice assembly 6410 shown in FIGS. 64 and 65 is in its expanded state, ready for implantation such that it accommodates to the natural geometry of the implantation site. Connected at least to the three jack assemblies 6430 at an interior side of one or both of the distal and proximal drive blocks 6432, 6434 is an exemplary embodiment of a three-leaf valve assembly 6440 (e.g., an aortic valve assembly). The valve assembly 6440 can be made of any desired material and, in an exemplary configuration, is made of bovine pericardial tissue or latex.

[0350] An exemplary embodiment of a delivery system and method shown in FIGS. 66 to 70 and disclosed herein can be used to percutaneously deploy the inventive aortic valve assembly 6440 in what is currently referred to as Transcatheter Aortic-Valve Implantation, known in the art under the acronym TAVI. As set forth above, this system and method can equally be used to deploy replacement pulmonary, mitral and tricuspid valves as well. The configuration of the delivery system and the valve assembly 6440 as an aortic valve assembly provide significant advantages over the prior art. As is known, current TAVI procedures have a risk of leak between an implanted device and the aortic valve annulus, referred to as paravalvular leak. Other disadvantages of prior art TAVI procedures include both migration (partial movement) and embolism (complete release). The reason for such movement is because, before use and entry into the patient, the prior art

replacement aortic valves are required to be crushed manually by the surgeon onto an interior balloon that will be used to expand that valve again when ready for implantation. Because the native annulus of the implantation site is not circular, and due to the fact that the balloon forces the implanted pre-crushed valve to take a final shape of the circular balloon, prior art implants do not conform to the native annulus. Not only are such prior art systems hard to use, they provide no possibility of repositioning the implanted valve once the balloon has expanded.

[0351] The progression of FIGS. **66** to **70** illustrates an exemplary implantation of the inventive aortic valve assembly **6440**. Various features of the delivery system are not illustrated in these figures for reasons of clarity. Specifically, these figures show only the guidewire **6610** and the nose cone **6620** of the delivery system. FIG. **66** shows the guidewire **6610** already positioned and the aortic valve assembly **6440** in a collapsed state resting in the delivery system just distal of the nose cone **6620**. In this illustration, the aortic valve assembly **6440** and nose cone **6620** are disposed in the right iliac artery. FIG. **67** depicts the aortic valve assembly **6440** and nose cone **6620** in an advanced position on the guidewire **6610** within the abdominal aorta adjacent the renal arteries. FIG. **68** shows the aortic valve assembly **6440** just adjacent the aortic valve implantation site. Finally, FIGS. **69** and **70** show the aortic valve assembly **6440** implanted in the heart before the nose cone **6620** and/or the guidewire **6610** are retracted.

[0352] The inventive delivery system and aortic valve assembly **6440** eliminate each of the disadvantageous features of the prior art. First, there is no need for the surgeon to manually crush the implanted prosthesis. Before the aortic valve assembly **6440** is inserted into the patient, the delivery system simply reduces the circumference of the lattice **6410** automatically and evenly to whatever diameter is desired by the surgeon, the delivery system requires, or is reduced in diameter by the manufacturer and loaded into the delivery system for later implantation. The stent and valve assemblies described herein can be reduced to a loading diameter of between 4 mm and 8 mm, and, in particular, 6 mm, to fit inside a 16-20 French sheath, in particular, an 18 French or smaller delivery sheath. When the aortic valve assembly **6440** reaches the implantation site, the surgeon causes the delivery system to evenly and automatically expand the aortic valve assembly **6440**. As this expansion is slow and even into the implant position, it is gentle on calcification at the implant site. Likewise, the even expansion allows the lattice structure to assume the native, non-circular perimeter of the implant site not only due to the way the delivery system expands the lattice assembly **6410**, but also because the hinged connections of each of the struts **6412** allows the lattice assembly **6410** to bend and flex naturally after implantation dependent upon the corresponding non-uniform tissue wall adjacent to each pivoting strut **6412** (assumption of the natural shape of the implantation wall also occurs with the alternative non-hinged embodiments disclosed herein). Due to these facts, a better seating of the implant occurs, which leads axiomatically to a better paravalvular seal. The inventive delivery system sizes the prosthesis precisely, instead of the gross adjustment and installation present in the prior art. Another significant disadvantage of the prior art is that a balloon is used within the central opening of the valve to expand the valve, thus completely occluding the aorta and causing tremendous backpressure on the heart, which can be hazardous to the patient. The valves described herein, in contrast, remain open during deployment to, thereby, allow continuous blood flow during initial deployment and subsequent repositioning during the procedure and also start the process of acting as a valve even when the implant is not fully seated at the implantation site.

[0353] Significantly, prior art TAVI systems require a laborious sizing process that requires the replacement valve to be sized directly to the particular patient's annulus, which sizing is not absolutely correct. With the delivery system and aortic valve assemblies described herein, however, the need to size the valve assembly beforehand no longer exists—all that is needed is to select an implant having somewhere within an intermediate position of the implant's expansion range the approximate diameter of the annulus at the implantation site. Additionally, with regard to both stent graft and valve systems described herein, because the stent assemblies are adjustable, they can be

adjusted even after being implanted within a vessel for a long period of time. For example, when conducting a TAVI process in children having congenital defects, there is a need to remove and implant a new valve after a few years because of the patient's growth. The assemblies described herein, in contrast to the prior art, can be re-docked well after implantation and further expanded, either at regular intervals or periodically, to adjust for the patient's growth.

[0354] The aortic valve assembly **6440** is configured to have a valve leaf overlap **6542** (see FIG. **65**) that is more than sufficient when the aortic valve assembly **6440** is at its greatest diameter and, when the aortic valve assembly **6440** is smaller than the greatest diameter, the valve leaf overlap **6542** merely increases accordingly. An exemplary range for this overlap can be between approximately 1 mm and approximately 3 mm.

[0355] A further significant advantage not provided by prior art TAVI systems is that the inventive delivery system and valve assembly can be expanded, contracted, and re-positioned as many times operatively as desired, but also the inventive delivery system and valve assembly can be re-docked post-operatively and re-positioned as desired. Likewise, the learning curve for using the inventive delivery system and valve assembly is drastically reduced for the surgeon because an automatic control handle (described in further detail below) performs each operation of extending, retracting, adjusting, tilting, expanding, and/or contracting at a mere touch of a button (see, e.g., FIGS. **105** to **107**).

[0356] Another exemplary use of the inventive lattice assembly and delivery system is for a latticework-actuated basket filter, which can be either added to the devices, systems, and methods disclosed herein or which can stand-alone. Such an embolic umbrella can perform better than, for example, the EMBOL-X® Glide Protection System produced by Edward Lifesciences. Such a filter would be attached to the docking jacks so that it expands in place automatically as the device is expanded but would be removed with the delivery system without any additional efforts on the part of the surgeon.

[0357] Another exemplary embodiment of a replacement heart valve assembly **7100** according to the invention is shown in FIGS. **71** to **83**. Even though the exemplary embodiment is shown for an aortic valve, the invention is not limited thereto. This embodiment is equally applicable to pulmonary, mitral and tricuspid valves with appropriate changes to the valve leaflets, for example. The replacement heart valve assembly **7100** shown in various views in FIGS. **71** to **75** is comprised of a stent lattice **7110**, graft enclosures **7120**, jack assemblies **3000**, graft material **7130**, valve leaflets **7140**, and commissure plates **7150**. Operation and construction of the replacement heart valve assembly **7100** is explained with reference to FIGS. **76** to **83** with various views therein having the graft material **7130** and/or the valve leaflets **7140** removed. In FIGS. **75** and **76**, the replacement heart valve assembly **7100** is in an expanded state (when used herein, “expanded state” does not mean that the state shown is the greatest expanded state of the prosthesis; it means that the prosthesis is expanded sufficiently enough to be sized for an implantation in some anatomic site) such that it accommodates to the natural geometry of the implantation site. With the graft material removed (see, e.g., FIG. **76**), the structure around the three valve leaflets **7140** is easily viewed. The proximal and distal drive blocks **3020**, **3010** have internal configurations and the support rod **3080**, the drive screw **740**, and the distal drive screw coupler part **3052** disposed therein.

[0358] The stent lattice **7110** is similar to previous embodiments described herein except for the center pivot points of each strut **7112** of the stent lattice **7110** and the graft enclosures **7120**. In the exemplary embodiment shown, the center pivot points are not merely pivoting connections of two struts **7112** of the stent lattice **7110**. In addition, the outer-most circumferential surface of the pivoting connection comprises a tissue anchor **7114**, for example, in the form of a pointed cone in this exemplary embodiment. Other external tissue anchoring shapes are equally possible, including spikes, hooks, posts, and columns, to name a few. The exterior point of the tissue anchor **7114** supplements the outward external force imposed by the actively expanded stent lattice **7110** by

providing structures that insert into the adjacent tissue, thereby further inhibiting migration and embolism.

[0359] The graft enclosures **7120** also supplement the outward external force imposed by the actively expanded stent lattice **7110** as explained below. A first characteristic of the graft enclosures **7120**, however, is to secure the graft material **7130** to the replacement heart valve assembly **7100**. The graft material **7130** needs to be very secure with respect to the stent lattice **7110**. If the graft material **7130** was attached, for example, directly to the outer struts **7112** of the stent lattice **7110**, the scissoring action that the adjacent struts **7112** perform as the stent lattice **7110** is expanded and contracted could adversely affect the security of the graft material **7130** thereto—this is especially true if the graft material **730** was sewn to the outer struts **7112** and the thread passed therethrough to the inside surface of the outer strut **7112**, against which the outer surface of the inner strut **7112** scissors in use. Accordingly, the graft enclosures **7120** are provided at a plurality of the outer struts **7112** of the stent lattice **7110** as shown in FIG. **71** to **87**. Each graft enclosure **7120** is fixedly attached at one end of its ends to a corresponding end of an outer strut **7112**. Then, the opposing, free end of the graft enclosure **7120** is woven through the inner side of the graft material **7130** and then back from the outer side of the graft material **7130** to the inner side thereof as shown in FIGS. **71** to **75**, for example. The opposing, free end of the graft enclosure **7120** is fixedly attached to the other end of the outer strut **7112**. This weaving reliably secures the outer circumferential side of the graft material **7130** to the stent lattice **7110**.

[0360] As mentioned above, graft enclosures **7120** simultaneously supplement the outward external force imposed by the actively expanded stent lattice **7110** with edges and protrusions that secure the replacement heart valve assembly **7100** at the implantation site. More specifically, the graft enclosures **7120** are not linear as are the exemplary embodiment of the outer struts **7112** of the stent lattice **7110**. Instead, they are formed with a central offset **7622**, which can take any form and, in these exemplary embodiments, are wave-shaped. This central offset **7622** first allows the graft enclosure **7120** to not interfere with the tissue anchor **7114**. The central offset **7622** also raises the central portion of the graft enclosure **7120** away from the stent lattice **7110**, as can be seen, for example, to the right of FIGS. **76** and **77** and, in particular, in the views of FIGS. **82** and **83**. The radially outward protrusion of the central offset **7622** inserts and/or digs into adjacent implantation site tissue to, thereby, inhibit any migration or embolism of the replacement heart valve assembly **7100**. By shaping the central offset **7622** appropriately, a shelf **7624** is formed and provides a linear edge that traverses a line perpendicular to the flow of blood within the replacement heart valve assembly **7100**. In the exemplary embodiment of the central offset **7622** shown in FIGS. **76**, **77**, and **79** to **81**, the shelf **7624** is facing downstream and, therefore, substantially inhibits migration of the replacement heart valve assembly **7100** in the downstream direction when exposed to systolic pressure. Alternatively, the central offset **7622** can be shaped with the shelf **7624** is facing upstream and, therefore, substantially inhibits migration of the replacement heart valve assembly **7100** in the upstream direction when exposed to diastolic pressure. The graft material needs to be able to stay intimately attached to the lattice throughout a desired range of terminal implantable diameters. To accomplish this, the graft material is made from a structure of material that moves in a fashion like that of the lattice. That is to say, as its diameter increases, its length decreases. This kind of movement can be accomplished with a braid of yarns or through the fabrication of graft material where its smallest scale fibers are oriented similarly to a braid, allowing them to go through a scissoring action similar to the lattice. One exemplary embodiment of the material is a high end-count braid made with polyester yarns (e.g., 288 ends using 40-120 denier yarn). This braid can, then, be coated with polyurethane, silicone, or similar materials to create stability and reduce permeability by joining all the yarns together. These coatings can be doped or filled with radiopaque material to improve visibility under fluoroscopy. The amount of coating (polyurethane, for example) can be varied to be increased where high-wear or trimming occurs. If the braid is trimmed by laser-cutting, for example, the cutting process seals the cut edge to prevent fraying and

allows for reduction or elimination of the need for a coating, such as polyurethane. Likewise, a spun-fiber tube can be made with similar polymers forming strands from approximately 2-10 microns in diameter. These inventive graft fabrication methods provide for a material that will be about 0.005" to 0.0015" (0.127 mm to 0.381 mm) thick and have all the physical properties necessary. See, for example, FIGS. **195-199**. A thin material is desirable to reduce the compacted diameter for easier introduction into the patient. This material is also important in a stent graft prosthesis where the lattice is required to seal over a large range of possible terminal diameters. The adjustable material is able to make the transition from the final terminal diameter of the upstream cuff to the main body of the graft.

[0361] As best shown in FIG. **73**, the valve leaflets **7140** are connected by commissure plates **7150** to the jack assemblies **3000**. Fixed connection of the commissure plates **7150** to the jack assemblies **3000** is best shown in FIGS. **82** and **83**. Each valve leaflet **7140** is connected between two adjacent commissure plates **7150**. Each commissure plate **7150** is comprised of two vertically disposed flat plates having rounded edges connected, for example, by pins projecting orthogonally to the flat plates. Pinching of the flat plates against the two adjacent valve leaflets **7140** securely retains the valve leaflets **7140** therein while, at the same time, does not form sharp edges that would tend to tear the captured valve leaflets **7140** therein during prolonged use. This configuration, however, is merely exemplary. This could be replaced with a simpler rod design around which the leaflets are wrapped and stitched into place.

[0362] Even though each valve leaflet **7140** can be a structure separate from the other valve leaflets **7140**, FIGS. **71** to **78** illustrate the three leaflets **7140** as one piece of leaf-forming material pinched, respectively, between each of the three sets of commissure plates **7150** (the material can, alternatively, pinch around the commissure plate or plates). The upstream end of the valve leaflets **7140** must be secured for the replacement heart valve assembly **7100** to function. Therefore, in an exemplary embodiment, the upstream end of the graft material **7130** is wrapped around and fixedly connected to the replacement heart valve assembly **7100** at the upstream side of the valve leaflets **7140**, as shown in FIG. **78**. In such a configuration, the upstream edge of the valve leaflets **7140** is secured to the graft material **7130** entirely around the circumference of the stent lattice **7110**. Stitches can pass through the two layers of graft and the upstream edge of the leaflet material to form a hemmed edge.

[0363] FIGS. **79** to **81** show the stent lattice **7110** in various expanded and contracted states with both the graft material **7130** and the valve leaflets **7140** removed. FIG. **79** illustrates the stent lattice **7110** and jack assemblies **3000** in an expanded state where the tissue anchor **7114** and the central offset **7622** protrude radially out from the outer circumferential surface of the stent lattice **7110** such that the stent lattice **7110** accommodates to the natural geometry of the implantation site. FIG. **80** illustrates the stent lattice **7110** and the jack assemblies **3000** in an intermediate expanded state and FIG. **81** illustrates the stent lattice **7110** and the jack assemblies **3000** in a substantially contracted state.

[0364] FIGS. **84** and **85** show an exemplary embodiment of a support system **8400** of the delivery system and method according to the invention for both supporting the jack assemblies **3000** and protecting the various control wires **750**, **770**, **2182**, **3098** of the jack assemblies **3000**. In these figures, the support bands **8410** are shown as linear. This orientation is merely due to the limitations of the computer drafting software used to create the figures. These support bands **8410** would only be linear as shown when unconnected to the remainder of the delivery system for the replacement heart valve assembly **7100**. When connected to the distal end of the delivery system, as diagrammatically shown, for example, in FIGS. **1**, **3**, **4**, and **9** with a wire-guide block **116**, all control wires **750**, **770**, **2182**, **3098** will be directed inwardly and held thereby. Similarly, the proximal ends **8412** of the support bands **8410** will be secured to the wire-guide block **116** and, therefore, will bend radially inward. In the exemplary embodiment of the support bands **8410** shown in FIGS. **84** and **85**, the distal ends **8414** thereof are fixedly secured to the disconnectors

drive block **3030** by an exemplary hinge assembly **8416**. In this exemplary embodiment, therefore, the support bands **8410** are of a material and thickness that allows the delivery system to function. For example, while traveling towards the implantation site, the replacement heart valve assembly **7100** will traverse through a curved architecture. Accordingly, the support bands **8410** will have to bend correspondingly to the curved architecture while, at the same time, providing enough support for the control wires **750, 770, 2182, 3098** to function in any orientation or curvature of the delivery system.

[0365] An alternative exemplary connection assembly of the support bands **8610** according to the invention is shown in FIGS. **86** and **87**. The distal end **8614** of each support band **8610** is connected to the disconnect drive block **3030** by a hinge assembly **8416**. The hinge assembly **8416**, for example, can be formed by a cylindrical fork at the distal end **8614** of the support band **8610**, an axle (not illustrated), and a radially extending boss of the disconnect drive block **3030** defining an axle bore for the axle to connect the cylindrical fork to the boss. In such a configuration, the support bands **8610** can have different material or physical properties than the support bands **8410** because bending movements are adjusted for with the hinge assembly **8416** instead of with the bending of the support bands **8410** themselves. The proximal end of the support bands **8610** are not shown in either FIG. **86** or **87**. Nonetheless, the proximal ends can be the same as the distal end of the support bands **8610** or can be like the distal end **8614** of the support bands **8410**. By pre-biasing the support bands to the outside, they can help reduce or eliminate the force required to deflect the control wires.

[0366] An embodiment of the replacement heart valve assembly **7100** as an aortic valve is shown implanted within the diseased valve leaflets of a patient's heart in FIG. **88**. As can be seen in this figure, the natural valve takes up some room at the midline of the replacement heart valve assembly **7100**. Therefore, the stent lattice of the replacement heart valve assembly **7100** can be made to have a waistline, i.e., a narrower midline, to an hourglass shape instead of the barrel shape. In such a configuration, the replacement heart valve assembly **7100** is naturally positioned and held in place.

[0367] A further exemplary embodiment of the inventive actively controllable stent lattice and the delivery system and method for delivering the stent lattice are shown in FIGS. **89** to **93**. In this embodiment, the prosthesis **8900** includes a stent lattice **110, 3810, 4200, 4600, 6410, 7110** and three jack assemblies **700, 2100, 3000, 6430**. These figures also illustrate a distal portion of an exemplary embodiment of a delivery system **8910** for the inventive prosthesis **8900**. Shown with each jack assembly **700, 2100, 3000, 6430** are the drive and disconnect wires **750, 770**, which are illustrated as extending proximally from the respective jack assembly **700, 2100, 3000, 6430** into a wire guide block **116**. Due to the limitations of the program generating the drawing figures, these wires **750, 770** have angular bends when traversing from the respective jack assembly **700, 2100, 3000, 6430** towards the wire guide block **116**. These wires, however, do not have such angled bends in the invention. Instead, these wires **750, 770** form a gradual and flattened S-shape that is illustrated diagrammatically in FIG. **89** with a dashed line **8920**. Operation of the prosthesis **8900** is as described above in all respects except for a tilting feature regarding the wires **750, 770**.

Specifically, rotation of the drive wire **750** in respective directions will contract and expand the stent lattice **110, 3810, 4200, 4600, 6410, 7110**. Then, when the stent lattice **110, 3810, 4200, 4600, 6410, 7110** is implanted correctly in the desired anatomy, the disconnect wire **770** will be rotated to uncouple the proximal disconnect drive block and, thereby, allow removal of the delivery system **8910**. This embodiment provides the delivery system **8910** with a prosthesis-tilting function. More specifically, in the non-illustrated handle portion of the delivery system **8910**, each pair of drive and disconnect wires **750, 770** are able to be longitudinally fixed to one another and, when all of the pairs are fixed respectively, each pair can be moved distally and/or proximally.

[0368] In such a configuration, therefore, if the wires **750, 770** labeled with the letter X are moved proximally together and the other two pairs of wires Y and Z are moved distally, then the entire

prosthesis **8900** will tilt into the configuration shown in FIG. **90**. Alternatively, if the wires X are kept in position, the wires Y are moved proximally and the wires Z are moved distally, then the entire prosthesis **8900** will tilt into the configuration shown in FIG. **91**. Likewise, if the wires X are moved distally and the wires Y and Z are moved proximally, then the entire prosthesis **8900** will tilt into the configuration shown in FIG. **92**. Finally, if the wires X are extended distally, the wires Y are extended further distally, and the wires Z are moved proximally, then the entire prosthesis **8900** will tilt into the configuration shown in FIG. **93**.

[0369] Still a further exemplary embodiment of the inventive actively controllable stent lattice and the delivery system and method for delivering the stent lattice are shown in FIGS. **94** to **102**. In this embodiment, the prosthesis **9400** is a stent graft having a proximal, actively controlled stent lattice **110**, **3810**, **4200**, **4600**, **6410**, **7110** and only two opposing jack assemblies **700**, **2100**, **3000**, **6430**. Instead of two additional jack assemblies **700**, **2100**, **3000**, **6430**, this embodiment contains two opposing pivoting disconnecter drive blocks **9430**. These disconnecter drive blocks **9430**, as shown for example in the view of FIG. **96** rotated circumferentially ninety degrees, have bosses **9432** extending radially outward and forming the central pivot joint for the two crossing struts **9410**. The two disconnecter drive blocks **9430** act as pivots to allow the prosthesis **9400** to tilt in the manner of a swashplate when the two opposing sets of control wires **750**, **770** are moved in opposing distal and proximal directions. FIG. **94** shows the near set of control wires **750**, **770** moved proximally and the far set moved distally. In FIG. **95**, the swashplate of the prosthesis **9400** is untilted, as is the prosthesis **9400** in FIGS. **96** and **97**, the latter of which is merely rotated ninety degrees as compared to the former. FIGS. **98** and **99** depict the prosthesis **9400** as part of a stent graft having the stent lattice **9810** inside a proximal end of a tubular shaped graft **9820**.

[0370] The prosthesis **9400** in FIGS. **100** to **102** is also a stent graft but, in this exemplary embodiment, the graft **10010** is bifurcated, for example, to be implanted in an abdominal aorta. FIGS. **101** and **102** show how the proximal end of the prosthesis **9400** can be tilted with the swashplate assembly of the invention, for example, in order to traverse a tortuous vessel in which the prosthesis **9400** is to be implanted, such as a proximal neck of abdominal aortic aneurysm.

[0371] The exemplary embodiment of the prosthesis **10300** shown in FIGS. **103** and **104** does not include the swashplate assembly. Instead, the delivery system includes a distal support structure **10310** that ties all of the support bands **10312** to a cylindrical support base **10314** connected at the distal end of the delivery catheter **10316**.

[0372] An exemplary embodiment of the entire delivery system **10500** for the prosthesis **10300** is depicted in FIGS. **105** to **107**. In FIG. **105**, the delivery system is entirely self-contained and self-powered and includes the actively controllable stent lattice with an integral control system **10510**. The prosthesis **10300** is in an expanded state and the graft material is in cross-section to show a rear half. An alternative to the integral control system **10510** is a wireless control device **10600** that wirelessly communicates **10610** control commands to the system. Another alternative to the integral control system **10510** shown in FIG. **107** separates the control device **10700** with a cord **10710** for communicating control commands to the system. In this exemplary embodiment, the controls comprise four rocker switches **10712**, **10714**, **10716**, **10718** arranged in a square, each of the switches having a forward position, a neutral central position, and a rearward position.

[0373] Yet another exemplary embodiment of a control handle **10800** for operating a prosthesis having the actively controllable stent lattice according to the invention is depicted in FIGS. **108** to **118**. The views of FIGS. **108** and **109** show various sub-assemblies contained within the control handle **10800**. A user-interface sub-assembly **10810** includes a circuit board **10812** having circuitry programmed to carry out operation of the systems and methods according to the invention.

Electronics of the user-interface sub-assembly **10810** comprise a display **10814** and various user input devices **10816**, such as buttons, switches, levers, toggles, and the like. A sheath-movement sub-assembly **11000** includes a sheath-movement motor **11010**, a sheath movement transmission **11020**, a sheath movement driveshaft **11030**, and a translatable delivery sheath **11040**. A strain

relief **11042** is provided to support the delivery sheath **11040** at the handle shell **10802**. A power sub-assembly **11200** is sized to fit within the handle **10800** in a power cell compartment **11210** containing therein power contacts **11220** that are electrically connected to at least the circuit board **10812** for supplying power to all electronics on the control handle **10800** including all of the motors. A needle-movement sub-assembly **11300** controls deployment of the needles and keeps tension on the needles continuously even when the delivery sheath **11040** is bent through tortuous anatomy and different bends are being imposed on each of the needles. The needles are three in number in this exemplary embodiment. Finally, a jack engine **11600** controls all movements with regard to the jack assemblies.

[0374] The user-interface sub-assembly **10810** allows the surgeon to obtain real-time data on all aspects of the delivery system **10800**. For example, the display **10814** is programmed to show the user, among other information, deployment status of the stent lattices, the current diameter of the stent lattices, any swashplate articulation angle of the stent lattice to better approximate an actual curved landing site, all data from various sensors in the system, and to give audio feedback associated with any of the information. One informational feedback to user can be an indicator on the display **10814** that the delivery sheath **11040** is retracted sufficiently far to completely unsheath the prosthesis. Other information can be a force feedback indicator showing how much force is being imparted on the lattice from the vessel wall, e.g., through a torque meter, a graphical change in resistance to the stepper motor, a mechanical slip clutch, direct load/pressure sensors on lattice. With such information, the prosthesis can have Optimal Lattice Expansion (OLE), achieve its best seal, have a decrease in migration and embolization, and have an amount of outward force limited (i.e., a force ceiling) to stop expansion before tissue damage occurs. A suitably sized visual indicator can even show in a 1:1 ratio the actual diameter position of the stent lattice. Other possible sensors for taking measurements inside and/or outside the prosthesis (e.g., above and below touchdown points of lattice) can be added into the inventive powered delivery system or handle. These devices include, for example, a video camera, a flow wire to detect flow showing blood passing around prosthesis/double lumen catheter and showing pressure gradients, a Doppler device, an intrinsic pressure sensor/transducer, an impedance of touchdown zone, fractional flow reserve, and an intracardiac/intravascular ultrasound. For an example of the latter sensor, an ultrasound device could be incorporated into the nose cone of the delivery system and can be extended or retracted to provide assistance in positioning the implant. Additionally and/or alternatively, by measuring pressure above and below the implant, pressure sensors or lumens to pressure sensors located within the handle can provide a pressure gradient useable to calculate orifice area when coupled with cardiac output.

[0375] Having all of the user interface actuators **10816** within reach of a single finger of the user provides unique and significant advantages by allowing the surgeon to have one-hand operation of the entire system throughout the entire implantation procedure. In all mechanical prior art systems when torque is applied, the second hand is needed. Pushing of single button or toggling a multi-part switch eliminates any need for the user's second hand. Using different kinds of buttons/switches allows the user to be provided with advanced controls, such as the ability to have coarse and fine adjustments for any sub-procedure. For example, expansion of the lattice can, initially, be coarse by automatically directly expanded out to a given, pre-defined diameter. Then, further expansion can be with fine control, such as a millimeter at a time. The varying of diameter can be both in the open and close directions. If the prosthesis needs to be angled, before, during, and/or after varying the expansion diameter, the user can individually manipulate each jack screw or control wires to gimbal the upstream end of implant so that it complies with vessel orientation; both during diameter/articulation changes, the physician can inject contrast to confirm leak-tightness. Even though the exemplary embodiment of the needle deployment shown is manual, this deployment can be made automatic so that, once the prosthesis is implanted, and only after the user indicates implantation is final, an automatic deployment of the engaging anchors can be made. With regard

to undoing the delivery system, this release can be with a single touch, for example, of a push button. Also, with an integrated contrast injection assembly, a single touch can cause injection of contrast media at the implantation site.

[0376] The sheath-movement sub-assembly **11000** also can be controlled by a single button or switch on the circuit board **10812**. If the user interface is a two-position toggle, distal depression can correspond with sheath extension and proximal depression can correspond with sheath retraction. Such a switch is operable to actuate the sheath movement motor **11010** in the two rotation directions. Rotation of the motor axle **11022**, therefore, causes the transmission **11024**, **11026** to correspondingly rotate, thereby forcing the threaded sheath movement driveshaft **11030** to either extend distally or retract proximally. The exemplary embodiment of the transmission includes a first gear **11024** directly connected to the motor axle **11022**. The first gear **11024** is meshed with the outside teeth of a larger, hollow, driveshaft gear. The interior bore of the driveshaft gear **11026** has threads corresponding to the exterior threads of the sheath movement driveshaft **11030**. As such, when the driveshaft gear **11026** rotates, the sheath movement driveshaft **11030** translates. The driveshaft gear **11026** is surrounded by a bushing **11028** to allow rotation within the housing shell **10802**. In order to prevent rotation of the sheath movement driveshaft **11030**, as shown in FIG. **111**, the sheath movement driveshaft **11030** has a longitudinal keyway **11032** that has a cross-sectional shape corresponding to a key that is grounded to the handle shell **10802**. The sheath movement driveshaft **11030** also is hollow to accommodate a multi-lumen rod **10804** (shown best in FIG. **112**) housing, within each respective lumen, any of the control wires **750**, **770**, **2182**, **3098** and the guidewire **6610**, these lumens corresponding to those within the wire guide block **116** at the distal end of the delivery sheath **10040**.

[0377] The size and shape of the power sub-assembly **11200** is limited in shape only by the power cell compartment **11210** and the various wires and rods that traverse from the needle-movement sub-assembly **11300** and the jack engine **11600** therethrough until they enter the lumens of the multi-lumen rod **10804**. Some of these wires and rods are illustrated with dashed lines in FIG. **112**. Power distribution to the circuit board **10812** and/or to the motors is carried out through power contacts **11220**. Such power distribution lines are not illustrated for reasons of clarity. This method or similar such as a rack and pinion or drag wheels can be used to drive the sheath extension and retraction.

[0378] The needle-movement sub-assembly **11300** is described with reference to FIGS. **113** to **115**, and best with regard to FIG. **113**. Each of the needle rods **11302** that connect to the needles in the prosthesis to the needle-movement sub-assembly **11300** is associated with a tension spring **11310**, an overstroke spring **11320**, and a control tube **11332**. The three control tubes **11332** are longitudinally held with respect to a control slider **11330** by the overstroke spring **11320**. As long as the force on the needles is not greater than the force of the overstroke spring **11320**, movement of the needle rod **11302** will follow the control slider **11330**. A needle deployment yoke **11340** slides with respect to the shell **10802** of the control handle **10800**. When the needle deployment yoke **11340** contacts the control slider **11330** as it moves distally, the needle deployment yoke **11340** carries the control slider **11330** and the needle rods **11302** distally to, thereby, deploy the needles. The transition from FIGS. **113** to **114** shows how the tension spring **11310** keeps tension on the needles by biasing the control slider **11330** proximally. Deployment of the needles is shown by the transition from FIGS. **114** to **115**. As mentioned above, the needles **3070** each have a bent needle tip **3072**. In a configuration where the needles **3070** are connected directly all the way back to the needle-movement sub-assembly **11300**, there is a high likelihood that bending of the delivery catheter **11040** will impart various different forces on the needle rods **11302**. These forces will tend to pull or push the needle rods **11302** and, thereby possibly extend the needles **3070** when not desired. Accordingly, each tension spring **11310** is longitudinally connected to the needle rod **11302** to compensate for these movements and keep the bent needle tip **3072** within the needle tip groove of the **3013** distal drive block **3010**.

[0379] Because deployment of the needles is intended (ideally) to be a one-time occurrence, a yoke capture **11350** is provided at the end of the yoke stroke. Capture of the yoke **11340** can be seen in FIG. **116**. Of course, this capture can be released by the user if such release is desired. Finally, if too much force is imparted on the needles when being deployed, the force of the overstroke spring **11320** is overcome and the control tube **11332** is allowed to move with respect to the control slider **11330**. The compression of the overstroke spring **11320** cannot be shown in FIG. **115** because of the limitation of the software that created FIG. **115**.

[0380] The jack engine **11600** is configured to control all rotation of parts within the various jack assemblies **700**, **2100**, **3000**, **6430**. The exemplary embodiment of the control handle **10800** shown in FIGS. **108** to **118** utilizes three jack assemblies similar to jack assemblies **3000** and **6430**. In other words, the needles are separate from the proximal drive blocks of both assemblies and only two rotational control wires **750**, **770** are needed. Therefore, for the three jack assemblies, six total control wires are required—three for the drive wires **750** and three for the disconnect wires **770**. These control wires **750**, **770** are guided respectively through six throughbores **10806** (surrounding the central guidewire throughbore **10807** in FIG. **115**) and proximally end and are longitudinally fixed to a distal part **11512** of each of six telescoping wire control columns **11510**, shown in FIGS. **115** and **116**. All control wires, even the needle rods **11302**, terminate at and are fixed longitudinally to a distal part **11512** of a respective telescoping wire control column **11510**. Each part of these telescoping wire control columns **11510**, **11512** are rigid so that rotation of the proximal part thereof causes a corresponding rotation of the distal part **11512** and, thereby, rotation of the corresponding control wire **750** or **770**. The reason why all control wires, even the needle rods **11302**, terminate at and are fixed longitudinally to a distal part **11512** of a respective telescoping wire control column **11510** is because tortious curving of the wires/rods from their proximal ends to the distal ends longitudinally fixed at the stent assembly to be implanted will cause the wires to move longitudinally. If there is no play, the wires/rods will impart a longitudinal force on any parts to which they are grounded, for example, to the threaded connection at the stent assembly at the distal end. This longitudinal force is undesirable and is to be avoided to prevent, for example, the drive screws from breaking loose of their threads. To avoid this potential problem, the proximal end of each wire/rod is longitudinally fixed to the distal part **11512** of a respective telescoping wire control column **11510**. The distal part **11512** is keyed to the wire control column **11510**, for example, by having an outer square rod shape slidably movable inside a corresponding interior square rod-shaped lumen of the proximal part of the wire control column **11510**. In this configuration, therefore, any longitudinal force on any wire/rod will be taken up by the respective distal part **11512** moving longitudinally proximal or distal depending on the force being exerted on the respective wire/rod.

[0381] Torque limiting is required to prevent breaking the lattice or stripping the threads of the drive screw. This can be accomplished in software by current limiting or through a clutch mechanism disposed between the drive motors and the sun gears. An integral contrast injection system can be incorporated into the handle of the delivery system through another lumen. With a powered handle, therefore, a powered injection as part of handle is made possible.

[0382] Because all of the drive wires **750** need to rotate simultaneously, and due to the fact that all of the disconnect wires also need to rotate simultaneously, the jack engine **11600** includes a separate control motor **11650**, **11670** (see FIG. **115**) and separate transmission for each set of wires **750**, **770**. The view of FIG. **117** illustrates the transmission for the drive-screw control motor **11650**. At the output shaft **11651** of the drive-screw control motor **11650** is a first drive gear **11652** interconnected with a larger second drive gear **11653**. The second drive gear **11653** is part of a coaxial planetary gear assembly and has a central bore therein for passing therethrough the guidewire **6610**. A hollow rod **11654** is fixedly connected in the central bore and extends through a transmission housing **11610** to a distal side thereof, at which is a third drive gear **11655**, as shown in FIG. **118**. The third drive gear **11655** is interconnected with three final drive gears **11656**, each of

the final drive gears **11656** being fixedly connected to a respective proximal part of one of the three telescoping wire control columns **11510** associated with each drive wire **750**. Accordingly, when the drive-screw control motor **11650** rotates, the three final drive gears **11656** rotate the control columns **11510** that rotate the drive screws of the jack assemblies **3000, 6430**.

[0383] The disconnect control motor **11670** operates in a similar manner. More specifically and with regard to FIG. **116**, the output shaft **11671** of the disconnect control motor **11670** is a first disconnect gear **11672** interconnected with a larger second disconnect gear **11673**. The second disconnect gear **11673** is part of a coaxial planetary gear assembly and has a central bore therein for passing therethrough the guidewire **6610**. A hollow rod **11674** is fixedly connected in the central bore about the hollow rod **11654** and extends through the transmission housing **11610** to the distal side thereof, at which is a third disconnect gear **11675** (also disposed about the hollow rod **11654**), as shown in FIG. **118**. The third disconnect gear **11675** is interconnected with three final disconnect gears (not illustrated), each of the final disconnect gears being fixedly connected to a respective proximal part of one of the three telescoping wire control columns **11510** associated with each disconnect wire **770**. Accordingly, when the disconnect control motor **11670** rotates, the three final disconnect gears rotate the control columns **11710** that rotate the retainer screws of the jack assemblies **3000, 6430**. The activation of the disconnect drive also unscrews the needle connections when included. One exemplary embodiment for having the needles disconnect before the entire implant is set free from the docking jacks provides a lower number of threads on the needle disconnects.

[0384] Not illustrated herein is the presence of a manual release for all actuations of the delivery system. Such manual releases allow for either override of any or all of the electronic actuations or aborting the implantation procedure at any time during the surgery. Manual release sub-assemblies are present for retraction and extension of the delivery sheath, expansion and contraction of all stent lattices, and undocking of all disconnect drive blocks. One exemplary embodiment for each of the manual releases is a lever and ratcheting assembly that permits rotation in only one direction. Manual releases are utilized, for example, when something is detected as being wrong, if there is a failure of the electronics or software, or if the battery dies, and the user desires to remove the delivery system from the patient without implanting the stent lattice or other embodiments of the stent assemblies. With regard to the delivery sheath, for example, manual release means that it is desired to either retract or extend the delivery sheath. In the extension direction, the delivery sheath is extended as much as possible so that resheathing can possibly be accomplished, at which time the distal nose cone is retracted into the delivery sheath. In such a situation, the mechanism will progressively extend the delivery sheath distally until the user determines that the force required to further extend the delivery sheath is too large or the delivery sheath has extended as far as designed. In each case of the various manual release mechanisms described, the mechanisms will have torque and/or force limiting devices that prevent the user from inputting too much force that would break the system. In the retraction direction, the delivery sheath is retracted as much as possible so that implantation of the implant can still be accomplished. In such a situation, the mechanism will progressively retract the delivery sheath distally until the user determines that the force required to further extend the delivery sheath is too large or the delivery sheath has retracted as far as needed for implantation to occur. With regard to the stent lattice embodiments, manual release means that it is desired to contract the stent lattice as much as possible. Accordingly, the ratchet will progressively rotate all drive screws in the direction that causes contraction of the stent lattice. It is equally possible to have this manual release be bi-directional so that forcible expansion of the stent lattice can occur. Likewise, a separate manual release can be uni-directional to only rotate in a direction to expand the stent lattice. With regard to undocking of the disconnect drive blocks, the manual release would only be used in a situation where implantation was acceptable and desired but, for some reason, one or more of the control tubes preventing disconnect was not allowing the disconnect to occur. In such a situation, a ratchet could progressively retract the

coils/wires connected to the control tubes. As this movement is only longitudinal and is in the nature of a few millimeters, the ratchet could be replaced by a lever or pull knob. Finally, with regard to retraction of the nose cone (and its control lumen) manual release means that it is desired to move the nose cone as much as possible proximally so that resheathing can possibly be accomplished. In such a situation, the ratchet will progressively retract the nose cone control lumen proximally until the user determines that the force required to further retract the nose cone is too large or the nose cone has retracted as far as desired.

[0385] Based upon the above, therefore, the delivery system control handle **10800** is entirely self-contained and self-powered and is able to actively control any prosthesis having the stent lattice and jack assemblies of the invention. An alternative embodiment to the combined drive of multiple control wires with a single motor as described above is a configuration providing a single drive motor for each of the control wires. With such a configuration, for example with respect to the jack screw drives, each motor is monitored for the amount of turns and synchronized with the other motors so that substantially simultaneous rotation of the jack screws occurs. The same monitoring is possible for the multiple control wires for disconnecting the implant. A beneficial result of independent driving of control wires is that it becomes possible to monitor the torque requirements and position of each drive wire. If a particular drive wire experiences a variation, the software can have built in allowances (based on testing potential faults, such as one drive screw being jammed or rubbing on the implantation site) to either allow the implantation to continue or to notify the user that some aspect is in fault. In such a case, the user can attempt a contraction/re-expansion to clear the fault, or, if needed or desired, a repositioning or recapture of the prosthesis.

[0386] An exemplary embodiment of a process for delivering an abdominal aortic stent graft of the invention as shown in FIG. **107** with the stent lattice as a proximal stent is described with regard to the flow chart of FIG. **119**. The procedure is started in Step **11900** where the lattice has been translated through the femoral artery to the implantation site just downstream of the renal arteries. Actuation of the upper left button rearward in Step **11902** causes the delivery sheath **10720** to unsheath from the AAA implant **10730** sufficient to expose the actuatable end (e.g., stent lattice) of the implant **10730** (which can be shown, for example, by the progression from FIG. **217** to FIG. **212**—a direction opposite the re-sheathing progression which is shown in the progression from FIG. **212** to **217**). In Step **11904**, visualization, such as through fluoroscopy, provides the user with feedback to show where the distal end **10732** of the prosthesis **10730** is situated. In this position, the stent lattice is in a contracted state (the expanded state is shown in the view of FIG. **107**). Radiopaque markers on the prosthesis **10730** are visible to show the proximal most points of the prosthesis **10730**. In Step **11906**, another surgery staff, typically, has marked the location of the renal arteries on the screen (on which the surgeon sees the markers) with a pen or marker. In Step **11908**, the surgeon translates the lattice of the prosthesis **10730** with the radiopaque markers to a location targeted below the renal arteries. The physician begins to expand the lattice in Step **11910** by pressing the upper right button forward (i.e., forward=open and rearward=close). Depending upon the desire of the surgeon or as set in the programming of the control device **10700**, the lattice can open incrementally (which is desirable due to blood flow issues) or can be expanded fluidly outward. Implantation occurs in Step **11912** and has three phases. In the first phase of implantation, the physician performs a gross orientation of the proximal end of the prosthesis **10730** until touchdown in the abdominal aorta. In the second phase, the physician fine-tunes the implantation using intermittent expansion prior to coaptation in all three dimensions and, in the third phase, the proximal end of the implant **10730** is either satisfactorily coapted or, if the physician is not satisfied with the coaptation, then the physician reduces the diameter of the stent lattice and starts, again, with phase two. It is noted that the control device **10700** can be programmed to, at the first touch of the upper right button, to go to a particular diameter opening. For example, if the implantation site is predetermined to be approximately 20 mm, then the control device **10700** can be programmed to expand directly to 15 mm and, for each touch of the upper right button thereafter, expansion will

only occur by 1 mm increments no matter how long the upper right button is pushed forward. During Step **11912**, the physician is able to view all of the various feedback devices on the control handle, such as the real time diameter of the prosthesis, the angulation thereof, a comparison to a predetermined aortic diameter of the touchdown point, an intravascular ultrasound assessing proximity to wall, and when wall touch occurs. With the digital display **10711** of the invention, the physician can even see an actual representation of the expanding lattice demonstrating all of the characteristics above. During the various implantation phases, the physician can pause at any time to change implant position. Angulation of the stent lattice can be done actively or while paused. As the outer graft material approaches the wall, adjustment of the entire delivery device continues until complete coaptation of the prosthesis **10730**, where it is insured that the location with respect to the renal arteries is good, along with proper angulation. As the stent graft touches the aortic wall, the physician can analyze all of the feedback devices and information to make implantation changes. At any time if the physician questions the implantation, then restart occurs to readjust the stent lattice along with a return to phase two. Further, as coaptation occurs, any other fixation devices can be utilized, for example, passive tines/barbs, a outwardly moving flex-band that presses retention device (e.g., through graft) and into aortic wall, the tissue anchor **7114**, and the graft enclosures **7120**. For such devices, there is no secondary action required to disengage/retract tines that are engaged. In Step **11914**, the physician performs an angiogram to determine positioning of the implantation (the angiogram can be either separate or integral with the delivery system **10700**), and if the positioning is not as desired (e.g., endoleak), the physician can retract the stent lattice and use the sheath **10720** to re-collapse the stent lattice using the graft material to ease the delivery sheath **1020** back over the lattice. However, if the physician determines that there is good positioning, the physician retracts the delivery sheath **10720** by pressing the upper left button rearward until at least contralateral gate is exposed. It is noted that stabilization of the ipsilateral graft material with the delivery system **10700** allows for better cannulization of the contralateral gate for a secondary prosthesis.

[0387] In Step **11916**, the contralateral limb is deployed as is known in the art. However, if desired, the contralateral limb can also include the actively expanded stent lattice according to the invention. It is also desirable to perform a balloon expansion at a graft-to-graft junction if the contralateral limb utilizes a self-expanding distal stent. If the actively controllable stent lattice is used, then Steps **11900** to **11914** are repeated but for the contralateral limb. In Step **11918**, the delivery sheath **10720** is retracted by pressing the upper left button rearward until ipsilateral limb is deployed. The prosthesis **10730** is, now, ready to be finally deployed.

[0388] In Step **11920**, the physician actuates the lower left button rearward to unscrew the retainer screws and, thereby undock the disconnect drive blocks from the prosthesis **10730**. One significant advantage of the delivery system **10700** is that there is no surge either distal or proximal when undocking occurs and finally releases the prosthesis because the entire undocking movement is merely an unscrewing of a rod from a threaded hole. Torque imposed on the stent lattice is also minimized by using counter rotating screws that result in a zero imposed torque when even in number. It is noted that, for all of the exemplary embodiment of the stent lattices that utilize the jack screws described herein, the delivery system does not impart an actuation force either on or to the stent lattices. In other words, the force for changing the configuration of the stent lattice is generated entirely within the stent lattice itself. More specifically, the forces that are used to actuate the configuration change of the stent lattice are imposed by the distal and proximal jack struts. This means that the actuation force causing expansion of the stent lattice is delivered and countered within the implant independent of the delivery tool.

[0389] The upper left button is pressed forward to extend the delivery sheath **10720** so that it connects with the distal end of nose cone **10740** while making sure that the open distal end of the delivery sheath **10720** does not catch any part of the ipsilateral distal stent or the actively controlled proximal stent. It is at this point where a manual override could be employed if the surgeon wanted

to feel the re-docking of the delivery sheath **10720** to the nose cone **10740**. If desired, using the lower right button pressing rearward, the physician can retract the nose cone **10740** into the distal end of the delivery sheath **10720** with the lower right button. In Step **11922**, if the ipsilateral distal stent is self-expanding, the physician performs a final balloon expansion. However, if the ipsilateral distal stent utilizes the actively controllable stent lattice of the invention, Steps **11900** to **11914** are repeated but for the ipsilateral limb. A completion angiogram is performed in Step **11924** to make sure the prosthesis did not shift and that all leak possibilities have been ruled out. In an exemplary embodiment where the control system **10700** includes an integral dye system, the physician would extend the system proximal to the proximal active lattice to perform this angiogram. Finally, in Step **11926**, the lower right button is pressed rearward to retract the delivery system as much as possible into the handle and, in Step **11928**, the delivery system **10700** is removed from the patient.

[0390] FIG. **120** shows an exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly **12000** having nine lattice segments **12010** in a self-expanded native position as will be described below. In one exemplary embodiment, each of the nine lattice segments is formed with one-half of either a threaded or smooth bore **12012** for respective coordination with either a threaded or smooth portion of a jack screw **12020**. In another exemplary embodiment, the nine lattice segments are formed from one integral piece of a shape memory metal (e.g., Nitinol or other super-elastic material) and with a jack screw **12020** disposed between adjacent pairs of repeating portions of the lattice and through the wall of the stent lattice. In the views shown in FIGS. **120** and **121**, each jack screw **12020** is placed in a non-engaged state to allow crimp of the stent lattice for loading into a stent delivery system. In this regard, FIG. **121** illustrates the stent assembly **12000** in a contracted/crimped state for loading into the stent delivery system, an example of which is illustrated in the progression of FIGS. **217** to **212**. In this non-engaged state, as the stent assembly **12000** is crimped for delivery, the proximal jack strut **12014** surrounding the non-threaded portion of each jack screw **12020** can slide thereabout with play between the two positions shown in FIGS. **120** and **121** without hindrance or bottoming out the distal drive screw coupler part **12052** while the lattice expands longitudinally when contracted by the delivery sheath of the delivery system. When the stent assembly **12000** is allowed to self-expand back to the position shown in FIG. **120**, the jack screw **12020** moves into the bore of the distal jack strut **12014** until the distal drive screw coupler part **12052** hits the proximal end of the proximal jack strut **12014**. Accordingly, with rotation of the jack screw **12020** in the stent-expansion direction, after the distal drive screw coupler part **12052** hits the proximal end of the proximal jack strut **12012**, further lattice-expanding movement of the drive screw **12020** starts moving the proximal jack strut **12014** towards the distal jack strut **12013** to expand the stent assembly **12000**.

[0391] Longitudinally, the stent assembly **12000** is provided with pairs of jack struts **12013**, **12014** connected by a respective jack screw **12020** and intermediate non-moving struts **12030**. In the exemplary embodiment of the stent assembly **12000** shown, there are nine pairs of jack struts **12013**, **12014** and nine non-moving struts **12030**. This number is merely exemplary and there can be, for example, only six of each or any other number desired. Connecting the pairs of jack struts **12013**, **12014** and the non-moving struts **12030** are laterally extending arms **12040**. As the stent assembly **12000** is either contracted or expanded, the arms **12040** each flex at their two endpoints, one at a respective non-moving strut **12030** and the other at a respective one of a pair of jack struts **12013**, **12014**. As can be seen from the configuration shown in FIG. **121**, when the stent assembly **12000** is contracted (e.g., for installation into the delivery sheath), the arms **12040** move towards a longitudinal orientation. Conversely, when the stent assembly **12000** is expanded (e.g., for implantation), the arms **12040** move towards a radial orientation.

[0392] FIG. **122** shows the lattice after being allowed to return to its native position, for example, at a deployment site. Each jack screw **12020** is in an engaged state for controlled further outward expansion of the lattice. As the lattice is sized for implantation, the delivery system forcibly

expands the lattice, as shown in the progression of FIGS. **123**, **124**, and **125**. In the view of FIG. **125**, the lattice is about to enter a maximum expansion state, which occurs when the proximal surface of the distal jack strut **12013** contacts the distal surface of the proximal jack strut **12014**. It is noted that this exemplary embodiment does not show features of a valve sub-assembly. Valve sub-assemblies, such as shown in FIGS. **135** to **136** are envisioned to be used with this stent assembly **12000** but are not shown for reasons of clarity.

[0393] This exemplary embodiment and other exemplary embodiments of the self-expanding, forcibly expanding stent lattices described herein circumscribe cells **12310** that are comprised of either the distal jack or proximal jack part, a portion of the non-moving strut, and two arms, which together define a parallelogram (one of which is outlined in FIG. **123** with dashed lines). This shape is beneficial because it keeps the moving struts (e.g., **12013**, **12014**) parallel as they expand and contract, thus keeping the distal and proximal jack parts aligned with the jack screw **12020** to insure stability of the lattice. More specifically, any pitch, roll, or yaw movements of either of the jack parts (e.g., **12013**, **12014**) or the non-moving struts (e.g., **12030**) is substantially prevented by this configuration. The configuration of adjacent cells provides significant benefits and differences from the above-mentioned scissor lattice or braided lattice structures, respectively, where mechanical pivots are present at crossings of two members or where two wires of a stent lattice or braid cross and change angles relative to one another to form a scissor when changing geometrically. In the embodiments of stent lattices described herein, no wires or members cross. Thus, without the presence of any scissoring members, the ability to sew to the stent lattice becomes possible. Further, the stent lattices are able to use all of the wall thickness for strength (there is no point where two thinner members are crossing one another). Additionally, the entire stent lattices have no points of instability and are more stable because the lattice is one continuous piece of material (with the exception of the configurations similar to FIG. **126**, where the continuity is formed by fixation, such as welding or soldering).

[0394] FIG. **126** is an alternative exemplary embodiment of a portion of a self-expanding/forcibly-expanding lattice of an implantable stent assembly **12600**. In the portion of the configuration shown, a separate jack screw assembly **12610** connects the two adjacent lattice segments (here the non-moving strut **12616** is shown in a vertical cross-section passing through the mid-line thereof). Separate jack tube halves **12612**, **12613** are connected respectively to upper and lower jack-contact struts **12614** of the two adjacent lattice segments. To fix these tubes to the nitinol lattice, the tubes can, for example, be made of Niobium. In the exemplary embodiment shown, the external threads of the jack screw **12620** are engaged with the interior threads of the distal jack tube half **12612**. A lattice-disconnect tube **12630** of the stent delivery system is engaged to cover a pair of drive screw coupler parts therein. FIG. **127** shows the lattice-disconnect tube **12630** disengaged from an exemplary embodiment of a pair of drive screw coupler parts **12752**, **12754**. This connected state of the pair of drive screw coupler parts **12752**, **12754** is idealized because, due to the natural lateral/radial forces existing in the disconnect joint, once the lattice-disconnect tube **12630** retracts proximally past the coupling of the drive screw coupler parts **12752**, **12754**, the two drive screw coupler parts **12752**, **12754** will naturally separate, as shown in the view of FIG. **128**. In the disconnected view of FIG. **128**, the proximal member of the pair of drive screw coupler parts **12752**, **12754**, which is part of the delivery system, is partially retracted into the central bore of the lattice-disconnect tube **12630**.

[0395] FIG. **129** illustrates another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly. This assembly also has nine separate lattice segments, but more or less in number is equally possible, for example, six segments. In this embodiment, a proximal disconnect block **12930** and disconnect subassemblies **12931**, **12932** of a stent delivery system is an alternative to the lattice-disconnect tubes **12630** of the embodiment of FIGS. **126** to **128**. Here, a proximal disconnect block **12930** is in an engaged state covering the pair of drive screw coupler parts **13052**, **13054** therein. After the disconnect block **12930** is retracted in a

proximal direction, all of the lattice-disconnect arms **12932** are removed from covering the pair of drive screw coupler parts **13052**, **13054**, thereby allowing disconnect of the lattice **12900** from the delivery system, as shown in FIG. **130**. The proximal disconnect block **12930** allows all of the pairs of drive screw coupler parts **13052**, **13054** to be coupled together for substantially simultaneous release.

[0396] FIGS. **131** and **132** show an alternative to the exemplary embodiment of the self-expanding/forcibly-expanding lattice of FIGS. **126** to **130**. Here, the intermediate jack tubes halves **13112**, **13113** for receiving one jack screw **13120** therein are connected to the adjacent lattice segments with the adjacent lattice segments **13114** not directly on opposing sides of the jack tubes **13112**, **13113**. The angle that the two adjacent lattice segments make is less than 180 degrees and greater than 90 degrees. In particular, the angle is between 130 degrees and 150 degrees and, more specifically, is about 140 degrees, as shown in FIG. **132**.

[0397] FIG. **133** is another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly **13300**. In this embodiment, there are nine lattice segments but more or less is equally possible, for example, six segments. Here, the distal and proximal jack struts **13313**, **13314** of the lattice are locally thicker to accommodate and connect to non-illustrated jack screw assemblies. One possible method of fabricating the stent lattice with locally thicker sections is to start with a tube of material that is at least as thick as the thickest region and to wire-EDM one of or both of the inside and outside surfaces to cut out the narrower sections and either form or leave the locally thicker sections. This applies to all of the exemplary embodiments of the stent lattice herein where locally thicker sections appear.

[0398] FIG. **134** is another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly **13400**. In this embodiment, there are nine lattice segments but more or less is equally possible, for example, six segments. Instead of having the non-illustrated jack screws pass entirely through the material of the lattice as shown in previous embodiments, here, the jack struts of the lattice are elongated and the elongated portions are bent-over to form tabs **13413**, **13414** for connecting to non-illustrated jack screw assemblies. The tabs **13413**, **13414** are shown here as bent inwards, but they can also be bent to face outwards. To operate the jacks, various ones of each of the set of longitudinal tabs are threaded or smooth.

[0399] FIGS. **135** to **137** show another exemplary embodiment of the self-expanding/forcibly-expanding lattice of an implantable valve assembly **13500**. The jack assemblies are similar to the embodiment of FIGS. **120** to **125**. Here, however, there are six lattice segments. The intermediate non-moving struts **13530** between the jacks **13520** form commissure connections and include through-bores **13532** for connecting the valve end points of the intermediate valve **13540** to the lattice. In this embodiment, the upper plane of the valve **13540** is in line with the upper end of the non-moving struts **13530**, which are at the same plane as the upper end of the jacks **13520**. The valve **13540** here is shown with three leaflets **13542** and, therefore, three commissure connections exist at three of the non-moving struts **13530**. The valve assembly is shown in FIGS. **135** and **136** in an expanded position that can be commensurate with an implantation position of the valve assembly. FIG. **137**, in comparison, shows the lattice of the valve assembly **13500** in a natural or pre-set, non-forcibly-expanded state. There exist some reasons for having the lattice of the valve assembly **13500** be set to a larger natural diameter than the compressed, pre-implantation size as shown in FIG. **121** or **139**. For example, in such a configuration, it is desirable to have the angle of the arms of the lattice, when starting to force drive, not be shallow because it is desired to have all of the force of the screw to drive the diameter change of the stent assembly. Also, when driving the stent assembly from a natural position to a final, forcibly open position, strain is induced in the material. It is understood that, the higher the induced strain, the greater the installed strain, increasing the possibility of failure. Therefore, the closer the final position is to the natural position, the lower is the installed strain. Further, the natural position cannot be made so large because there will be too much strain to collapse the lattice for delivery. In this regard, the heat-set diameter of

the stent lattices of the disclosed exemplary embodiments is optimized so that the strain imparted when moving the stent lattice from its heat-set diameter to the crimped diameter is maximized within the allowable super-elastic range of the material, therefore, minimizing the installed strain when the stent lattice is moved from its heat-set diameter to the forcibly expanded implantation diameter.

[0400] FIGS. **138** to **142** show another exemplary embodiment of the self-expanding/forcibly-expanding lattice of a stent assembly **13800**. As in the above embodiments, this exemplary embodiment does not show features of a valve sub-assembly for reasons of clarity even though valve sub-assemblies, such as shown in FIGS. **135** to **136**, are envisioned to be used with this stent assembly **13800**. Here, the lattice of the stent assembly **13800** has six lattice segments. Instead of having the jack screws contact longitudinal bores in the wall of the lattice, pairs of jack tubes **13812**, **13813** are connected (e.g., laser welded) to respective longitudinal pairs of jack connection struts **13822**, **13823**. This embodiment shows the jack tubes **13812**, **13813** connected on the interior of the lattice but they can also be connected on the exterior, or the pairs can even be staggered on the interior and exterior in any way and in any number. The jack tubes **13812**, **13813** are formed with interior threads or interior smooth bores.

[0401] After being forcibly contracted, the lattice of FIG. **138** can be further compressed within the delivery sheath of the delivery system, an orientation that is shown in FIG. **139**. After delivery to the implantation site, the lattice is expanded for implementation, first naturally and then forcibly. FIGS. **140** to **142** show various expansion stages of the lattice in various perspective views with FIG. **142** showing the lattice expanded near a maximum expansion extent.

[0402] An alternative to forming interior-threaded, longitudinal through-bores in the lattice is shown in the exemplary embodiment of FIGS. **143** to **154**. Here, the self-expanding/forcibly-expanding lattice of an implantable stent assembly **14300** has nine lattice segments. FIG. **143** shows the lattice in a native, self-expanded position. Both of the distal and proximal jack struts **14313**, **14314** in each of the nine segments have smooth bores. The distal jack strut **14313** in each lattice segment has a proximal end formed as a first connection part **14315** shaped to receive thereat a second connection part **14330**. It is this second connection part **14330** that contains the interior threads for threadingly mating with the exterior threads of the jack screw **14320**. In an exemplary embodiment, the second connection part described herein can be made curved from tube-stock to fit within wall of the stent assembly when collapsed. Additionally, the drive screws and/or the second connection part can be made of materials that do not have galvanic corrosion, such as Titanium. By locking the two connection parts **14315**, **14330** at least in the longitudinal direction of the lattice (for example by the opposing T-shaped tongue and grooves shown in FIG. **143**), when the jack screw **14320** (longitudinally secured at the proximal side of the proximal jack strut **14314** by the distal drive screw coupler part **12052**) is threaded into the second connection part **14330**, a two-sided connection is formed that allows each jack assembly to function by moving the respective pair of longitudinally aligned jack struts **14313**, **14314** towards or away from one another. In the exemplary embodiment, the first connection part **14315** has a T-shape and the second connection part **14330** is a nut having a bore with an interior shape corresponding to the exterior of the T-shape of the first connection part **14315**. This configuration, therefore, forms a form-locking connection. A form-locking or form-fitting connection is one that connects two elements together due to the shape of the elements themselves, as opposed to a force-locking connection, which locks the elements together by force external to the elements. By creating a threaded interior bore in the second connection part **14330** that is co-axial with the smooth bores of the distal and proximal jack struts **14313**, **14314**, once the jack screw **14320** is threaded completely through the second connection part **14330** and enters the bore of the distal jack strut **14313**, the jack screw **14320** prevents the second connection part **14330** from being removed. FIG. **153** illustrates the interior threads of the nut and shows how the jack screw **14320** prevents removal of the nut **14330** after the jack screw **14320** passes completely therethrough.

[0403] In these figures, an exemplary embodiment of commissure connector pads **14350** for each of three commissure points of the valve leaflets are provided. An exemplary form for the commissure connector pads **14350** is a rectangle with four through-bores for suturing the commissure point thereto. Outer lattice fixation paddles **14360** are provided at the ends of the non-moving struts **14316** to improve fixation of the stent assembly **14300** in the implantation site.

[0404] As can be seen by the progression from FIG. **145** to **146**, the lattice-disconnect tubes **14340** move proximally in order to disconnect the pair of drive screw coupler parts **14652**, **14654**. FIG. **146** shows the lattice of FIG. **143** with the connector control tubes **14340** of the delivery system in a non-engaged state after disconnection of the stent assembly **14300** by the delivery system has occurred. FIG. **149** shows the lattice expanded by the jack screw assemblies almost up to the fullest expanded extent possible, where the two jack struts **14313**, **14314** of the lattice almost touch. Various other views of the stent assembly **14300** are shown in FIG. **144** (top view), FIGS. **147**, **148** and **152** (enlarged views of the connection parts expanded and contracted), and FIGS. **150** and **151** (contracted stent assembly). FIG. **144** reveals that, in an exemplary embodiment of the jack struts **14652**, **14654**, the radial thickness **14410** is greater than a thickness **14420** of the remainder of the stent lattice, in this case, it is thicker from the interior of the lattice. If desired, it can be thicker from the exterior of the lattice.

[0405] FIG. **154** shows one exemplary embodiment of the lattice **14300'** of FIG. **143** in an intermediate manufacturing step before the cylindrical stent assembly **14300** is created. For example, the lattice of the stent assembly can be laser cut from a sheet of Nitinol, wrapped around a mandrel, and welded at the two ends to form the shape of the lattice **14300** shown in FIG. **143**.

[0406] FIGS. **155** to **166** illustrate another exemplary embodiment of a self-expanding/forcibly-expanding lattice of an implantable stent assembly **15500** having six lattice segments. As best seen in FIGS. **156** and **159**, the jack struts **15513**, **15514** have keyhole slots to accommodate the jack screws therein. This exemplary embodiment, therefore, can create the keyhole slots using a wire-EDM (electric discharge machining) process. This exemplary embodiment shows the keyhole slots open to the inside but, as described below, the keyhole slots can also be open to the outside.

[0407] The configuration of the stent assembly **15500** has similar features of the stent assembly **14300**, all of which are not repeated for the sake of brevity. One feature of the stent assembly, for example, similarly has commissure connector paddles **15550** for each of three commissure points of the valve leaflets. This exemplary form for the commissure connector paddles **15550** is a waffle pattern with five through-bores and dimpled sides for suturing the commissure point thereto. The stent assembly **15500** also has some differences from the stent assembly **14300**. A first difference is that outer lattice fixation pads **15560** on the non-moving struts **15516** are bent outwards to shape the lattice into a longitudinal hourglass. As particularly shown in the views of FIGS. **159** and **160**, the outer lattice fixation pads **15560** each provide a beneficial location at the a distal end thereof for placing a radiopaque marker **15562**.

[0408] FIG. **155** shows the stent assembly **15500** partially expanded state with each of the jack screws **15520** in a thread-engaged state for further outward expansion. As can be seen in FIG. **155**, turning the jack screws **15520** so that they enter the second connection part **15530** further pulls the distal jack strut **15513** towards the proximal jack strut **15514** (because the distal drive screw coupler part **14652** is prevented from further distal longitudinal movement after hitting the proximal side of the proximal jack strut **15514**). It is noted, however, that this pulling does not occur until the jack screws **15520** enter the distal jack strut **15513** to eliminate any slack that exists between the distal drive screw coupler part **14652** and the proximal most surface of the proximal jack strut **15514**. When the distal drive screw coupler part **14652** finally touches the proximal end of the proximal jack strut **15514**, further rotation of the jack screws **15520** cause the distal and proximal jack struts **15513**, **15514** to move towards one another because the threads of the jack screws **15520** are connected with the internal threads of the second connection part **11530**. As is apparent, because the stent assembly **15500** is forcibly expanded in this state, reversing the jack

screws **15520** allows the stent assembly **15500** to retract radially inwards towards its natural state, which is shown, for example, in FIG. **161**. FIGS. **157**, **158**, **160**, **165**, and **166** also show the stent assembly **15500** in various forcibly expanded configuration states.

[0409] However, when it is desired to forcibly contract the stent assembly **15500**, further reversal of the jack screws **15520** will merely turn the screws out of the respective second connection parts **11530**. To prevent this removal from occurring (because removal of the jack screws **15522** from the second connection part **15530** would allow the latter to fall off the stent lattice), each of the jack screws **15520** is provided with a back drive sleeve **15570** that is disposed fixedly on the outside of each of the jack screws **15520** at a location between the second connection part **15530** and the distal surface of the proximal jack strut **15514**. To fix the back drive sleeve **15570** in place, it can be, for example, machined directly with the screw or laser welded on as secondary process. Use of the back drive sleeve **15570** to cause forcible contraction of the stent assembly **15500** can be seen in the transition from FIG. **161** to FIG. **162** to FIG. **163**. In FIG. **161**, the jack screws **15520** are in a position where the lattice is in the natural, self-expanded state but in a position where the distal drive screw coupler part **14652** touches the proximal-most surface of the proximal jack strut **15514**. In this position an unscrewing movement of the jack screws **15520** causes no movement of the lattice until the back drive sleeve **15570** touches the distal-most surface of the proximal jack strut **15514**, which is the position shown in FIG. **162**. Any further reversal of the jack screws **15520** causes the distal portion of the jack screws **15520** to begin moving away from the distal jack strut **15513** but the back drive sleeve **15570** prevents any longitudinal movement of the jack screws **15520** with respect to the proximal jack strut **15514**. As a result, the distal and proximal jack struts **15513**, **15514** are forced apart to cause inward contraction of the stent assembly **15550**. Once the stent assembly **15550** is contracted sufficiently to be loaded into the delivery sheath (indicated diagrammatically in with dashed lines **16400** in FIG. **164**), any further inward contraction can be effected by forcibly loading the lattice into the delivery sheath of the delivery system for implantation. Loading the lattice into the delivery sheath can be seen in the progression from FIG. **217** to FIG. **212**.

[0410] FIGS. **167** and **168** show another exemplary embodiment of a self-expanding/forcibly-expanding lattice of the implantable stent assembly of FIGS. **155** to **166**. Here, the lattice also has six lattice segments but the longitudinal location of each intermediate jack screw nut to a respective one of the second connection parts **16730** is staggered longitudinally about the circumference of the lattice. This exemplary embodiment shows the staggering of the second connection parts **16730** in only two longitudinal positions (i.e., at two cross-sectional planes). However, three or more different longitudinal positions are also envisioned. FIG. **168** illustrates how the lattice is able to be collapsed even further than a configuration where all of the second connection parts **16730** are in the same radial plane (i.e., cross-sectional plane). As can be seen, the second connection parts **16730** do not hit one another during contraction. Not only does the staggered orientation reduce the impact of the second connection parts **16730** on the cross-section for available space, this configuration also reduces impact on the circumferential length where, with regard to FIG. **168**, it can be seen that the second connection parts **16730** touch metal in the lattice directly adjacent the second connection parts **16730**, e.g. at the lattice arms.

[0411] FIGS. **169** to **173** show a distal end of an exemplary embodiment of a delivery system containing the stent assembly **15500** of FIGS. **155** to **166**, which is only shown as a stent lattice. Of course any of the stent assemblies described herein can be substituted for this stent lattice, including valve assemblies. Use of the stent assembly **15500** in this exemplary embodiment is merely for illustration purposes. The view of FIG. **169** shows the state of the lattice after the delivery sheath **11040** of the delivery system has been withdrawn to an implantation position and after the lattice has been forcibly expanded, for example, to an implantation size of the lattice.

[0412] In addition to the stent assembly **15500**, distal portions of the stent delivery system are shown. First, lattice-disconnect tubes **16940** are connected to disconnect wires **770** which, in this

embodiment, take the form of hollow flexible coils. Accordingly, proximal movement of the disconnect coils **770** causes movement of the lattice-disconnect tubes **16940** as shown in the progression of FIGS. **170** to **171** to **172**. The disconnect coils **770** each have disposed therein a respective one of the drive wires **750** that cause rotation of each of the proximal drive screw coupler parts **14654** for expansion and contraction of the stent assembly **15500**. Also shown is a distal nose cone **16920** defining therein at least a longitudinal guidewire lumen (not illustrated). Connecting the nose cone **16920** to the remainder of the delivery system is a hollow guide wire tube **16922** having a guidewire lumen coaxial with the guidewire lumen of the nose cone **16920** for a guidewire that is used to guide the delivery system to the implantation site (see, e.g., FIGS. **66** to **69**). Containing the stent assembly **15500** during delivery is the delivery sheath **11040**, which is shown in the retracted state in FIG. **169** for implantation of the stent assembly **15500**.

[0413] FIG. **170** shows the interior of the delivery system of FIG. **169** proximal of the stent assembly to be implanted by the delivery system by the removal of the delivery sheath in this figure to a point proximal of the connector control sub-assembly **17000** of the delivery system. The connector control sub-assembly **17000** is shown fragmented in FIG. **170** for clarity. The progression of the connector control sub-assembly **17000** in FIGS. **170** to **171** to **172** illustrates how the stent assembly **15500** is actively disconnected (arrow A) from the delivery system.

[0414] In the connector control sub-assembly **17000**, each of the disconnect coils **770** is grounded to a disconnect puck **17020**. Accordingly, control of the stent disconnect by the delivery system is effected by retracting the disconnect puck **17020** a proximal distance sufficient to remove the lattice-disconnect tubes **16940** from covering the drive screw coupler parts **14652**, **14654**, which movement is illustrated in the progression of FIGS. **171** and **172**. In FIG. **170**, the connector control sub-assembly **17000** is in a lattice-connected state with the lattice-disconnect tubes **16940** over the drive screw coupler parts **14652**, **14654**. Proximal retraction of the disconnect coils **770** causes all of the lattice-disconnect tubes **16940** to move proximally into a lattice-disconnected state, shown in FIG. **171**. In this figure, each of the lattice-disconnect tubes **16940** is respectively retracted proximally from each of the drive screw coupler parts **14652**, **14654** an instant before all of the drive screw coupler parts **14652**, **14654** disconnect from one another. It is noted that the connection of the drive screw coupler parts **14652**, **14654** illustrated in FIG. **171** with the lattice-disconnect tubes **16940** disengaged from the pair of drive screw coupler parts **14652**, **14654** is idealized because, due to the natural lateral/radial forces existing in the disconnect joint, once a lattice-disconnect tube **16940** retracts proximally past the coupling of these drive screw coupler parts **14652**, **14654**, the two drive screw coupler parts **14652**, **14654** will naturally separate, as shown in the view of FIG. **172**. In actuality, manufacturing tolerances and variable resistance will have the jack-screw-connector pairs disconnect at different times, even if they are microsecond from one another. FIG. **172** shows the connector control sub-assembly **17000** in a lattice-disconnected state where each of the drive screw coupler parts **14652**, **14654** are disconnected from one another.

[0415] FIGS. **170** and **173** are enlarged views of various details illustrating how the disconnect coils **770** are retracted simultaneously for substantially simultaneous disconnection of the drive screw coupler parts **14652**, **14654**. More specifically, each of the lattice-disconnect tubes **16940** is longitudinally fixed to a respective disconnect coil **770** at the distal end of the disconnect coil **770**. Two sleeves **17022**, **17024** are fixed to the proximal end of each disconnect coil **770**. The disconnect puck **17020** has a number of passages **17021** equal to the number of disconnect coils **770** (which also is equal to the number of jack screw assemblies). As can be seen in FIGS. **170** and **173**, attachment of the proximal end of the disconnect coils **770** to the disconnect puck **17020** occurs by first placing the distal sleeve **17022** in a respective distal counterbore of one passage **17021** of the disconnect puck **17020**. The passages each comprise the distal counterbore, an intermediate groove, and a proximal counterbore. At this point, the proximal end of the disconnect coil **770** and the proximal sleeve **17024** stick out from the side of the disconnect puck **17020**. Then,

the coil **770** is slightly stretched so that the proximal sleeve **17024** moves over the proximal corner of the disconnect puck **17020** and is allowed to move down and rest inside the proximal counterbore, shown in FIGS. **171** to **173**. Even though longitudinally fixed after such a connection, the entire sub-assembly at the proximal end of the coils **770**, including the distal and proximal sleeves **17022**, **17024**, is freely rotatable within its respective passage **17021**. When torque is transmitted through the drive screw coupler parts **14652**, **14654**, a strong outward radial force separating the drive screw coupler parts **14652**, **14654** exists. This separating force is counteracted by the sleeve **16940**. To prevent drag upon the rotating drive mechanism that houses the drive screw coupler parts **14652**, **14654** therein, the sleeve **16940** is allowed to spin freely within the puck passages **17021**.

[0416] The disconnect puck **17020** is longitudinally slidable about the central hollow shaft of the guidewire tube **16922**. Proximal of the disconnect puck **17020** but longitudinally fixed to the central guidewire tube **16922** is a control spool **17030**. The control spool **17030** has puck control screws **17032** rotationally freely connected thereto but threaded in respective internally threaded bores of the disconnect puck **17020**. These puck control screws **17032** are connected proximally to the disconnect drive subsystem in the delivery system handle through the delivery sheath **11040**. In this way, rotation of the puck control screws **17032** allows distal and proximal movement of the disconnect puck **17020**, which corresponds to distal and proximal movement of the lattice-disconnect tubes **16940**. Not illustrated in FIGS. **170** to **173** but shown in FIG. **178** is an O-ring **17800** that is pierced by all of the wires passing through the control spool **17030** and is made of a polymer to provide a fluid-tight seal preventing flow of blood into the delivery sheath and/or delivery system handle.

[0417] It is noted that the position of the delivery sheath **11040** in the view of FIG. **169** covers the disconnect puck **17020** and prevents the disconnect coils **770** from coming out of the disconnect puck **17020**. Accordingly, in use, the delivery sheath **11040** is retracted proximally no further than is shown in FIG. **169**.

[0418] As in previous embodiments described herein, each jack assembly of the stent assembly utilizes one set of control wires, one driving wire **750** (here, rotational) and one disconnect wire **770** (here, longitudinally actuated). Also described in this exemplary embodiment are puck control screws **17032**. Each of the driving wires **750** and the puck control screws **17032** extends from the connector control sub-assembly **17000** all the way distal to the control handle of the delivery system. Because the delivery sheath **11040** is flexible and is intended to move through tortuous anatomy, it is understood that all of these wires/rods will experience longitudinal forces and will move longitudinally as the delivery sheath **11040** bends. As any longitudinal force exerting on these wires/rods is undesirable, especially with the disconnect wire **770**—which causes removal of the lattice-disconnect tubes **16940** and complete disconnection of the stent assembly when the lattice-disconnect tubes **16940** are moved proximally, it is important to minimize any affect that such forces would have on any of the wires/rods.

[0419] To eliminate action of such forces on the distal end of the device, the wires/rods are all grounded to the control spool **17030**. As shown in FIGS. **170** to **172**, and in particular FIG. **173**, each of the puck control screws **17032** have puck grounding cuffs **17033** on either side of the control spool **17030** that allow rotation movement but prevent longitudinal movement. Similarly, the drive wires **750** each have drive grounding cuffs **751** on either side of the control spool **17030** that allow rotation movement but prevent longitudinal movement.

[0420] All of these control wires/rods terminate at and are fixed longitudinally to a distal part **11512** of a respective telescoping wire control column **11510**. Each part of these telescoping wire control columns **11510**, **11512** are rigid so that rotation of the proximal part thereof **11510** causes a corresponding rotation of the distal part **11512** and, thereby, rotation of the corresponding control wire **750**, **17032**. The distal part **11512** is keyed to the wire control column **11510**, for example, by having an outer square rod shape slidably movable inside a corresponding interior square rod-

shaped lumen of the proximal part of the wire control column **11510**. In this configuration, therefore, any longitudinal force on any wire/rod will be taken up by the respective distal part **11512** moving longitudinally proximal or distal depending on the force being exerted on the respective wire/rod and virtually no longitudinal force will be imparted distal of the control spool **17030**.

[0421] FIGS. **174** to **177** are photographs of an exemplary embodiment of a delivery system and stent assembly lattice similar to the configuration shown in FIGS. **167** to **168**. These views show the lattice in various rotational views and in a forcibly expanded state with the back drive sleeve **15570** clearly appearing on the jack screw **15520**. FIGS. **178** to **180** show additional views of the connector control sub-assembly **17000** of the delivery system. In addition to the outer delivery sheath **11040** surrounding all of the control wires/rods, also provided is a flexible multi-lumen extrusion **17810**, shown in FIG. **178**, which provides a separate, independent lumen for each of the driving wires **750** and the puck control screws **17032**.

[0422] FIGS. **181** to **194** are photographs of various different exemplary embodiments of self-expanding/forcibly-expanding implantable heart valve assemblies. FIGS. **181** to **186** show a heart valve assembly having nine lattice segments in an expanded state and with valve leaflets in an open state. In this embodiment, the outer lattice fixation pads **15560** on the non-moving struts **15516** are bent outwards to shape the lattice into a longitudinal hourglass. The valve leaflets **18110** are connected by commissure plates **18120** to the non-moving struts **15516**. The proximal ends of the drive screw coupler parts **14652** are shown in FIG. **186**.

[0423] FIGS. **187** to **194** show a heart valve assembly having six lattice segments in an expanded state and with valve leaflets in an open state. The view of FIG. **188** shows only the valve leaflet sub-assembly **18800** removed from the lattice. Easily viewed in FIG. **188** is an exemplary embodiment of a commissure connector **18810** that is shown installed within the stent lattice of FIGS. **189**, **190**, and **208**. This commissure connector **18810** allows for easier connection of a single surface valve sub-assembly **18800**. When so used, the valve sub-assembly **18800** traverses approximately the dashed line shown in FIG. **208** as seen in FIGS. **189** and **190**. The commissures of the various embodiments described herein are attached to the non-moving or rigid portions of the lattice, for example, at the non-moving strut or at the proximal jack strut adjacent the downstream end of the valve sub-assembly.

[0424] In the exemplary embodiment of FIGS. **190** to **192**, the upper plane of the valve leaflet sub-assembly **19000** is in line with the upper end of the commissure connector paddles **15550** on the non-moving struts **15516**, which are significantly longer and are not at the same plane as the upper end of the proximal jack struts **15514**. Here, the plane of the upper end of the proximal jack struts **15514** is in line with the downstream end of the graft material **19010**.

[0425] In an alternative, non-illustrated configuration of the self-expanding and forcibly expanding stent lattice, the commissures are fixed to the non-moving strut **19016** at a point **19020** between the outer row of arms **19040** extending away from either side of the non-moving strut **19016** and the first row of arms **19042** closer to the center of the stent lattice. In such a configuration, the loads applied by the valve leaflet sub-assembly **19000** are spread to a greater number of support areas, thereby reducing the stress and strain upon the arms **19040**, **19042**. In particular, in such a configuration, the forces are spread to four arms **19040**, **19042**, whereas in previously described configurations, the forces are born principally by the two outer-most arms **19040**. One reason for this is because each arm has a mean installed strain that is generated by the forcibly expanding portion of the implantation. The same areas will experience additional strain associated with supporting the valve leaflets. In order for the stent lattice to survive long-term fatigue associated with high cycles of use, the alternating strain should be below a threshold and this process of spreading the forces to more arms allows the threshold to be maintained.

[0426] FIG. **191** shows an exemplary embodiment of a heart valve assembly connected to an exemplary embodiment of a delivery system and forcibly expanded. Here, the shape of the graft

material is shown as corresponding to the shape of the upstream arms as in a saw tooth pattern.

[0427] The views of FIGS. **192** to **194** show an exemplary embodiment of how the graft and how the leaflet sub-assembly are connected to the lattice. FIGS. **193** to **194** show how the braid angle of the braided graft material **19310** closely matches the angle of the arm portions of the self-expanding/forcibly-expanding implantable heart valve. In this way, the braid is able to expand longitudinally as the sections of the frame move away from each other during collapse.

Substantially simultaneously, the graft material and the stent are reducing in diameter and staying at a similar angle to reduce the stress of fixedly attaching the graft to the stent with a plurality of stitches. These figures also show one exemplary embodiment of how the graft material **19310** is stitched to the stent lattice.

[0428] One exemplary embodiment of the graft for the heart valve assemblies disclosed herein comprises nanofiber polyurethane spun into a braid-like form back and forth on a central mandrel. The view of FIG. **195** is a microscopic view of this graft as so fabricated when it is first laid down and when no stretch is being imparted; in other words, the graft is in its natural state. FIGS. **196** and **197** are close-up views that illustrate contact points between each nanofiber and show how the nanofibers adhere together. The braid angle of the graft is matched to the angle of the central arms of the stent lattice so that, as the lattice transforms, the braid goes through a matching transformation. This means that, when the graft is stretched longitudinally, it reduces in diameter, so it behaves like a braided structure. FIG. **198** shows the graft material when stretched in its length by 100%. FIG. **199** shows the graft material returning to its natural state after being stretched in FIG. **198**. The braid-like form is so tightly packed, fluid does not pass through, therefore, the graft material is fluid-tight for purposes of use in blood vessels. An amount of polyurethane is added to the braid-like form of this graft material. A minimal amount is added throughout the material but a heavier amount is used at the trimmed ends of the graft where the graft, when installed on the valve assembly, might be at risk of fraying.

[0429] It is noted that the various exemplary embodiments of the graft sub-assemblies described herein show the graft material on the inside of the lattice. Placing the graft material on the outside surface of the lattice is also envisioned. In such an exemplary embodiment, the exposed drive screws are now covered and protected on one side by the graft material and, if a valve assembly is present, on the other side by the valve leaflets. Also the key-holes forming the bores for the drive screws (see, e.g., FIGS. **156** to **160** and **204** to **209**) are protected by placing the graft material on the outside surfaces. An alternative to placing the graft material on the outside surfaces to protect the drive screws is a non-illustrated cover or sleeve that can be placed about the drive screws. Such a cover can be, for example, corrugated or bellows-like or smooth or any other variation.

[0430] As the lattice enlarges circumferentially, the valve leaflets, which are fixed in size, change the size of the overlap at the downstream ends of the leaflets. It may be desirable to adjust the size of this overlap. Furthermore, the leaflet length is also a factor with regard to longevity as undesirable wear can occur the more that the leaflet contacts any appreciably hard surface, including the stent lattice, the graft material, the sutures, etc. Therefore, adjustment of the leaflets to minimize or prevent such contact is desirable. With these embodiments, therefore, the leaflet size can be insured to maximize orifice area while assuring coaptation of the leaflet edges but also preventing undesirable wear.

[0431] Accordingly, FIG. **200** shows an exemplary embodiment of a device that adjusts the valve leaflet sub-assembly in the heart valve assembly. In this embodiment, the ends of the valve leaflets near the commissures are wound about a mandrel. If more overlap is desired, then the mandrel will be spun in one direction and, if less overlap is desired, the mandrel will be spun in the other direction.

[0432] FIG. **201** show another exemplary embodiment of an adjustable valve leaflet sub-assembly where each commissure has two mandrels for winding individual ends of each leaflet. In this embodiment, each mandrel of each pair of mandrels is shown as having to wind in opposing

directions in order to take in the leaflets or let out the leaflets.

[0433] The view of FIGS. **202** and **203** show another exemplary embodiment of an adjustable valve leaflet sub-assembly of a heart valve assembly where a longitudinally moving adjustment shim **20300**, when moved longitudinally (into or out from the view of FIG. **202**), takes up more or lets out more of the valve leaflet edge to shorten or lengthen the overlap portions of the valve leaflets. [0434] In use, the exemplary adjustable valves described herein are deployed with a minimal amount of released leaflet. This deployment configuration will likely cause some amount of central regurgitation, assuming that the valve is sized to an amount above a minimum deployed diameter. The leaflets are then released (played out) while monitoring with a transesophageal echocardiogram, for example. Once sufficient material is released to cause complete coaptation of the leaflets, the central regurgitation will cease, which can be easily confirmed with TEE Doppler evaluation.

[0435] FIG. **204** is another exemplary embodiment of a self-expanding/forcibly-expanding implantable stent assembly **20400**. In contrast to previously described lattices of the stent assemblies, the distal and proximal jack struts **20412**, **20414** of this embodiment have the wire-EDM jack screw bores machined from the outside surface of the stent assembly **20400**. As shown in FIG. **205**, the second connection part **14330** contains the interior threads for threadingly mating with the exterior threads of the jack screw **14320** and the second connection parts **14330** are staggered longitudinally about the circumference of the stent assembly **20400**. As in previous exemplary embodiments, the first connection part **20615** at the distal jack strut has a T-shape and the second connection part **14330** is a nut having a cutout with an interior shape corresponding to the exterior of the T-shape of the first connection part **20615**. The beneficial difference between the wire-EDM jack bores being machined from the outside surface of the stent assembly **20400** instead of the inside surface can be explained with regard to FIGS. **206**, **207A**, and **207B**. In particular, if the cross-section of the distal jack strut **20412** was square, then, when the T-shape is formed at the proximal end for receiving the second connector part **14330** (the jack nut), there would be three spans of material connecting the proximal end to the remainder of the distal jack strut to provide three columns of support therebetween, in particular, **20610**, **20612**, **20614**. However, the outer cross-section of the distal jack strut is not square and, instead, is trapezoidal in order to allow the circular stent assembly to be constricted to the smallest possible diameter as shown in FIG. **207**. Thus, when the bore of the distal jack strut **20412** is machined from the inside surface (as shown in FIGS. **156** to **160** and **207**) and the T-shape is formed at the proximal end **20615** for receiving the second connection part **14330**, the first and second spans **20710**, **20712** shrink to the point of disappearing, depending on the depth of the groove for receiving the tongues of the interior of the second connection part **14330**. In such a case, only the third span **20714** remains, which, in such a small manufactured part, allows for the possibility of deformation of the proximal end **20615** with respect to the remainder of the distal jack strut **20412**, which deformation is disadvantageous and could cause the jack screw(s) **14320** to malfunction or break. In comparison, when the jack bore is machined from the outside surface, as shown in FIG. **206**, the two outer spans **20610**, **20612** within the T-shape are large enough to remain because they are on the larger side of the trapezoid and, therefore, support the proximal end **20615** receiving the second connection part **14330**.

[0436] FIG. **209** illustrates an exemplary embodiment of a self-expanding/forcibly-expanding implantable valve assembly **20900** utilizing the lattice of the stent assembly of FIGS. **204** to **206**. Here, both the valve sub-assembly **20950** and the valve graft sub-assembly **20960** are connected using sutures **20970**. Here, the upstream side of the valve sub-assembly **20950** (left in FIG. **209**) is not connected at only its upstream-most end at the upstream circumference of lattice arms **20902**. In addition, a suture line **20972** is created that follows two separate arms **20904** of the lattice. With this suture line **20972**, pockets **20962** that might be created between the valve graft sub-assembly **20960** and the valve sub-assembly **20950** are minimized and closed off to diastole flow. This suture line **20972** can be seen in the interior of the valve sub-assembly **20950** in the views of FIGS. **210**

and **211**.

[0437] Also shown in FIG. **209** is a sawtooth proximal edge **20964** of the valve graft sub-assembly **20960** in addition to a sawtooth distal edge **20966**. By trimming the proximal edge **20964**, the possibility of obscuring either of the coronary arteries after implantation of the valve assembly **20900** is minimized. Also, having no graft material at the distal end decreases the overall amount of graft material, increasing the ability of the valve assembly to collapse and increasing the ease of recapture within the delivery sheath.

[0438] As set forth above, this application is a continuation-in-part of U.S. Pat. No. 8,252,036, U.S. patent application Ser. Nos. 12/822,291, 13/339,236, and 13/544,379, each of which have been incorporated herein. Described therein are various exemplary embodiments of aortic implants, including thoracic, abdominal, and valvular. Even though many of the exemplary embodiments of the stent lattices described herein are described as stents or valve replacements, they are equally applicable to stent grafts for treating the thoracic and abdominal aorta, including the treatment of thoracic and abdominal aortic aneurysms, whether for just the upstream end, the distal end, or both ends thereof. The specific incorporation of the exemplary stent lattices described herein into the thoracic and abdominal applications, therefore, is not repeated for the sake of brevity but is to be construed as applying to each of the embodiments described in all related and parent applications.

[0439] FIGS. **212** to **217** illustrate a process for either unsheathing and expanding a stent/valve assembly from the delivery sheath **10040**, **10720**, **11040** or reducing and re-sheathing the stent/valve assembly back into the delivery sheath **10040**, **10720**, **11040**. Re-sheathing the stent/valve assembly is shown by the transition through all of these figures starting from FIG. **212** and ending at FIG. **217**. Unsheathing the stent/valve assembly is shown by the transition through all of these figures in the reverse order starting from FIG. **217** and ending at FIG. **212**. In these figures, the nose cone and its catheter are not illustrated for clarity.

[0440] Beginning with a re-sheathing process, FIG. **212** shows an exemplary embodiment of a valve assembly similar to the valve assemblies of FIGS. **120**, **143**, **169**, and **191** connected to the distal end of the delivery system with the valve assembly expanded for delivering the valve assembly with the delivery system in an implantation state. As the delivery sheath **10040**, **10720**, **11040** is extended distally, the sheath entry device **21200** at the distal end of the delivery sheath **10040**, **10720**, **11040** slides upon and over the disconnect coils **770**. As the delivery sheath **10040**, **10720**, **11040** is extended further, FIG. **213** shows the sheath entry device **21200** partially re-sheathing the lattice-disconnector tubes **12630**, **14340**, **16940**. Continuing the re-sheathing process, FIG. **214** illustrates the valve assembly in an intermediate re-sheathing state where a proximal portion of the valve assembly, including the proximal jack struts **12012**, **13314**, **14314**, **15514** is re-sheathed. It is noted here that the outer radial corner of the proximal end of the proximal jack struts **12013**, **13314**, **14314**, **15514** can be chamfered in order to ease the entry of the proximal jack struts into the delivery sheath. If a wire-EDM process is used to create the keyhole bores for the drive screws as described herein, during the wire-EDM process of forming those key-holes, the outer radial corner of the proximal end of the proximal jack struts can be chamfered for the same reason. Further in the re-sheathing process, FIG. **215** shows the valve assembly re-sheathed half way across the exposed portions of the jack screws **12020**, **12620**, **14320**, **15520**. FIG. **216** shows the re-sheathing process almost complete where the distal jack struts **12013**, **13313**, **14313**, **15513** are partially re-sheathed in the sheath entry device **21200**. The re-sheathing process is complete in FIG. **217**, where the valve assembly is entirely contained in the delivery sheath **10040**, **10720**, **11040**. At this point, the entire system can be removed from the patient or repositioned and, then, unsheathing at an improved implantation site.

[0441] It is noted that the sheath entry device **21200**, as shown in FIG. **217**, is slightly conical with the distal end slightly larger in area than the outer circumference of the delivery sheath **10040**, **10720**, **11040**. It is noted that this shape would be disadvantageous if in this orientation when the system is being extended up to the delivery site as providing the delivery sheath with its smallest

outer circumference is most desirable. In order to minimize the outer diameter of the sheath entry device **21200**, the material of the sheath entry device **21200** is selected so that the sheath entry device **21200** can be heat-shrunk or otherwise collapsed. In one exemplary process for minimizing this outer circumferential diameter, the sheath entry device **21200** is shrunk about the disconnect coils **770** as shown in FIG. **218**. Thus, a maximum outer circumference of the sheath entry device **21200**, even though it may be greater than 18 French, for example, can easily fit within an 18 French orifice after be so processed. FIG. **219** shows the distal end of the sheath entry device **21200** after this process is performed but without all internal components of the delivery system and FIG. **220** shows the distal end of the sheath entry device **21200** fully expanded after retraction of the implant by the delivery system as explained above.

[0442] Unique aspects of the various embodiments of the self-expanding/forcibly-expanding implantable stent/valve assemblies of the present invention include an ability to better correspond with natural geometry that is very different from prior art devices, which only expand to an ideal circular shape of an inflating balloon. FIGS. **221** to **224** show the stent assembly of FIG. **191** expanding progressively inside an irregularly shaped mock-up implantation site that is hardened so that the stent assembly does not move the mock-up from its irregular shape. As can be seen as the stent assembly expands, the mock-up implantation site does not move and the stent assembly orients itself automatically to the particular irregular interior cross-sectional shape in which it is being implanted. As clearly shown in FIG. **224**, the stent assembly is implanted within the irregular-shaped implantation site with little or no room between the lattice and interior walls of the mock-up implantation site. Therefore, the invention can be used to conform to any shape, circumference, perimeter, diameter, cross-section, or other geometric configuration in two or three dimensions.

[0443] One exemplary embodiment of a process for implanting any of the stent/valve assemblies described herein is described with reference to the flow diagram FIG. **225**. A handle, such as the one illustrated in FIGS. **108** to **118** and **226** to **230** includes a display **10814**, **23010** and various user interface actuators **10816** and **23011** to **23017**, such as buttons. The following exemplary implantation process assumes that the user interface actuators are seven in number, including a solid orange “center” button **23011**, two retroflex buttons (flex **23012** and unflex **23013**), expand and contract buttons **23014**, **23015**, and extend and retract buttons **23016**, **23017**. Additionally, the process flow steps shown in FIG. **225** are exemplary display screens that occur at each stage of the implantation procedure after a few preliminary steps have occurred. First, the system is opened from its dry package. The system can be pre-loaded with a stent assembly (e.g., 23 mm) at its native size and includes a stent-loading funnel. Once the device is turned on, the first screen shows the status “Ready to Collapse” and gives the user instructions on how to collapse with “Hold center button to collapse.” When the button is held, the drive screws move to collapse the stent assembly and, while the collapsing occurs, the screen of the display **23010** indicates “release button to abort.” The display **23010** can show, simultaneously, a progress bar indicating the progress of stent assembly collapsing. When fully collapsed by the system, the display **23010** indicates that the user should now “Manually Collapse Stent” and indicates how this is accomplished by showing “Hold center to advance funnel/sheath.” Moving of the sheath starts slow then moves faster. As advancing of the sheath/funnel occurs, the display **23010** indicates “Release Button to Abort.” The display **23010** simultaneously shows a progress bar indicating how much sheathing is left. When sheathed, the display shows “Sheathing Complete Remove Funnel” and the user removes the sheathing funnel from the end of the delivery sheath. The display **23010** shows that the next step can occur after the user “Hold[s the] center button to continue” and confirms that the funnel was removed. The display **23010** now indicates that the user should “Flush Stent” and to confirm this flush by “Hold[ing the] center button to continue.” At this point, the “Device Ready for Patient” is displayed and the user is instructed to “Hold center button to continue.”

[0444] Now the distal end of the device can be guided to the implantation site. If retroflexing is

desired, then the display **23010** indicates that the Left/Right buttons show retroflexing along with a progress bar. The Up/Down buttons control sheath retraction. The display **23010** now indicates that the user should “Hold down button to commit device” and indicates that the user should “Press center button to return to previous screen” if not ready for implantation. The sheath is retracted slowly for the first half of the stent assembly with an option to reverse direction if desired. The display **23010** shows “Retracting Sheath” and gives the user the direction to “Release button to abort.” If desired, the button can be held for full sheath retraction along with a displayed countdown to full retraction. The device takes up play in the drive screws and presents the stent assembly in a pre-defined position and/or a native position. For example, the stent assembly can be expanded to 15 mm in diameter and progress bars can indicate status of stent assembly along with showing diameter with a circle that is the same diameter as the stent assembly. When done, the display indicates “Sheath Fully Retracted” and the user should “Hold center button to continue.” [0445] Now, the stent assembly is ready for implantation. Both radial force and the diameter of the stent assembly can be displayed along with progress bars if desired. The diameter can be indicated with a circle that is the same diameter of the stent assembly. Any of the buttons can be used to limit the maximum radial force that is desired by the user and the display **23010** can show, with a status bar, a radial force limit indicator. The software can account for the loads required to expand the stent lattice and the friction loads of the drive system so that an accurate representation of the force being imparted to the tissue can be communicated to the user, for example, in pounds/kg. Given the time-dependent nature of tissue's reaction to load, the software can continue to apply a known target force to the tissue for a period of time after initially meeting the load. As the tissue remodels, the stent lattice will continue to expand and continue to apply the target force up to the maximum limit of the lattice's circumference. The software can track the response of the tissue and, once the rate of change of expansion declines below a threshold, it can stop expansion of the stent lattice. In an alternative embodiment, all expansion of the implant can occur synchronized with the heart's contractions to coincide with any particular portion of the sinus rhythm. Diameter control buttons **23014**, **23015** are used to control the diameter, which, for this example, starts at 15 mm. Each press of a button can cause, for example, a 0.5 mm increment or decrement of the stent assembly along with showing the diameter in millimeters and/or with a circle of the same diameter. At a point where the nominal radial force detected from the implant has stabilized, the display **23010** can indicate to the user that a first angiogram or ultrasound (for example) should be carried out to see if any paravalvular leak exists. If a leak exists, the user can increase the target force level up to the maximum that the implant can take as designed and further or re-expand with the same rate of change detection repeated until paravalvular leaks are sealed or maximum expansion of the stent lattice results. If the latter occurs, the user will be notified if the radial force measured will be sufficient to prevent embolization of the implant if deployed as implanted. If not, the user will be advised or prevented from deploying the implant. When ready to implant, one of the buttons lights green (e.g., the center button **23011**) to indicate that pressing that button will disconnect the implant from the delivery system. To perform the disconnect, the user holds the button for a period of time (e.g., five seconds), during which time the button flashes and the handle produced an audio indicating the automated disconnect sequence is about to begin. When first pressed, the display **21010** shows “Disconnecting” along with a countdown number for seconds to disconnect. At the end of the countdown, the automated disconnect sequence starts. One step in this sequence is to remove any wind up that has been built up in the drive screw wires. This can be done in one of two ways. A first way to perform wind up release is to reverse the drive motor with a low level of torque sufficient to remove wind up but not sufficient to create additional reverse wind up or drive the screws in reverse. Alternatively, by having known the last input torque, a look-up table can be accessed that contains the fixed number of turns that are required to remove the wind up associated with that torque. At the end of the automated disconnect sequence, the disconnect wires **770** are withdrawn from the drive screw coupler parts **14652**, **14654** to complete disconnection of the

implant.

[0446] When disconnected, the display shows “Disconnected” and directs the user to “Hold center button to continue.” Any retroflex can be released and the user is shown that the device is “Ready to Re-Sheath” the distal parts of the delivery system into the delivery sheath and is shown that, to do so, the user must “Hold center button to re-sheath.” With that button press, the display **23010** shows that “Re-Sheath” is occurring and that the user can “Release button to abort” the re-sheathing of the delivery components. When the exposed parts of the delivery system are safely located in the delivery sheath, the display indicates “Device Terminated” and, therefore, the delivery sheath is “Ready to remove from patient.”

[0447] Another exemplary embodiment of a delivery system control handle is shown in FIGS. **226** to **230**. The housing of the delivery system control handle **22600** is, in this exemplary embodiment, a clam-shell having top and bottom halves. The top half of the housing contains most of the electronic elements, including the electronic control circuitry, the display, and user control buttons (see FIGS. **229** and **230**). The bottom half of the housing contains most of the mechanical elements. Regarding the latter, a grounding base **22610** is fixedly connected to the bottom half of the housing. This forms the ground for most of the mechanical components.

[0448] First, a retroflex support tube **22710** is secured at its proximal end to the grounding base **22610**. The retroflex support tube **22710** surrounds a multilumen shaft **22630** having lumens therethrough for all control and guide wires. In this exemplary embodiment, the number of lumens is nine in total. Not illustrated is a distal retroflex knee fixedly attached to the distal end of the retroflex support tube **22710** about the multilumen shaft **22630**. The knee is cylindrical in shape and has slits along one lateral side thereof. Also attached to the distal end of the knee is a distal end of a retroflex wire **22622**. The proximal end of the retroflex wire **22622** is attached translatably within a retroflex trolley **22624**. When the retroflex trolley **22624** translates proximally, retroflex of the distal end of the delivery system occurs. It is desirable to allow physical flexing or retroflexing of the distal end of the delivery system without use of the retroflex trolley **22624**. Accordingly, the retroflex wire **22622** has a non-illustrated collar connected within a slot of the retroflex trolley **22624** that is free to move proximally but constrained to move distally within the slot. Thus, if a bend in the delivery sheath causes tension in the retroflex wire **22622**, that wire is free to move proximally through the retroflex trolley **22624**. Movement of the retroflex trolley **22624** upon a retroflex guide pin **22625** is caused by interaction of the retroflex trolley **22624** with rotation of a retroflex shaft **22626**, for example, by a follower nut, a cam, or other similar connection that allows the retroflex trolley **22624** to translate longitudinally as the retroflex shaft **22626** rotates. Rotation of the retroflex shaft **22626** is controlled by a retroflex motor **22628**.

[0449] The delivery sheath **11040** is coaxial with and surrounds the retroflex support tube **22710**. The proximal end of the delivery sheath **11040** is fixed to a sheath trolley **22744**, which rides along a sheath guide pin **22746** based on movement caused by rotation of a sheath drive screw **22748**. Interaction of the sheath trolley **22744** occurs with rotation of a sheath drive screw **22748**, for example, by a follower nut, a cam, or other similar connection that allows the sheath trolley **22744** to translate longitudinally as the sheath drive screw **22748** rotates. Rotation of the sheath drive screw **22748** is controlled by a sheath motor **22749**. Because the sheath guide pin **22746** is fixed to the grounding base **22610**, translation of the sheath trolley **22744** causes distal or proximal translation of the delivery sheath **11040** with respect to the control handle **22600**.

[0450] It is desirable to have the entire distal delivery portion of the implant and the implant (as shown for example in FIGS. **105**, **107**, and **169** to **173**) translate longitudinally when at the implantation site. But, it is not simply desirable to have the user control that longitudinal placement. Once the implant is near the implantation site, it would be desirable to fix the control handle **22600** with respect to the patient and mechanically extend and/or retract the implant for best positioning. However, because the drive wires **750** for turning the jack screws and the disconnect drive wires connected to the puck control screws **17032** for translating the puck **17020** and thereby

the disconnect wires **770** must be rotated from within the handle, translation of those elements requires translation of at least the transmission imparting such rotation, if not also translation of the respective motors. The control handle **22600** provides such translation of both the motors and the respective transmissions. In particular, a translation ground **22800** is fixedly connected to the control handle **22600**. The translation ground **22800** has a threaded bore that guides a translation drive screw **22810** therein. The translation drive screw **22810** is fixedly connected at one end to a transmission of a non-illustrated translation motor (located under a self-contained motor/transmission sub-assembly **22820**). The translation motor rotates the translation drive screw **22810** when actuated and is also fixed to the control handle **22600**. The motor/transmission sub-assembly **22820** is translationally associated with the translation drive screw **22810** such that the entire motor/transmission sub-assembly **22820** translates longitudinally when the translation drive screw **22810** rotates. For example, under the motor/transmission sub-assembly **22820** is a bracket attached thereto and having a threaded bore receiving therein the translation drive screw **22810** such that, when the translation drive screw **22810** rotates, the bracket with the motor/transmission sub-assembly **22820** translates. Within the motor/transmission sub-assembly **22820** are all of the motors and/or transmissions for rotating each of the, for example, six drive wires **750** for the drive screws (whether in groups or individually) and for rotating each of the, for example, two disconnect drive wires **22840**, **22842**. At the other end of the translation drive screw **22810** is a multilumen ground **22830** fixed to the proximal end of the multilumen shaft **22630**. Like the motor/transmission sub-assembly **22820**, the multilumen ground **22830** has a threaded bore shaped to fit therewithin the translation drive screw **22810**. In such a configuration, any rotation of the translation drive screw **22810** simultaneously and synchronously moves the multilumen ground **22830** and the motor/transmission sub-assembly **22820** together, all the while keeping the two separated by a fixed distance. In this way, translation of the implant with respect to the control handle **22600** and the delivery sheath **11040** can be effected, for example, by the positioning buttons **23016**, **23017** located on the side of the control handle. Such positioning can be analog and smooth without any particular steps or it can be programmable to move a given fixed distance every time one of these buttons **23016**, **23017** is pressed.

[0451] The exemplary embodiments of the valve assemblies described herein seek to have a valve that is sized and formed for a minimum deployment diameter. This valve is secured inside the stent lattice/frame that is capable of expanding to a much larger final diameter than the internal valve. The commissures of the valve are secured to the frame with a mechanical linkage that allows the frame to expand and keep the valve at a proper size to minimize regurgitation. A lower skirt of the valve is attached to the stent through a loose connection of the variable diameter braided graft or a similar device. This configuration allows the stent frame to continue to grow and fit into a variety of native annuli that are larger than the valve carried within the device.

[0452] Even though the exemplary embodiments shown above relate primarily towards an aortic valve, these embodiments are not limited thereto. As stated before, the invention is equally applicable to pulmonary, mitral and tricuspid valves. Additionally, the invention is equally usable in any tubular anatomic structure, some embodiments of which will be disclosed herein.

[0453] Known to surgeons, physicians, and biomedical engineering and medical personnel is that, with regard to devices to be implanted in a tubular or hollow structure, whether for expanding (e.g., angioplasty), for occluding, for wall patency (e.g., stents, stent grafts), or for replacing a physical structure (e.g., replacement valve), size mismatches lead to problems. Simply put, sizing symmetrical devices to match non-ideal anatomy is an issue in medicine. Where the surgeon has no way to control the prior art devices, implantation flaws and a certain amount of para-device leaking just must be accepted. Real-time feedback (the elements of which are integrated to the platform, the delivery system, the device itself, or a combination thereof) of these flaws, imperfections, and or leaks would be beneficial for current procedures but are simply not available for, for example, balloon-installed or self-expanding devices. Significant to the embodiments disclosed and equally

applicable to other tubular structures or orifices, the self-expanding and forcibly-expanding exemplary lattices described herein in contrast to the prior art has absolute controllability, allowing it to become a platform for many procedures, operations, and/or anatomies. As described herein, the various devices have actuatable geometry on all or specific locations of the device in which the self-expanding and forcibly-expanding exemplary lattice is used.

[0454] With regard to other structures of the heart, for example, the described embodiments are applicable to mitral valve replacement or repair. The mitral valve annulus is more pliable than the aortic valve annulus. It is also very close to the atrioventricular (AV) groove and the fibrous trigone (the thickened area of tissue between the aortic ring and the atrioventricular ring), which can be damaged when dealing with the mitral valve. The mitral valve also has significant sensitivity to both sizing and radial force. Undersizing of an implant causes leaks and embolization. Oversizing of the implant causes damage to the heart, not just by tearing the valve seat but also by changing the overall geometry of the left ventricle, rendering the heart less efficient for cardiac cycling. Accordingly, precise attachment and fine adjustment to the anatomy is needed, which is easily accomplished by the embodiments disclosed. Adjustability of the self-expanding and forcibly-expanding lattices of the described embodiments allows for precise and concomitant sizing and sealing, the significant issues associated with mitral valve replacement or repair. Additionally, the self-expanding and forcibly-expanding lattices described herein can be modified for mitral valve replacement to be D-shaped as an alternative to the circular shapes described herein such that a portion of the circumference is expandable (and contractable) and another portion remains constant (i.e., the flat of the D-shape). The self-expanding and forcibly-expanding lattice gives precise sizing and imparts a precise and controllable force to reduce damage and create a near-perfect seal. In mitral valve repair, modulation of the mitral valve annulus is the cornerstone of annuloplasty. In this exemplary embodiment, the device can be fixed to the mitral valve annulus directly, percutaneously, or minimal invasively, the implanted device being actuatable to change all or portions of the shape, diameter, perimeter, or overall configuration or a combination thereof to achieve proper coaptation of the native mitral valve leaflets. Other embodiments allow concomitant or independent actuation of the mitral sub-valvular apparatus. In yet other embodiments, this device can effect mitral valve repair by placement within the coronary sinus or epicardially in plane with the mitral annulus allowing for similar actuation of the native mitral annulus. Adjustability of the self-expanding and forcibly-expanding lattice, therefore, resolves both the issues of sizing and radial force. Another notable item is that the mitral valve has an axis that is very offset from the entrance vector during operation. Therefore, the device must be very steerable. The swashplate embodiments described herein, for example, aid in replacement of the mitral valve even where placement tolerance is very narrow.

[0455] Likewise, the described embodiments are applicable to tricuspid valve replacement or repair. All of the features described above for the mitral valve are equally applicable for the tricuspid valve replacement or repair and, therefore, they are not repeated for reasons of brevity. In recent times, there has been an increase in tricuspid valve disease. Significant to tricuspid valve disease is that the patients are “high-risk”; they are very sick when the disease has progressed enough for the patient to show symptoms. With regard to the tricuspid valve, it is close to the conduction region of the heart. Currently, tricuspid valve disease is repaired with a split ring so that the conduction region is not injured in any way. In comparison to the mitral valve, the axis of the tricuspid valve is ninety (90) degrees offset from the vena cava. Therefore, significant steerability is needed. The radial force that can be imparted is also limited in tricuspid valve replacement or repair. In particular, the maximum radial force that can be imparted is limited due to the proximity of the heart's conduction system. Accordingly, precise attachment and fine adjustment to the anatomy is needed, which is easily accomplished by the embodiments disclosed. The adjustability of the self-expanding and forcibly-expanding lattice of the embodiments allows for precise and concomitant sizing and sealing, the significant issues associated with tricuspid valve replacement

or repair. The self-expanding and forcibly-expanding lattice gives precise sizing and imparts a precise and controllable force to reduce damage and create a near-perfect seal. Adjustability of the self-expanding and forcibly-expanding lattice, therefore, resolves both the issues of sizing and radial force. Like the mitral valve repair or replacement the swashplate embodiments, for example, can be used to aid in replacement of the tricuspid valve even where placement tolerance is very narrow.

[0456] The described embodiments are applicable also to pulmonic valve replacement or repair. The adjustability of the self-expanding and forcibly-expanding lattice of the embodiments allows is an important advantage with regard to pulmonic valves. This is because disease of the pulmonic valve is often congenital and, therefore, commonly presents in childhood. For Pulmonic Atresia, for example, the pulmonic valve becomes narrow and muscular. Therefore, not only must the surgeon replace the valve, but the surgeon must also replace some or all of tract for the pulmonic valve. The most significant problem with pulmonic valve disease is that the typical patient keeps growing after the first operation. Typically, each patient has four to six (4-6) operations during his or her life. This is because, as the patient grows, a larger and larger valve needs to be provided. Each of these surgeries has its own associated risks, but multiple surgeries just compound those risks.

[0457] In contrast to the prior art, the embodiments of the self-expanding and forcibly-expanding lattice described entirely eliminate surgeries after the first valve implantation. In particular, instead of performing open surgery on the patient for the second and subsequent surgeries, the adjustable self-expanding and forcibly-expanding lattices described herein are able to be enlarged percutaneously by simply re-docking an adjustment device to a portion of the implant's structural platform (for example, one or more jack screws) and further expanding the lattice to accommodate the growth of the patient. Not only does this save significant costs by virtually eliminating the most costly parts of the subsequent surgeries, the non-invasive expansion greatly decreases the probability of injury caused by such open surgeries.

[0458] The self-expanding and forcibly-expanding lattice embodiments described herein have been mostly cylindrical for purposes of clarity and brevity. They are not limited to this configuration, however. As shown, for example, in FIG. 155, variations to the outer shape of the self-expanding and forcibly-expanding lattices, here the lattice being hourglass-shaped, cause different and procedure-dependent improvements in the efficacy of the implant. Other shapes having geometries depending upon the implant location, including half of an hourglass, tapering, curving, and other geometries, are equally suitable for creating a seal and retention in other anatomical location. In another exemplary embodiment, an implant **23100** has a self-expanding and forcibly-expanding central lattice **23102**. One or both of the open ends of the lattice **23102** (here, both ends) have a self-expanding, conical stent structure **23104**, **23106** outwardly expanding away from the central lattice **23102**. In the exemplary embodiment shown, all ends, including the non-moving strut **23116** and the two parts of the moving strut (the distal jack strut **23113** and the proximal jack strut **23114**), extend longitudinally with respect to the implant **23100**. These extended ends are curved outward to form the hourglass shape. Depending on the desired configuration of the distal and proximal ends of the implant **23100**, any combination of extensions is possible. Some exemplary configurations include: both ends of only the non-moving strut **23116** extending longitudinally; only one end of the non-moving strut **23116** extending longitudinally; only the distal jack strut **23113** extending longitudinally; only the proximal jack strut **23114** extending longitudinally; and both the distal jack strut **23113** and the jack strut **23114** extending longitudinally. The illustrated exemplary embodiment also shows extensions **23123**, **23124** of the distal and proximal jack struts **23113**, **23114** shorter than the extension **23126** of the non-moving strut **23116**. The lengths of the extensions **23123**, **23124**, **23126** can be the same or reversed to the configuration shown. For example, because the central lattice **23102** will shorten longitudinally when expanded, the length of the extensions **23123**, **23124** of the distal and proximal jack struts **23113**, **23114** can be longer than the extension **23126** of the non-moving strut **23116** so that, when circumferentially expanded to the

desired size for implantation, all of the ends of the extensions **23123**, **23124**, **23126** are substantially aligned along a single circumferential plane perpendicular to the longitudinal axis of the implant **23100**. While the surgeon may not be able to know the exact implantation diameter/perimeter/shape that would correspond to a particular geometry of the patient's implantation site, pre-surgery measurement (e.g., by trans-esophageal echocardiogram, CT scan, MRI, intra-cardiac echocardiogram, nuclear scanning, or fluoroscopic visualization) could provide the surgeon with enough information to select an implant **23100** having extensions **23123**, **23124**, **23126** that will be substantially aligned about a given circumference when expanded to the pre-measured geometry in the patient. Thus, before surgery, the surgeon measures the diameter of the implant desired and selects one implant **23100** from a set of differently sized implants **23100**. Slightly larger or smaller expansion of the central lattice **23102** with respect to the measured diameter, therefore, only means that the ends of the extensions **23123**, **23124**, **23126** do not reside in the same circumferential plane. As geometry of an implantation site is irregular, it is envisioned that such an ideal configuration of ends in a single plane will be difficult to achieve even with a perfectly symmetrical device.

[0459] Between the extensions **23123**, **23124**, **23126** of the distal and proximal ends **23104**, **23106** are intermediate arms or webs **23128** that connect adjacent pairs of extensions **23123-23126**, **23124-23126** in any way desired. Here, the intermediate webs **23128** circumferentially connect adjacent pairs of extensions **23123-23126**, **23124-23126** approximately at the midpoint of the extensions **23123**, **23124**, **23126** and at approximately the endpoints of the extensions **23123**, **23124**, **23126**. This, however, is merely one exemplary embodiment and more or less webs **23128** can be used with any geometry, with any angle, and with any length. For example, the webs **23128** can follow the angles of the arms of the stent lattice **23102**.

[0460] FIGS. **232** and **233** show, respectively, the implant **23100** before it is completely implanted in a heart valve **23201** and in a vessel **23301**. In FIG. **232**, the central lattice **23102** is partially expanded to start collapsing the diseased leaflets of the patient's valve **23201**, but the distal and proximal ends **23104**, **23106** have not yet touched the walls on either side of the valve **23201**. In contrast, in FIG. **233**, the central lattice **23102** is partially expanded but still not touching the walls of the diseased vessel **23301**. The distal and proximal ends **23104**, **23106** of the implant in FIG. **233**, however, have already touched the vessel walls. In this configuration, connection of the ends to the wall of the vessel **23301** before connection of the central lattice **23102** occurs prevents shuttling of the implant **23100** while the central lattice **23102** continues to expand to the desired implantation circumference, which can be approximately equal to, smaller than, or greater than the diameter **23302** of the vessel **23301**.

[0461] The ends **23104**, **23106** of the central lattice **23102** or attached to the central lattice **23102** can be made of a shape-memory material such as Nitinol that is integral or connected to (e.g., fused) the central lattice **23102**. These ends **23104**, **23106** can also be configured in various shapes. In FIG. **231**, each end **23104**, **23106** is shown as being an exponentially increasing expanding cone. Alternative shapes, some described below, can include the shape of a barbell, such as the distal end **23402** of the implant **23400**, or the shape of a bulb, such as the distal end **23504** of the implant **23500**. The two ends of a self-expanding and forcibly-expanding implant can be identical or different, depending on the requirements of the particular implant. In each case, these extensions have a given final memory shape to suit the particular surgical procedure being conducted. When captured in the delivery sheath, as in other self-expanding memory-shape devices, the ends compress radially inward to permit loading into a delivery catheter. When allowed to release therefrom, the ends expand to the pre-defined memory shape (which can be constrained partially if the self-expanding and forcibly-expanding portion of the implant is not yet forcibly expanded to allow the self-expanding end(s) to completely self-expand.

[0462] The hourglass shape is particularly suited for various surgical applications where acute retention to prevent shuttling in the anatomy is desired. Not only is this shape beneficial for

replacement of all valves of the heart, it is also advantageous in procedures treating, for example, atrial septal defects (ASD), ventricular septal defects (VSD), patent ductus arteriosus (PDA), ventricular aneurysms, patent foramen ovale (PFO), arteriovenous fistulae, paravalvular leaks, and left atrial appendage (LAA) ligation, and in performing embolization and angioplasty of vessels. Each of these procedures and conditions benefit from having self-expanding flares that establish the geometry of the anatomy and the self-expanding and forcibly-expanding central platform that precisely adjusts the waist between the flared ends. Even though an hourglass shape as described herein might be used for these procedures for various reasons, they are not described herein as being limited to and any of the implant shapes are equally applicable to any of the exemplary surgical procedures described herein.

[0463] Depending on the procedure being performed, the implant can be covered by different polymers or by a matrix or mesh of material. The covering can be semi-porous for sealing over time with cellular in-growth and/or it can have portions that are non-porous to seal immediately upon implantation or even just before implantation. A non-porous covering over the entirety is also contemplated. The covering can be external or internal or both, and can be located anywhere on the implant including the distal side, the central lattice, the proximal side, and even within the central orifice of the implant at any longitudinal position within the lumen to occlude the lumen and prevent flow through the lumen. For example, an occlusion curtain can be disposed within the cross-section of the central orifice, in particular, within the waist, dependent on the effect that is desired. It can be beneficial if the material used is distensible so that it does not corrugate or pleat but, in particular circumstances, it can be non-distensible.

[0464] An atrial septal defect (ASD) is a form of congenital heart defect that enables blood flow between two compartments of the heart—the left and right atria. Normally, the right and left atria are separated by the interatrial septum **23610**. If this septum is defective or absent, then oxygen-rich blood can flow directly from the left side of the heart to mix with the oxygen-poor blood in the right side of the heart, or vice versa. Another potentially fatal consequence of an ASD is that blood clots are able to pass through the ASD and, instead of going to the lungs, where a clot might be harmless and dissolve over time, such clots travel to the brain, which can cause stroke and, in some instances, death.

[0465] During development of the fetus, the interatrial septum develops to separate the left and right atria. However, during early fetal development, a hole in the septum called the foramen ovale, naturally allows blood from the right atrium to enter the left atrium. This opening allows blood to bypass the nonfunctional fetal lungs while the fetus obtains its oxygen from the placenta. A layer of tissue called the septum primum acts as a valve over the foramen ovale during fetal development. After birth, the pressure in the right side of the heart drops as the lungs open and begin working, causing the foramen ovale to close entirely. In approximately twenty five percent (25%) of adults, the foramen ovale does not entirely seal. In these cases, any elevation of the pressure in the pulmonary circulatory system (due to pulmonary hypertension, temporarily while coughing, etc.) can cause the foramen ovale to remain open. This is known as a patent foramen ovale (PFO), a type of ASD.

[0466] One device for treating ASD currently is H-shaped. When inserted into the ASD orifice, the device expands and blocks both sides of the septum with two opposing plates, one on each side of the defect. These plates, however, are much larger than the defect. This is disadvantageous because the device places a large mass inside the atrium, decreasing atrial volume. Further, the cylindrical central connection of the two plates is not sized to fill the defect, thus, unless seal is complete across the ASD, blood can still traverse the ASD. Also disadvantageous is the fact that these plates require significant overlap to insure a seal of the defect; in other words, they are larger than the diameter of the ASD orifice.

[0467] The hourglass-shaped, self-expanding and forcibly-expanding implant of the embodiments described herein provides significant advantages over prior art devices and treatments. First, the

precisely actuatable waist is able to be expanded to match the size of the defect without imposing excessive and/or uncontrollable outward force on the wall edges of the defect. Second, the implant includes an internal, central, solid mass, curtain, or plate to prevent blood cross-over, this curtain is only as thin as it needs to be to last and adds no volume to either atria. The material of the curtain can be human tissue or it can be other mammalian tissue or a natural or synthetic fabric. The curtain can even be semiporous to create a natural, endothelialized wall after the implant has been in the defect for a while. The implant is flared into the hourglass shape on both sides to prevent migration. FIG. 236 illustrates a heart with an ASD and FIG. 237 illustrates this defect repaired with such a self-expanding and forcibly-expanding implant 23700. In some circumstances, the seal may breakdown and later re-adjustment and/or repositioning may be required.

[0468] Because implantation of the self-expanding and forcibly-expanding device is less invasive, it can be used more often to treat ASDs because accessing the superior vena cava is performed with an endovascular procedure and not open surgery. For example, it is believed that ASDs are one cause of migraines, due to the ASD allowing flow from the right atrium to be directed into the brain. The self-expanding and forcibly-expanding implant can treat such conditions. While VSDs occur less frequently than ASDs, VSDs can be treated in the same way as ASDs as described above.

[0469] The ductus arteriosus is a normal fetal blood vessel that closes soon after birth. Patent ductus arteriosus (PDA) is a congenital disorder in the heart wherein a neonate's ductus arteriosus fails to close after birth. In a PDA, the failure of the vessel to close results in an irregular transmission of blood between two of the most important arteries close to the heart, the aorta and the pulmonary artery. Specifically, a PDA allows a portion of the oxygenated blood from the left heart to flow back to the lungs by permitting flow from the aorta (which has higher pressure) into the pulmonary artery. The hourglass-shaped, self-expanding and forcibly-expanding device can be implanted inside the ductus arteriosus to close off the connection using a curtain within the lumen of the implant. One exemplary configuration of an internal curtain can be explained with regard to FIG. 232, 233, or 237. The inside lumen of the central lattice 23102 on one side of the jack screws 23113 (e.g., the distal side) is covered with a material that can expand along with the lattice 23102. The single sheet of material is connected to most or all of the distal struts 23113 and the non-moving struts 23116. If the material around and inside the implant is semiporous, natural ingrowth about the implant occurs, causing the implant to be completely covered and forming part of the atrial wall.

[0470] An arteriovenous fistula is an abnormal connection or passageway between an artery and a vein. It may be congenital, surgically created for hemodialysis treatments, or acquired due to pathologic process, such as trauma or erosion of an arterial aneurysm. These passageways, which may occur anywhere in the body including the brain or spinal cord, act like a short circuit diverting blood from fully circulating and delivering oxygen where it is needed. The hourglass-shaped, self-expanding and forcibly-expanding device can be implanted inside the arteriovenous fistula to close off the connection using a curtain within the lumen of the implant. The configuration illustrated in FIG. 232, 233, or 237 can form an example of a self-expanding and forcibly-expanding device that is covered about the exterior or interior and has an internal curtain within the lumen and is implanted in an arteriovenous fistula between the two vessels. If the material around and inside the implant is semiporous, natural ingrowth about the implant occurs, causing the implant to be completely covered and forming a new intermediate vessel wall.

[0471] While the cylindrical self-expanding and forcibly-expanding device can be used in vessels and other similarly shaped anatomy, in some circumstances, it may be desirable to use, instead, the hourglass-shaped, self-expanding and forcibly-expanding device. One such area includes embolization and angioplasty of vessels. Embolization is a minimally invasive treatment that occludes, or blocks, one or more blood vessels or vascular channels of malformations (abnormalities). In a catheter embolization procedure, medications or synthetic materials are placed

through a catheter into a blood vessel to prevent blood flow to the area. Catheter embolization is performed, for example, to control or prevent abnormal bleeding. This includes bleeding that results from an injury, tumor, or gastrointestinal tract lesions such as ulcer or diverticular disease. Embolization is commonly the first line of treatment in gastrointestinal bleeding of any cause. Controlling bleeding into the abdomen or pelvis from injuries caused in a motor vehicle crash is especially suitable for this treatment. Catheter embolization is also performed to occlude or close off the vessels that are supplying blood to a tumor, especially when the tumor is difficult or impossible to remove, such as a brain tumor. After embolization, a tumor may shrink or it may continue to grow but more slowly, making chemotherapy or surgery a more effective option. There exist many procedures in which a surgeon intervenes at a particular anatomy to use a catheter to access a vein/artery and strategically close it. For example, in a trauma situation such as a stab of a spleen or a liver. With regard to the latter, the right lobe of the liver can survive when flow to the left lobe of the liver is occluded. Where a patient is coughing blood, embolization can be used in a pulmonary artery. In each of these situations, the hourglass-shaped, self-expanding and forcibly-expanding device can be placed as shown, for example, in FIG. 233. With an exterior or interior covering along the peripheral cylinder and an appropriately placed internal curtain closing off the central lumen of the device, when implanted, the anatomy can be securely and removably occluded.

[0472] The inventive designs can also be used in other areas, such as the left atrial appendage. In current procedures for ligating the LAA, a device referred to as the WATCHMAN® is used. There exist significant disadvantages with such a device. Prior to implantation, the surgeon needs to measure the internal volume of the LAA, which is very hard to do because the LAA is a very compliant and soft structure. Thus, presizing is difficult. The LAA is typically not under pressure during sizing. Accordingly, problems occur with finding an accurate terminal diameter of the implant when the LAA is under pressure later caused by the device. Current treatment technology comes in certain sizes, one of which needing to be selected for the surgery. The current devices are hard to install due to the volumetric and shape mismatch. If a device is too small, then a space around the device occurs after implantation, making the space(s) ripe for undesirable clotting. With insufficient volume mismatch, existing clots in the LAA can be dislodged, which is dangerous as those clots will immediately enter the aorta. On the other hand, if the sized device is too large, implantation can cause acute traumatic disruption, requiring emergency surgery or even causing death. Additionally, even if the prior art device is successfully implanted, which is shown in FIGS. 238 and 239, if that device places too much force within the LAA, it will erode the wall, ultimately causing a rupture, with the resulting negative consequences. It is known that the LAA varies substantially in thickness along the appendage and some areas are so thin that a surgeon can see right through the wall. Accordingly, ruptures are likely when implants connect with that part of the wall.

[0473] Once a surgeon implants a device to treat the LAA, it is desirable to fill the space as much as possible but to not leave a volume that allows blood to enter, to exit, and/or to clot. If one is able to close the space within the LAA and seal the orifice as well, then this could eliminate any possibility of clot dislodgement. It would be, therefore, advantageous to be able to expand a device intermittently using visulation (e.g., by continuous fluoroscopy or echocardiogram) until complete occlusion/isolation of LAA from general circulation occurs, at which time, the surgeon would stop expanding the device. However, current devices are substantially circular and only fill LAA volumetrically; they are not intended to seal the atrial-appendage junction. Thus, current devices cannot both take up the volume and seal the orifice. Additionally, current devices cannot be expanded intermittently because they are all self-expanding—they either remain contracted before removal from the delivery catheter or expand completely all at once when the delivery catheter is removed.

[0474] All of these disadvantages of the prior art are resolved by the self-expanding and forcibly-

expanding devices described herein because these devices can both fill the volume of the LAA and, at the same time, create a seal at the atrial-appendage junction. FIG. 240 illustrates one exemplary embodiment of a device 24000 for treating the LAA. As before, the self-expanding and forcibly-expanding lattice 24002 forms the central portion of the device 24000. Extending distally from the distal end of the device 24000 is a bulb-shaped self-expanding extension 24004. Opposite the bulb extension 24004 is a self-expanding barbell extension 24006. Together, these two extensions 24004, 24006 work to fill the volume when the device 24000 is implanted. Sealing of the LAA occurs by placement of one or more curtains within the lumen of the device 24000. For example, if an occluding curtain is disposed within the lattice 24002 at the dashed line 24040, then, when the lattice 24002 is expanded within the atrial-appendage junction 24022 and the barbell extension is allowed to self expand at the LAA entrance, the interior of the LAA is shut off from the left atrium. Another possibility includes the curtain extending proximally into the barbell extension along the wall of the barbell extension and to the outer extremity thereof. Likewise, the curtain can extend distally into the extension 24004. If the interior catheter needs to pass through the curtain, then a valve is integral to the curtain which seals after withdrawal of the catheter or the material is self-healing to close after implantation. The curtain can also be a separately implanted device that expands into the lumen of the device to occlude the lumen after implantation of the device.

[0475] An exemplary procedure for ligating the LAA with the device 24000 starts with placing a guidewire into the left atrium 24010. The guidewire is, then, threaded into the LAA. The device 24000 is driven over the guidewire until the distal end enters the LAA cavity 24020. In its natural state, the LAA is floppy and bends from approximately ninety to approximately one-hundred and eighty degrees. The device 24000 is inserted sufficiently into the LAA cavity 24020 to place the distal bulb extension 24004 within the cavity 24020, the central lattice 24002 within the LAA orifice at the atrial-appendage junction 24022, and the barbell extension 24006 just proximal of the atrial-appendage junction 24022. In such a configuration, as shown in FIG. 240, when expanded, the device 24000 can both fill the LAA volumetrically and create a seal at the atrial-appendage junction 24022. The delivery catheter 24030 is retracted from the device 24000 to allow self-expansion of the device 24000 to a pre-set state, which, for example, can be the state shown in FIG. 240. The central lattice 24002 is set with a self-expansion diameter smaller than the diameter of the orifice at the atrial-appendage junction 24022. The distal bulb extension 24004 is partially self-expanded at this point because the lack of complete expansion of the central lattice 24002 prevents the bulb extension 24004 from self-expanding entirely. Alternatively, the bulb extension 24004 can be pre-set to be at full self-expansion when the central lattice 24002 is at its self-expanded and not-yet-forcibly-expanded state. Similarly, the proximal barbell extension 24006 is allowed to partially self-expand by being held slightly smaller due to the configuration of the central lattice 24002.

Likewise, an alternative of the barbell extension 24006 can allow full self-expansion when the central lattice 24002 is at its self-expanded diameter. The surgeon, then, positions the bulb extension 24004 under visualization further into the LAA cavity 24020 until the bulb extension 24004 seats in the LAA cavity 24020. The surgeon also moves the device 24000 under visualization in a yaw direction (in the plane of FIG. 240) to have the radial plane of the barbell extension 24004 approximate and align with the plane of the atrial-appendage junction 24022. At this point, the device 24000 is stabilized within the LAA cavity 24020. Expansion of the central lattice 24002 occurs when alignment is confirmed. It is noted that the expansion of the bulb extension 24004 indexes the relationship of the adjustable central lattice 24002 to the orifice at the atrial-appendage junction 24022. As the central lattice 24002 expands, the bulb extension 24004 will expand into and fill the adjacent wall of the LAA cavity 24020 to seal off the distal end of the LAA cavity 24020 from the proximal end. The central lattice 24002, again under visualization, is expanded until the outer surface thereof contacts the orifice at the atrial-appendage junction 24022, at which time, the barbell extension 24006 has expanded outward to contact as much of the atrial-appendage junction 24022 as possible, thereby sealing the LAA cavity 24020 off from the left

atrium **24010**. As the barbell extension **24006** is shaped to be larger in diameter than the largest expansion diameter of the central lattice **24002**, the barbell extension **24006** will contact the wall at the atrial-appendage junction **24022** to form a tight seal. Release of the device **24000** from the delivery catheter **24030** completes the operation. With such an implanted configuration, having circumferential contact at or inside the atrial-appendage junction **24022** allows endothelialization of the device at the LAA orifice **24022** to occur in a number of days.

[0476] Even though the device **24000** is shown without a covering in FIG. **240**, some or all of the external and/or internal surfaces are covered to create an entirely captured volume. The device **24000** can be covered with a semiporous material to encourage ingrowth and endothelialization that entirely fills up the LAA. Alternatively, the device **24000** can be entirely covered with non-porous material or some sections can be semiporous and some can be non-porous. By combining the barbell-shaped end shown in FIG. **234** and the bulb-shaped end shown in FIG. **235**, the self-expanding and forcibly-expanding device **24000** shown in FIG. **240** becomes ideal for ligating the LAA.

[0477] As mentioned above, the LAA naturally forms a curve that folds upon itself. If the LAA was pressed against the adjacent outer wall of the atrium **24012** and held there for a sufficient amount of time to seal the LAA cavity **24020** off from the left atrium **24010**, the LAA would naturally close. However, the time for securing this result is measured in days. The self-expanding and forcibly-expanding device described herein can be used to carry out such a procedure. In particular, if the outside surface of the LAA and the outside surface **24012** of the atrium are accessed by surgery, the self-expanding and forcibly-expanding device can be inflated against the side **24024** of the LAA opposite the atrium, thereby pressing the LAA towards the outside surface **24012** in the direction indicated by the arrow in FIG. **240**. As such, the self-expanding and forcibly-expanding device acts as a pillow that gently and slowly presses the folded appendage closed and remains there while the LAA seals off by endothelialization. Due to the floppiness of the LAA and because of the angle that the LAA naturally forms, the device easily closes up the LAA orifice **24022** in this way.

[0478] Repair of a ventricular aneurysm can also be effected using the device **24000**. Ventricular aneurysms are one of the many complications that may occur after a heart attack. They usually arise from a patch of weakened tissue in a ventricular wall, which swells into a bubble filled with blood. Such an aneurysm **24100** is shown, for example, in FIG. **241**. This, in turn, may block the passageways leading out of the heart, leading to severely constricted blood flow to the body. Ventricular aneurysms can be fatal. Treating such aneurysms can be done by either blocking the artery supplying the aneurysm or by closing the aneurysmal sac itself as an alternative to surgery. The device **24000** can be inserted into the aneurysmal sac of the aneurysm **24100** and can be expanded within the aneurysm **24100** to fill and close off the aneurysm **24100**.

[0479] Another vascular procedure that can be benefited by the repositionable and controlled expansion of the stent lattices of the exemplary embodiments described herein is the area of femoral bypass surgery. When a portion of the femoral artery becomes occluded, either partially or completely, one way to provide arterial blood to the portions of the leg receiving reduced blood flow is to create a shunt around the occlusion. If performed, such a shunt would require suturing of one end to a location in the femoral artery upstream of the occlusion and suturing the other end of the shunt to a location in the femoral artery downstream of the occlusion. A stent graft having one of the stent lattices described herein at each of the ends of the graft material can be utilized to create such a shunt without the need of suturing the stent graft to the femoral artery and with other benefits as described below. With regard to FIG. **242** (depicting the arterial and venous circulation of the legs), a stent graft is prepared with the graft material having a length longer than the occlusion in the artery and sufficient to traverse from the upstream landing point to the downstream landing point in the femoral artery. A first entry point is made into the popliteal artery and the entry catheter is led up into the femoral artery just below the occlusion. The entry catheter is, then, routed

out of the femoral artery either into the saphenous vein or into the subcutaneous fat of the thigh. In the former instance, the entry catheter is directed up the saphenous vein and then out thereof and into the femoral artery at a point upstream of the occlusion near the groin. In the latter instance, the entry catheter is directed through the subcutaneous fat of the thigh along the femoral artery (without accessing other vessels) and then back into the femoral artery at a point upstream of the occlusion near the groin. With a guidewire so placed, the delivery system of the various exemplary embodiments herein can be guided therealong until the distal implant is located (e.g., via fluoroscopy) at the upstream implantation site. There, the stent lattice, which is surrounded by graft material (either in the lumen or outside), is expanded to the necessary diameter (approximately between 8 and 10 mm) and is checked for perivascular leak. When seated without excessive outward pressure (which can be measured as described herein) and without leak, the upstream stent lattice is disconnected from the delivery system and upstream implantation is complete. At any time during the upstream implantation, the stent lattice can be expanded, contracted, and re-expanded and re-positioned as desired. The delivery sheath is further retracted over the graft material and the graft lumen fills with arterial blood as this retraction occurs. Again under visualization, the stent lattice at the downstream end of stent graft is positioned within the femoral artery or the upper popliteal artery just downstream of the occlusion. At any time during the downstream implantation, the stent lattice can be expanded, contracted, and re-expanded and re-positioned as desired until a beneficial orientation occurs. The downstream stent lattice is expanded to the necessary diameter and is checked for nominal pressure and no perivascular leak. When no leak is confirmed, the downstream stent lattice is disconnected from the delivery system, completing the shunt and allowing arterial flow from above the occlusion to below the occlusion. [0480] In this exemplary embodiment, the delivery system is longer than described above at least by the distance between the two stent lattices. Further, one of the stent lattices is loaded into the delivery sheath at an intermediate point and the delivery sheath is extended over the graft material substantially without corrugation or crumpling thereof until it reaches the upstream stent lattice. Loading of the upstream stent lattice occurs similarly as described herein and the nose cone is docked at the distal end adjacent the upstream stent lattice. As the stent lattices (and graft) for this procedure are much smaller in diameter than the stent lattices described for the aorta or heart, each stent lattice does not need as many expansion devices (e.g., the jack assemblies). In particular, 1, 2, or 3 of the jack assemblies are only needed for stent lattices of 8 to 10 mm in installed diameter. Therefore, the actual number of lumens and drive wires needed for such a small stent graft may be less than the ones illustrated herein, allowing for the multilumen and delivery sheath to be made much smaller in diameter for this procedure. Accordingly, all of the control features will be similar as described herein but reduced in number with half of the control wires extending to an intermediate point within the loaded delivery sheath to the downstream stent lattice and the other half of the control wires extending to the distal end of the delivery sheath to the upstream stent lattice in a similar manner as illustrated in FIG. 169.

[0481] This exemplary embodiment not only provides controllable expansion into the fragile artery, but also eliminates the need for any suturing as the sealed upstream and downstream lattices provide the assured seal of the artery. This procedure has the added benefit of entirely eliminating any need for use of a balloon to inflate a stent portion, which inflation commonly causes injury to such arterial locations.

[0482] Many of the embodiments described above for treating one or the other of the heart valves utilize femoral or lower aorta access to implant the replacement stent valve. These procedures require a significantly long delivery system.

[0483] There are a number of other surgical procedures that access valves of the heart with entry ports that are relatively much closer to the surgical site. These procedures include, for example, direct open surgery (such as sternotomy) or through port access (indirect), such as mini-sternotomy, mini-thoracotomy, thoracoscopy, or mini-thoracoscopy. In open surgery, access to the valves is very

direct, as the surgeon actually touches the repair site. When sewing in a replacement valve, such as the aortic valve, fifteen or more sutures are needed, which requires significant prowess by the surgeon to sew in such a small area. When operating with instruments in indirect procedures, significant skill is required as well.

[0484] The inventors recognize that it is equally possible to access the valves to be treated in these procedures with the systems, devices, and methods described herein. One advantage for doing so is because the distance to the implantation site is so much shorter. With a short distance, control of the handle and placement of the implant by the surgeon increases. As such, with this shorter device, the surgeon can deliver the prosthesis and implant it with one hand. Another significant benefit is that controlled expansion of the stent lattice described entirely eliminates all of the suturing, making the surgery easier and faster and negating the requirement of all surgeons to have significant technical surgical prowess. Further, as the delivery systems described herein are very small in diameter, they lend themselves to use with port access surgeries and their concomitant benefits and faster recovery times. With any of these shorter procedures, access from the incision site to the valve to be replaced also is not as curved as access through the aortic arch.

[0485] Understanding the above, a variation of the systems, devices, and methods described herein can be applied with reference to an alternative embodiment of a delivery system described with regard to FIGS. **243** to **255**. FIG. **243** illustrates this alternative embodiment of a hand-held delivery system **24300** from the side of the user, where a display surface **24310** contains a display screen **24312** having all of the functions and capabilities as described herein. Also provided are control buttons **24314**, **24316**, and **24318** for regulating movement and deployment as set forth herein. FIGS. **244** to **250** show various views of the handle.

[0486] Internal components of the handle **24300** are visible in FIGS. **251** to **255**. First, with the upper clamshell half removed, the motor and transmission assembly **25100** are visible. This assembly **25100** contains a drive screw motor **25102**, a puck control motor **25104**, and proximal and distal transmission covers **25106**, **25108**. As compared to the configuration shown in FIGS. **108** to **118**, many of the components are different and many are eliminated. This is because the features used to compensate for significant curvature are no longer needed with this short embodiment. As before six throughbores **25110** are present in the distal transmission cover **25108** for receiving the control columns **25410** for the six drive wires **750** (not illustrated) and three throughbores **25112** are present for receiving the puck control columns **25420** for the three puck control screws **17032**.

[0487] FIG. **252** illustrates the exterior gears **25210** of the transmission of the puck control motor **25104** proximal of the proximal transmission cover **25106**. With the distal transmission cover **25108** removed in FIG. **253**, the drive pinion **25310** and drive transmission gears **25312** for the six drive wires **750** are apparent. Also shown are the puck control gear **25320** and one puck transmission gear **25322** of the puck control screws **17032**.

[0488] With the proximal transmission cover **25106** removed in FIG. **254**, the drive control columns **25410** and the puck control columns **25420** are apparent. Also shown here are the internal transmission gears **25430** for the puck control columns **25420**.

[0489] Finally, FIG. **255** shows the configuration of the puck control gear **25320** driving the two exterior puck control columns **25420**.

[0490] Also, as the surgeon will be physically positioning the distal end of the delivery system (not illustrated) at the implantation site and can manually retract the sheath containing the stent lattice, most of the assembly for unsheathing the stent lattice is unnecessary. Present, instead, are catheter guides **25120** that retain the catheter in place. As compared to the previous embodiments, the delivery catheter **25130** (diagrammatically shown with a dashed line) does not need to be as flexible and, therefore, is retained by the catheter guides **25120**. An exemplary embodiment of the control circuitry **25140** is shown behind the display **24312**.

[0491] Based upon the above, many of the distal prosthesis delivery components described in the

other exemplary embodiments are reduced in length and some become entirely unnecessary.

[0492] An exemplary process for accessing the aortic valve is described using the handle **24300**. It is known that access to the aortic valve can be done in one of two ways. First, the aorta can be cross-clamped, the heart stopped, and the patient placed on bypass. The aorta is dissected and the aortic valve is visualized directly or with a scope. Alternatively, the aorta can be accessed directly (in a preferred port) or through one of the other arteries (e.g., the sub-clavians, the innominate, the brachial, or the axillary). The latter, being only a port access, does not require the patient to be on bypass. The latter is less invasive to the patient.

[0493] Because the delivery systems of the invention are sufficiently small to fit within a port access to the aorta, the inventive system can be used without bypass. Specifically, a sheath dilator accesses one of the arteries or the aorta and provides access for a guidewire. This guidewire is very short and very directly reaches the aortic valve. The system with the implant is run over the guidewire and visualized within the aortic valve. Implantation is visualized as described herein and the device is withdrawn by the surgeon.

[0494] When a replacement valve is to be implanted, the surgeon does not know with any assurance or precision the exact size of the native annulus. Presently, surgeons can determine an approximate size of the native annulus with fluoroscopy or with a CT scan, but both of these provide inexact results. As such, the surgeon can only guess what is the correct size of the implant to be used and trust that the self-expanding prior art devices will seal properly and, importantly, will not embolize.

[0495] With the systems, devices, and methods of the present invention described herein, however, exact determination of the native annulus size becomes possible. More particularly, as indicated herein, the inventive stent lattice implant is inserted into the annulus in which it is to be implanted. This stent lattice does not merely expand into the annulus as prior art self-expanding devices. Instead, as described herein, the system forcibly expands the stent lattice (e.g., the replacement valve) into the annulus.

[0496] Each of the mechanical devices used to expand the stent lattice has known properties and the system, overall, can be investigated for those characteristics. Some of the properties include current supplied to the motor and the torque of the motor. Because of the efficiencies of the screws used and due to the mechanical geometry of the stent lattice, the torque applied can be related to an outward radial force imposed by the expanding stent lattice. Each of these characteristics can be measured and/or calculated. Even further, the amount of torque required by the motors to open the stent lattice from at least the self-expanded state to the fully expanded state can be measured for one stent lattice as a baseline or it can be measured for every different stent lattice to be used.

[0497] It is known that, when unloaded and in a test-bench mode, the stent lattice to be implanted will require a certain amount of torque to expand over the entire lattice expansion range and the motors will draw a certain amount of current over that range. The required torque over the range can be measured because, with DC motors, torque relates proportionally to the amount of current needed by the motor to cause stent lattice expansion. The current, torque, or radial force curves can be recorded as a function of the stent lattice size and stored in a memory to define a characteristic or reference curve for that particular stent lattice. Then, during an actual (or simulated) implant, the current/torque can be compared to the characteristic curve and the deviation will indicate that there is force being applied to the tissue, which indicates first contact. From this, a process for remotely determining the exact size of a native annulus becomes possible. More particularly, a stent lattice is installed on the delivery system and is allowed to self-expand and then is caused to expand forcibly with the motors. The characteristics of this stent lattice are recorded and stored in the memory of the delivery system as a characteristic curve for that stent lattice. This curve is used during actual implantation (or during implantation of a simulated annulus).

[0498] Once the characteristic curve is stored for that stent lattice, the process for determining the native annulus can occur. The stent lattice is guided to the implantation annulus and is allowed to

self-expand. Then, the system forcibly expands the stent lattice and compares the current drawn to the characteristic curve. The system knows exactly the diameter of the stent lattice at each moment (based upon turns of the stent expansion control rods) and, therefore, the diameter and current can both be stored as a function of time. During this time, the force of the expanding stent lattice can be calculated and stored as a function of time as well.

[0499] In addition to torque, current, and diameter as variables, utilizing data processing routines and assuming a constant current, the velocity of expansion can be calculated as well and processing of these signals can be used to detect the first substantial contact by the stent lattice within the native annulus; this is because the expansion velocity will decrease in a substantial way after first contact is made, at which time the annulus begins restricting the outward expansion of the stent lattice. Depending upon the time constant for detecting, measuring, and calculating the native annulus value based upon this first contact, the detection could lag behind the time that the stent lattice actually reaches the native annulus diameter. This is not disadvantageous because the stent lattice needs to be implanted in the annulus, not just reach the native diameter. Accordingly, the system can indicate (for example, on the display) the native annulus size and that additional expansion is needed in order to complete implantation. The maximum expansion of the stent lattice along with the maximum amount of force imparted by the stent lattice upon the native annulus can be defined and the system can be set to prevent the user from exceeding these levels during implantation. For example, if the native annulus is detected as being 20 mm in diameter, then 24 mm can be set as the maximum size for expansion.

[0500] The system also provides implantation checks to ensure implantation force within a predefined minimum. For example, if a minimum threshold of implantation force is not met and the maximum diameter of the stent lattice has not been reached, then the system will not allow disconnection to occur because there could exist a risk of lattice embolization. Conversely, if a maximum implantation force is reached but expansion of the lattice is not within an operating range for the valve leaflets, then the system will not permit disconnection. This is because even with an excellent implantation, if the leaflets will not function appropriately, implantation should not occur.

[0501] FIGS. **261** to **263** are depictions of the simultaneous display of: [0502] an expansion velocity v. time curve (red); [0503] a force v. time curve (white); and [0504] a stent lattice diameter v. time curve (green)

for a simulated native annulus of 19 mm, 20 mm, and 19.5 mm, respectively.

[0505] In FIG. **261**, expansion of the stent lattice occurs at an 18 mm diameter at about time 21:06:14. Expansion occurs at an almost constant velocity and then starts to decrease because the model begins to resist the expansion force of the stent lattice. Then, at about 21:06:58 an implantation routine of the inventive system begins and is explained with the understanding that implantation of the stent lattice needs to create a seal at the native annulus but should not cause tissue damage. (The implantation routines for FIGS. **262** and **263** occur at about 02:34:02 and 01:16:18, respectively.) The needed input variable for preventing damage is the knowledge of the native annular diameter, which only the systems and methods described herein can provide. With this number, the surgeon can limit the expansion of the stent lattice and insure that it does not go beyond a force or size greater than pre-determined for a particular native annulus. In particular, a method for implanting a precision-actuated frame-based (PAF) stent (e.g., a replacement heart valve for any of the four valves) is now described.

[0506] The stent lattice is controllably expanded at a given velocity until the system detects contact with and application of pressure to the native annulus. At this point, the system is allowed to execute an inventive sealing routine, in which force is incrementally applied to expand and release the stent lattice, increasing the lattice diameter just a little bit each time. Therefore, as the tissue moves, pressure is applied but, then, the lattice is reduced somewhat for a short amount of time allowing the tissue to relax and to remove some pressure against the tissue. Application of this

expanding force to the stent lattice is repeated to open the lattice even more but then to release it again at a larger diameter/circumference. By measuring the incremental change in diameter between each cycle, the system can determine/plot an asymptote that will be able to detect where further force is unnecessary and implantation is complete. Simply put, if a force is applied too rapidly, it can cause tissue to tear. By expanding the stent lattice in stages over time, a lower installation force can achieve an implant that comparable to a higher instant force but without damaging the tissue of the annulus. This is referred to herein as “Tissue Remodeling” and originates from the characteristic that short-time pressure on tissue allows it to spring back quickly with no deformation but even a small amount of pressure on tissue, if lasting a long time (e.g., a rubber band on a wrist), will leave a mark (e.g., an indentation) that will last a while. From this it is understood that there is a time constant relating to the remodeling of the tissue because fast impartation does not leave a mark (high force/short time) but slow impartation (lower force/longer time) leaves a greater tissue remodeling “mark” or result. The time constant relates to intercellular desiccation.

[0507] After implantation, feedback parameters of the final diameter and the native annulus are used to determine absolute and percentage changes in area and diameter to give the surgeon information on satisfaction of implantation success. FIG. 264 shows what occurs on the display of FIGS. 261 to 263 when detection of the native annulus is being carried out. In the example of FIG. 264, the native annulus is 21 mm. From the start of the simulated implantation process, the stent lattice is reduced in size from just over 24 mm. Throughout the process, the green curve in the middle window is displayed. The green curve is the characteristic curve of the particular stent lattice being implanted, which was defined off-line before this implantation example. The characteristic curve is stored in memory of the system and is plotted as a function of the stent lattice diameter.

[0508] As can be seen in the four stages of reduction (starting at about 15:14:10), the velocity of reduction (shown in the bottom window in red) is substantially constant after reduction start-up. The diameter curve (yellow) shows the stent lattice being contracted down to about 18.5 mm. The stent lattice is advanced and placed in the annulus for implantation while in this contracted state. Expansion starts at approximately 15:14:45. It is noted that, as the stent lattice is contracting and expanding, the actual force (blue) applied to the stent lattice (which is related to torque/current) is plotted against the characteristic curve, which correlation can be seen from about 15:14:10 to about 15:14:43. When implantation is occurring, the force curve (blue) tracks the characteristic curve (green) until about 15:14:50, at which time, the force curve deviates from the characteristic curve. It is at this point where the native annulus of 21 mm is reached and is detected by that deviation.

[0509] While utilization of this deviation to determine native annulus size is the first time such a measurement could be taken with reliability and accuracy and without injury to the anatomy, it is known that the structure of the implantation site is pliable and, therefore, the native annulus determined in this manner (without more) could be slightly larger than the actual annulus size. Further, the time delay between expansion and measurement, while small, does impart a lag in the annulus detection conclusion. To improve upon this measurement, various signal processing routines can be implemented using the parameters being measured by the system. For example, improvement of the measurement can occur in the following manner. First, the stent lattice can be expanded throughout its range prior to use to define a force v. diameter characteristic curve for that lattice. Then, the force v. diameter curve can be measured dynamically during implantation in the native annulus. While this is occurring, the two curves can be compared. When the native annulus is detected, that time point on the force v. diameter curve can be stored and a tangent line can be determined. This tangent line will project backward in time and will intersect the stored force v. diameter characteristic curve at a point smaller in diameter than the detected native annulus. It is this smaller diameter point where the tangent line intersects that can be considered to be a “true” native annulus (before any force is imparted on the native annulus to give a larger than “true”

determination). As indicated, many signal processing routines can be used now that measurements from the device both on a test-bed and dynamically during an implantation can be obtained with the systems and methods described.

[0510] It is useful to have a single characteristic curve for every stent lattice. However, every stent lattice and delivery system is different on a micro-scale. Nonetheless, the systems and methods described can be used to determine a “single” characteristic curve of similar stent lattices based upon averages of measured curves, for example. But, if extreme accuracy is desired, every time a stent lattice is mated to a delivery device, the particular characteristic curve can be created on the fly for that particular implant and can be stored in the memory of that delivery system for use during implantation of that implant.

[0511] The systems shown in FIGS. **169** to **180** and **191** illustrate exemplary embodiments of connectivity between the drive-screw proximal ends of the stent lattice and the control subassemblies including the drive screws and the disconnect wires. If the distance between these two points is too small, then, even though the connections are coils and flexible wires, the wires will not bend easily as the system is guided around curved vessels. In contrast, if the distance is very long, then the flexible coils and wires will be very flexible and might not have enough columnar strength to push through curved vessels. Therefore, this distance is set (along with selection of the material properties of the disconnect coils and drive wires) to allow the stent to be extended out and retracted back within a curved anatomy while still having enough columnar strength to not buckle when pushed distally.

[0512] Even with these properties balanced and with excellent tracking through curved vessels, the length of each of the coil/wire subassemblies remains the same. This means that, when traversing through a curve, some of the lengths will be allowed to remain taut and some will shorten (i.e., the inside wires will be in compression and the outside wires will be in tension). Thus, the flexible coil/wire subassemblies will oppose any curving that axially aligns the cylinder of the expanded stent lattice with the cylinder of the curved anatomy in which the stent lattice is currently placed. The result is that the stent lattice is forced into an angled position in the native annulus that is not ideal.

[0513] One way to correct for this angled position is through active swashplating, for example, utilizing the exemplary systems illustrated in FIGS. **89** to **103**. As can be seen in the progression of FIGS. **265** to **268**, active swashplating of the subassemblies allows the partially expanded stent lattice in FIG. **265** to be rotated counter-clockwise with regard to the view of the figure (FIG. **266**), to be rotated clockwise (FIG. **267**), and to be implanted (FIG. **268**).

[0514] Without active swashplating, for implantation of a replacement aortic valve, there will be a difference in the distance from the distal ground on the delivery sheath to each of the connection points (e.g., six) on the replacement valve's stent lattice as the delivery device is flexed around the aortic arch. This difference in path length is a function of the difference in diameter of the arch that each of the connection points pass through. The path is shorter for the inside coils/wires and longer for those on the outside of the arch. Where all of the control coils/wires are the same length, the similar lengths will cause an angled orientation.

[0515] To overcome this difference in alignment, an exemplary angular correction device **26810** is shown in FIG. **269** with the disconnection coils **26920** and the screw wires **26930** lengthened to be even more flexible than in the above embodiments. As such, the coil/wire subassemblies are free to move into an orientation that does not impart this angling force on the stent lattice. The angular correction device **26910** is grounded at the connector control sub-assembly **17000** and has a band **26912** extending distally at least until the proximal end of the stent lattice. The band **26912** has guide sleeves **26814** that laterally hold one of the coil/wire subassembly **26920/26930** therein so that this one subassembly follows the band **26912**. In an exemplary embodiment, the band **26912** has guidewire sleeves **26916** on the side opposing the one subassembly to laterally hold the guidewire sleeve **26940** thereto.

[0516] With the angular correction device **26810**, the subassembly **26920/26930** held to the band **26912** will follow the superior axis of the aortic arch as shown in FIG. **270**, and, thereby, will cause the stent lattice to do the same. The flex of the band forces the band to the greater curvature of the arch and holds and supports the stent lattice in a position that is tangent to the end of the arch curve. This, in turn, places the lateral wall of the stent lattice on the superior side against the implantation site substantially aligned with (parallel to) the superior arch of the implantation site, which is a desirable implantation position for the stent lattice. As a result, the stent lattice is automatically placed coaxial to the axis on the native annulus, which is significant because coaxiality is the most desired orientation for limiting the depth into the left ventricular outflow tract and is required for lining up the sealing zone of the implant with the sealing area of the native annulus. An angular misalignment here can cause leaks or poor valve function. It can also contribute to lower gradients or negative impact on the hearts conduction system.

[0517] The band **26912** provides additional benefits: it supports the stent from rotating and gives the user an ability to apply longitudinal forces for precise positioning. Further, if the band **26912** is provided with forks or tines that engage the proximal end of the stent lattice, these tines can be used to rotate or translate the stent lattice as an additional measure for stent lattice repositioning.

[0518] With a parallel implantation orientation, expansion of the stent lattice (shown in FIG. **271**, will cause an implantation that is virtually parallel to the implantation site. An added benefit to this configuration is that the band **26912** (along with the delivery sheath) can be pulled proximally away from the implantation site, as shown in FIG. **272**, to rotate the plane of the stent lattice (here clockwise in the figure) to accommodate differing anatomy or for any other reason desired by the surgeon.

[0519] Because only one of the subassemblies is connected to the band **26912**, the other loose and flexible wires are free to reposition themselves within the arch and find a natural path that allows parallel implantation within the vessel, which paths are illustrated in FIG. **271**.

[0520] An exemplary embodiment of the band **26912** is a stainless steel strip that is much wider (e.g., between approximately 0.050 inches and approximately 0.2 inches) than it is thick (e.g., between approximately 0.005 inches and approximately 0.02 inches).

[0521] When the band **26912** presses the stent lattice in the implantation site, the orientation of the lateral wall of the stent lattice with respect to the implantation site will most likely be rotated too much in the clockwise direction of FIGS. **270** to **272**. In order to accommodate for this possibility, in an alternative and/or additional exemplary embodiment, the band **26912** can be provided with a pre-bend just proximal of the stent lattice. In such a configuration, the stent lattice will be rotated slightly counter-clockwise in the view of these figures. Another exemplary embodiment provides an active hinge at the distal end of the band **26912** just before the proximal end of the stent lattice and having a control rod that extends back towards the delivery handle. When actuated in the proximal direction, the end of the band will rotate counter-clockwise and, thereby, move the stent lattice similarly.

[0522] It is noted that various individual features of the inventive processes and systems may be described only in one exemplary embodiment herein. The particular choice for description herein with regard to a single exemplary embodiment is not to be taken as a limitation that the particular feature is only applicable to the embodiment in which it is described. All features described herein are equally applicable to, additive, or interchangeable with any or all of the other exemplary embodiments described herein and in any combination or grouping or arrangement. In particular, use of a single reference numeral herein to illustrate, define, or describe a particular feature does not mean that the feature cannot be associated or equated to another feature in another drawing figure or description. Further, where two or more reference numerals are used in the figures or in the drawings, this should not be construed as being limited to only those embodiments or features, they are equally applicable to similar features or not a reference numeral is used or another reference numeral is omitted.

[0523] The foregoing description and accompanying drawings illustrate the principles, exemplary embodiments, and modes of operation of the invention. However, the invention should not be construed as being limited to the particular embodiments discussed above. Additional variations of the embodiments discussed above will be appreciated by those skilled in the art and the above-described embodiments should be regarded as illustrative rather than restrictive. Accordingly, it should be appreciated that variations to those embodiments can be made by those skilled in the art without departing from the scope of the invention as defined by the following claims.

Claims

1. A stent assembly radially movable between a contracted configuration and a maximum expanded configuration, the stent assembly comprising: a plurality of longitudinally extending jack strut assemblies, wherein each jack strut assembly comprises: a proximal jack strut; a distal jack strut comprising: a smooth bore extending longitudinally at least partially therethrough; and a first connection part; a second connection part coupled to the first connection part of the distal jack strut and comprising a threaded bore longitudinally therethrough; and a jack screw connecting the proximal jack strut, the distal jack strut, and the second connection part, wherein the jack screw extends through the threaded bore of the second connection part and at least partially through the smooth bore of the distal jack strut.
2. The stent assembly of claim 1, wherein the stent assembly comprises nine longitudinally extending jack strut assemblies.
3. The stent assembly of claim 1, wherein the jack screw comprises external threads configured to threadingly mate with the threaded bore of the second connection part.
4. The stent assembly of claim 1, wherein the first connection part is disposed on a proximal end portion of the distal jack strut.
5. The stent assembly of claim 4, wherein the first connection part comprises a T-shaped head, wherein the second connection part comprises a nut, and wherein the nut comprises a cutout having an interior shape corresponding to an exterior of the T-shaped head of the first connection part.
6. The stent assembly of claim 4, wherein the distal jack strut comprises three spans connecting the first connection part to a distal part of the distal jack strut.
7. The stent assembly of claim 6, wherein the distal jack strut has a trapezoidal cross-section.
8. The stent assembly of claim 6, wherein the distal jack strut has a square cross-section.
9. The stent assembly of claim 6, wherein the three spans comprise two outer spans disposed radially outward of one inner span.
10. A stent assembly radially movable between a contracted configuration and a maximum expanded configuration, the stent assembly comprising: a plurality of longitudinally extending jack strut assemblies, wherein each jack strut assembly comprises: a proximal jack strut; a distal jack strut comprising a smooth bore extending longitudinally at least partially therethrough and a T-shaped proximal end portion; a connection portion comprising a threaded bore extending longitudinally therethrough and a cutout having an interior shape corresponding to an exterior of the T-shaped proximal end portion of the distal jack strut; and a jack screw coupling the proximal jack strut, the distal jack strut, and the connection portion, wherein the jack screw extends at least partially through the smooth bore of the distal jack strut and extends through the threaded bore of the connection portion.
11. The stent assembly of claim 10, wherein the connection portions of the plurality of longitudinally extending jack strut assemblies are staggered longitudinally about a circumference of the stent assembly.
12. The stent assembly of claim 10, wherein each jack strut assembly further comprises a back drive sleeve fixedly disposed on an outside of the jack screw.
13. The stent assembly of claim 12, wherein the proximal jack strut comprises a distal surface, and

wherein the back drive sleeve is disposed at a location longitudinally between the connection portion and the distal surface of the proximal jack strut.

14. The stent assembly of claim 12, wherein the back drive sleeve and the jack screw form a unitary component.

15. The stent assembly of claim 12, wherein the back drive sleeve is welded to the jack screw.

16. A prosthetic heart valve comprising: a stent assembly radially movable between a contracted configuration and an expanded configuration and comprising: a plurality of longitudinally extending jack strut assemblies, wherein each jack strut assembly comprises: a proximal jack strut comprising a bore extending longitudinally therethrough; a distal jack strut comprising a smooth bore extending longitudinally at least partially therethrough and a first connection part at a proximal end thereof; a second connection part coupled to the first connection part and comprising a threaded bore extending longitudinally therethrough; and a jack screw extending longitudinally through the bore of the proximal jack strut, the threaded bore of the second connection part, and at least partially through the smooth bore of the distal jack strut; and a plurality of leaflets coupled to the stent assembly.

17. The prosthetic heart valve of claim 16, further comprising a covering disposed around the stent assembly.

18. The prosthetic heart valve of claim 16, wherein each jack strut assembly further comprises a proximal drive screw coupler disposed at a proximal end portion of the jack screw.

19. The prosthetic heart valve of claim 18, wherein the proximal drive screw coupler is configured to engage a corresponding distal drive screw coupler of a delivery apparatus.

20. The prosthetic heart valve of claim 18, wherein the proximal drive screw coupler is configured to longitudinally secure the jack screw to a proximal end portion of the proximal jack strut.
